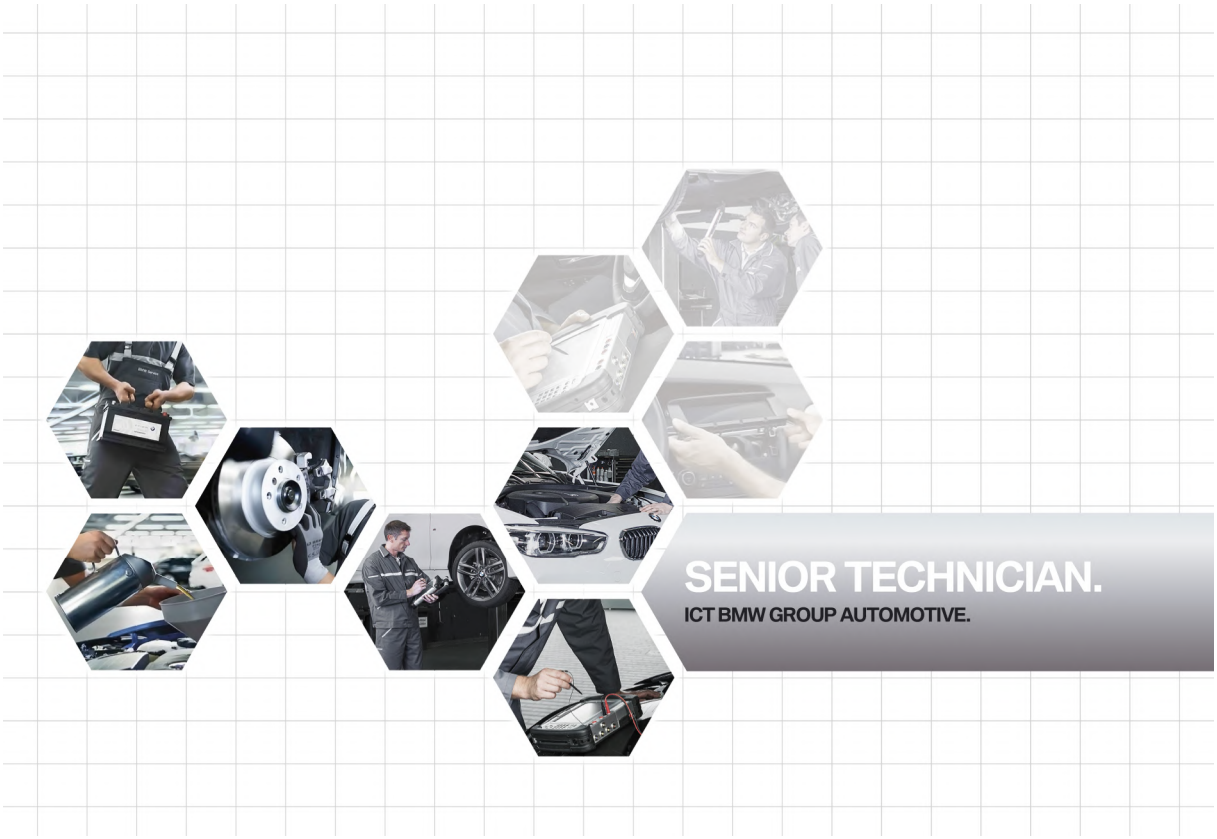
TECHNICAL QUALIFICATION.
PRODUCT INFORMATION.

SAFETY SYSTEMS.



BMW GROUP

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. This may result in differences compared with the content of this document.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

Contact: conceptinfo@bmw.de

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The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

Information status: **March 2017**
Technical qualification.

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SAFETY SYSTEMS.
1. INTRODUCTION.

1.1. Introduction

This product information is part of the “BMW Group Automotive Senior Technician" qualification programme and is a reference work for safety systems. The goal of this product information is to provide general background knowledge on the safety systems in current BMW and MINI vehicles.

Target group of qualification:

- BMW and MINI employees who have already successfully completed qualification as “BMW Group Automotive Technician”.
- New BMW and MINI employees who already have basic automotive engineering knowledge and are familiar with the basic knowledge relating to BMW and MINI vehicles.



These documents make information available that will serve as an overview and give you a better understanding of the qualification content. **As a rule, the instructions and other information in the workshop information system must be used for all maintenance and repair work.**



There are also extended regulations that apply to work on airbag systems in some countries. For example, certificates of competence are the minimum prerequisite for servicing and repairing airbag systems here.

1.2. Required competencies and knowledge

A number of competencies are required for the qualification programme “BMW Group Automotive Senior Technician”. The participants should already have these or have acquired them through qualification as “BMW Group Automotive Technician.

In general, the participants in the qualification programme should be able to perform general and simple repair work expertly and in accordance the specifications of the current repair instructions.

In addition, basic technical knowledge on the safety systems BMW and MINI vehicles is required:

Training module Electrical system I	
Advanced vehicle electrical system	• Introduction to control units • Door airbag sensor, acceleration sensor • Installation position of various pyrotechnic actuators • Active and passive pedestrian protection, sensor system • Task of ACSM • Rollover protection system

SAFETY SYSTEMS.
1. INTRODUCTION.

1.3. BMW guidelines



When working on the safety system, the BMW guidelines for handling pyrotechnic components must be observed in all cases. This applies in particular to disassembly and installation of pyrotechnic components. Particular attention must be paid to the regulations for straightening and welding work on the vehicle. The valid regulations must be observed in each case when vehicles are scrapped.

1.4. Pyrotechnic objects

Since pyrotechnic objects are filled with small quantities of explosives, they are subject to the law relating to the use of explosives in Germany. For this reason, the legal regulations of the respective country must be observed worldwide when handling pyrotechnic objects. In Germany, this includes nomination of a responsible person as well as compliance with regulations for storage and transport.



Failure to comply with warning and repair instructions can cause accidental activation and lead to injuries.

1.5. History of restraint systems at BMW

Many lives have been saved since the introduction of the seat belt in the 60s followed by the requirement for mandatory seat belt usage in Germany as from 1976. The three-point seat belt is recognised worldwide as the most effective protective system for vehicle occupants in the event of accidents.

Passive safety was revolutionised by the additional development of airbags in the 70s and 80s. Since BMW introduced the driver's airbag to vehicles in the middle of the 1980s, passive safety has been improved significantly.

Over the course of the years, further airbag systems have been developed for specific applications.

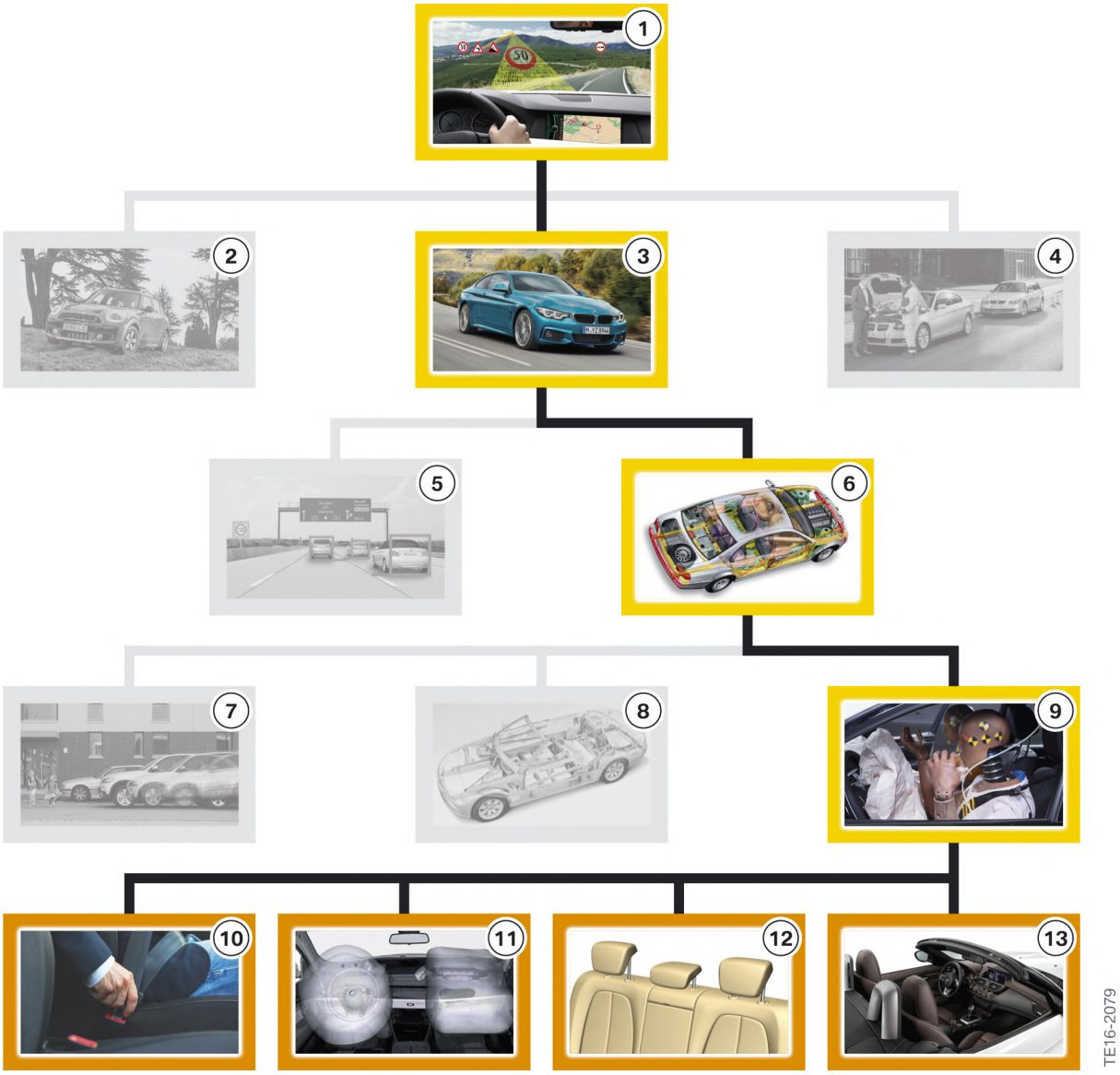
Among other things, the side airbags in the doors were replaced by the side airbags in the backrests. A further innovation from the BMW Development department was the introduction of the head airbag.

A pyrotechnic battery terminal clamp was also developed with disconnects the positive battery cable to the engine compartment from the battery in the event of a crash and thus prevents a short circuit.

As an innovative pioneer of passive safety systems, BMW was not satisfied with just the introduction of the airbag, but developed its own comprehensive safety concept with the name Fully Integrated Road Safety Technology, or **F.I.R.S.T.** for short.

The following graphic shows the structure of the F.I.R.S.T. safety concept:

SAFETY SYSTEMS.
1. INTRODUCTION.



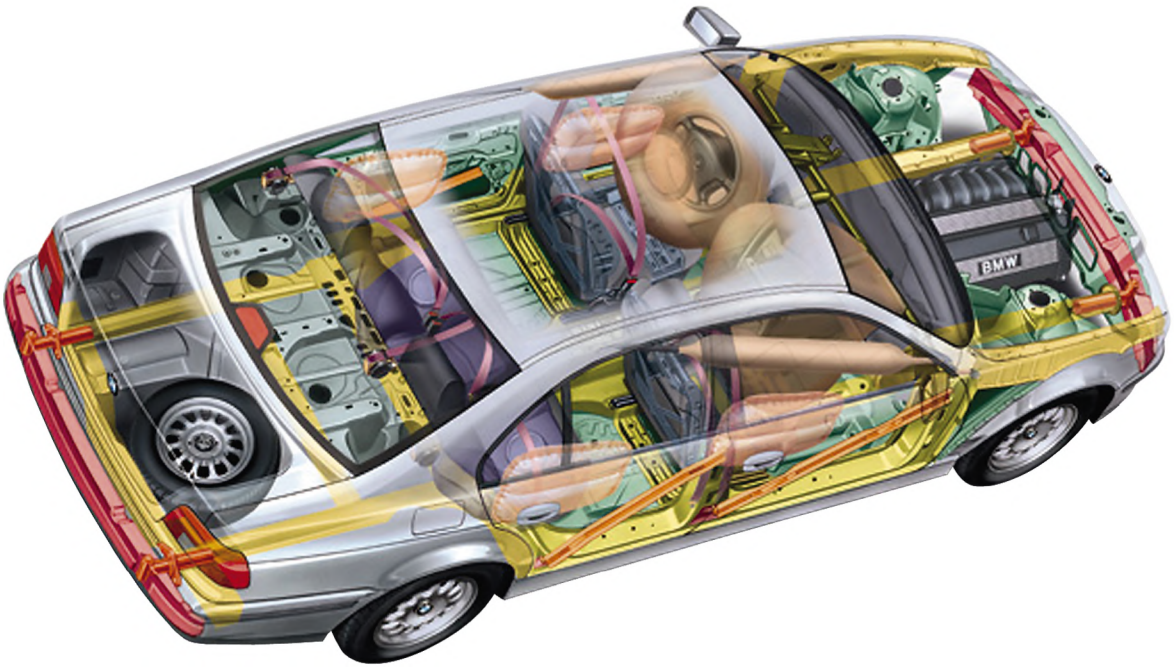
Safety concept F.I.R.S.T.

Index	Explanation
1	Safety in road traffic
2	Environment
3	Vehicle
4	Human
5	Active safety
6	Passive safety
7	Partner protection
8	Vehicle Protection

SAFETY SYSTEMS.
1. INTRODUCTION.

Index	Explanation
9	Occupant safety
10	Seat belt
11	Airbag
12	Head restraints
13	Overturn safety

The basis of the BMW F.I.R.S.T. safety concept is a passenger compartment which remains stable under the widest range of accident conditions possible. The combination of innovative restraint systems and deformation zones designed for real accidents offers effective protection for vehicle occupants. Thanks to intensive research and development work with involvement of BMW accident analysis department, the outstanding role of BMW in the field of vehicle safety is being continuously strengthened.



TE17-0327

Safety system of a E38

SAFETY SYSTEMS.
2. CONTROL UNITS.

2.1. Development of various safety systems

As a result of the rapid and continuous development of new passive safety components in the 90s, the number of interfaces on the control units has also multiplied. At the time, the increase in the number of sensors and actuators for control units was limited by the computing capacity of the micro controllers. In order to provide customers with the maximum possible passive safety, the sensors and actuators were therefore distributed among several control units. Among other things, this led to the development of the Intelligent Safety Integration System ISIS in the E65. The control units were connected with each other via a fast optical system, the **byteflight**. The Advanced Safety Electronics system (ASE) was used as the passive safety system with the introduction of the E60. The ASE is based on the technology of the Intelligent Safety Integration System ISIS with the **byteflight**.

Due to the rapid further development of micro controllers with every increasing computing capacity and speed, it is today possible to process and calculate all functions in 1 or 2 microcontrollers in one control unit.

For this reason, the **A**dvanced **C**rash and **S**afety **M**anagement **ACSM** has been used since 09/2005 in all models of the BMW 5 series and 6 series and since 01/2006 in the E85 LCI.

The ACSM is the successor of the **M**ultiple **R**estraint **S**ystem MRS. The MRS control unit is still installed in many vehicles today and includes the same functions as the ACSM.

The following table summarises the most important information on development of the multiple restraint systems from Cipro through to the MRS 4RD:

SAFETY SYSTEMS.
2. CONTROL UNITS.

Control unit	Description
	**Cipro/Siemens 2A/B** The first airbag system was introduced at BMW in 1985 and was from the company Cipro. The control unit was later replaced by the Siemens 2A/B. The airbag system comprised a single-stage driver's airbag with mechanical airbag front sensors close to the spring strut domes.
	**Siemens 2C** The Siemens 2C was introduced in 09/1992. The difference compared with the previous system 2A/B was extension by a front passenger airbag.
	**Euro airbag** The Euro airbag was developed especially for the European market. The volume of the Euro airbag is smaller than a full-size airbag so that it can be integrated in a sports steering wheel.
	**Basic gas generator BAE** The basic gas generator was used worldwide except for in US vehicles. A piezomechanical acceleration sensor is integrated in the basic gas generator. The mechanical airbag front sensors were omitted. The functional enhancements were the seat belt tensioners for driver and front passenger, as well as seat occupancy detection as from 09/1994. The basic gas generator was the first control unit with different triggering thresholds and was encoded on an equipment- and country-specific basis.
	**Multiple restraint system MRS 1** Changeover to the multiple restraint system MRS took place as from 03/1996. The MRS 1 was equipped with externally located multiple restraint system acceleration sensors for side impact detection. Side airbags were used for the first time in the driver and front passenger doors.

SAFETY SYSTEMS.
2. CONTROL UNITS.

Control unit	Description
 TE05-1483	**Multiple restraint system MRS 2** The MRS 2 was installed as from 05/1997. With the MRS 2, the functions were extended with head airbag ITS, rear side airbags and the safety battery terminal SBK.
 TE05-1484	**Multiple restraint system MRS 3/4** The MRS 3 increasingly replaces the MRS 2 as from 03/1999. The MRS 3 was the first airbag system with bus connection. It was extended by functions such as the two-stage driver's airbag, an additional head airbag for the C-pillar and a second safety battery terminal for the 750i. The electric fuel pump was additionally deactivated in the event of a crash. The MRS 4 was equipped with a more powerful processor, but included the same range of functions as the MRS 3.
 TE05-1485	**Multiple restraint system MRS 4 RD** The multiple restraint system 4 Redesigned, MRS 4 RD for short, was used only in the E83. It is the last airbag control unit with K-bus connection. The MRS 4RD was characterised by a newer sensor system. The multiple restraint system acceleration sensors were replaced by the airbag sensors on the B-pillar. In addition, pressure sensors (door) were used for side impact detection and airbag front sensors for frontal collision detection (only until 09/2004 in Europe).

2.2. Multiple restraint system MRS

Due to the continuous further development, more precise details will not be explained as from the 5th generation.

2.2.1. Multiple restraint system 5

The MRS 5 as the task of detecting a accident situations that are critical for the occupants and then selectively activating the necessary restraint systems depending on the crash severity.

SAFETY SYSTEMS.
2. CONTROL UNITS.

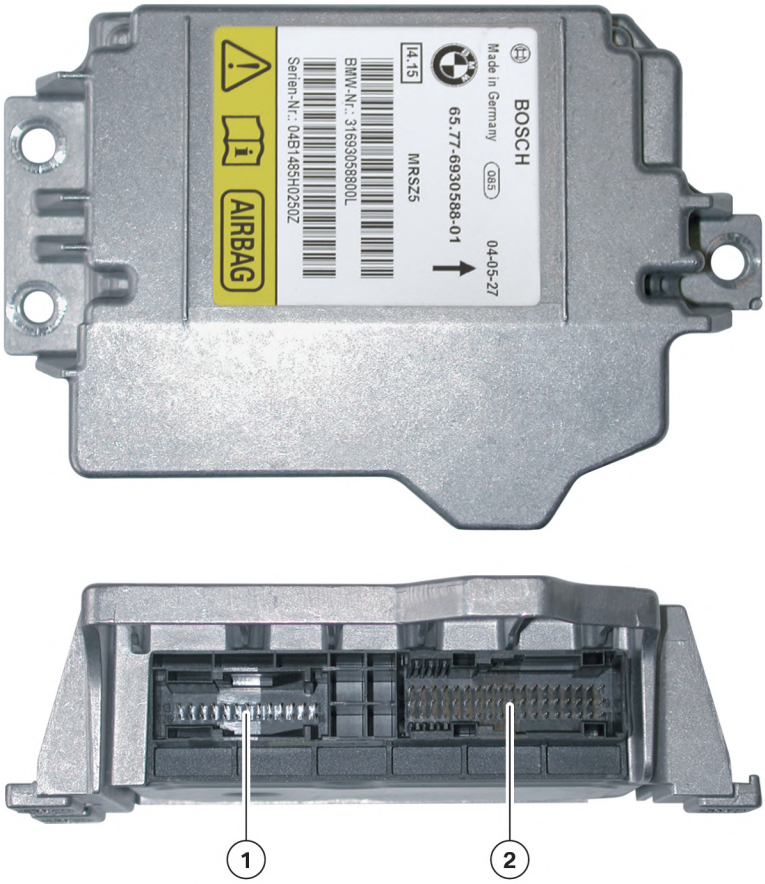
Changeover to the new vehicle electrical system structure with K-CAN connection took place with the introduction of the E87. The sensor concept of the MRS 4RD was adopted for the MRS 5 and configured depending on vehicle.

The multiple restraint system 5 consists of the following basic components:

- MRS5 control unit
- Sensors and switches
- Actuators.

The MRS5 control unit is located centrally in the vehicle on the transmission tunnel.

The following graphic shows the 5th generation multiple restraint system:



TE17-0319

Multiple restraint system 5

Index	Explanation
1	Connector, 26-pin, for cockpit wiring harness
2	Connector, 54-pin, for vehicle wiring harness

SAFETY SYSTEMS.
2. CONTROL UNITS.

Crash-relevant functions

The MRS 5 control unit must perform the following functions:

- Crash detection and ignition point determination
- Activation of the ignition output stages
- Documentation of crash data
- System self test
- Cyclic monitoring
- Display of system operability
- Fault display and fault storage
- Output of faults (diagnosis)
- Output of the crash telegram for other users in the communication network
- Activation of indicator lamp for front passenger airbag deactivation.

Sensor system

Airbag sensors are installed in the B-pillars in addition to the two acceleration sensors in the MRS 5 control unit. The airbag sensors consist of a longitudinal and lateral acceleration sensor in each case. The longitudinal and lateral acceleration sensors serve the purpose of side impact detection together with the acceleration sensors in the MRS control unit. The longitudinal acceleration sensors serve the purpose of front or rear crash detection.

Crash detection and ignition point determination

The MRS 5 control unit evaluates the values detected by the sensors and determines whether there is a crash. In order for a crash to be detected, the threshold values of 2 independent sensors must be detected. For example, in the case of a front-end crash, the acceleration values must be detected from the airbag sensor in the B-pillar and the acceleration sensor in the control unit. The values are used for determination of the crash severity and crash direction. An algorithm determines the ignition points of the restraint systems to be activated on the basis of the crash severity and direction.

Crash entries

In the event of a crash with activation of one or more actuators, the ignition commands must be recorded. The most important data for a crash is recorded in the control unit memory in the form of a crash telegram.

Cyclic monitoring

Once the system self-test has been successfully concluded and the system is ready for operation, a cyclic monitoring procedure is performed for fault monitoring purposes.

Fault display and fault storage

If there is a fault in the system, this is indicated by the indicator lamp for front passenger airbag deactivation. If a fault occurs in the MRS 5 system, this will be entered in the fault memory.

SAFETY SYSTEMS.
2. CONTROL UNITS.

Output of faults (diagnosis)

Using the BMW diagnostic system, the fault memory can be read out via the diagnostic interface. After rectification of the faults, the fault memory can be cleared by means of the diagnostic command “Delete fault memory”.

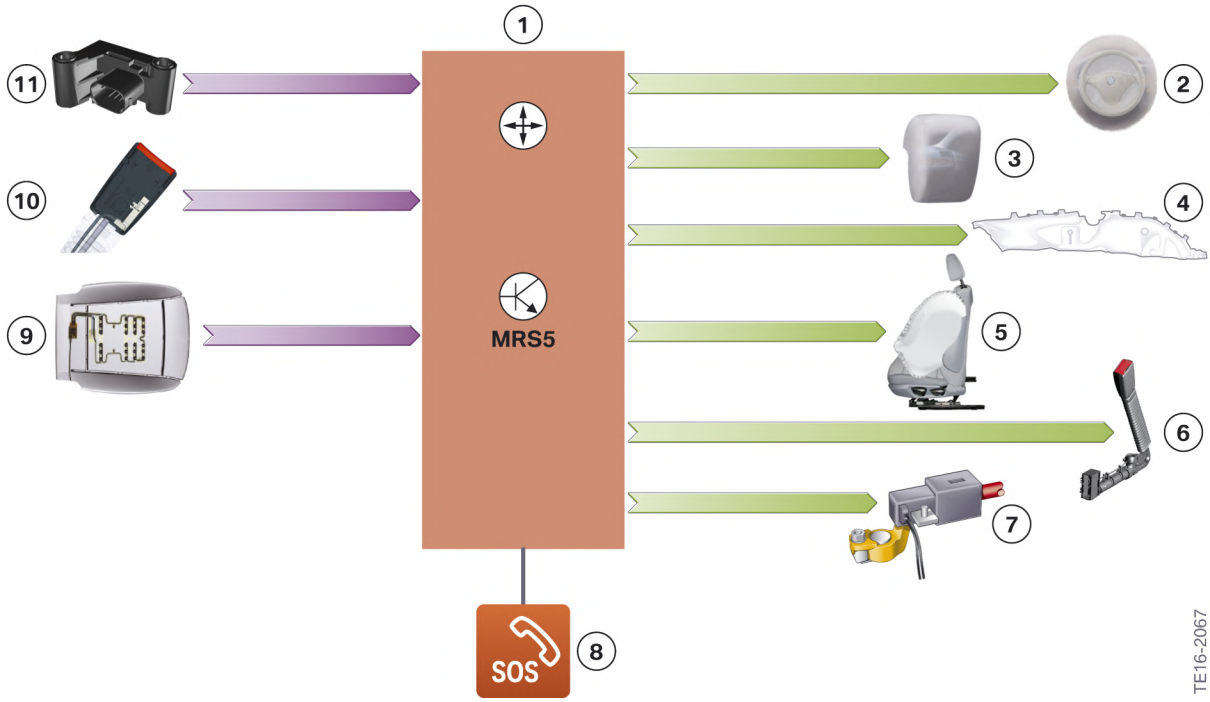
Output of crash telegram

In the event of a crash with activation of the restraint systems, the MRS 5 control unit sends a crash telegram to the users in the bus network. The respective control units then perform the following functions depending on the crash severity:

- Switch off electric fuel pump
- Switch off alternator
- Release central locking system
- Switch on hazard warning flashers
- Switch on interior light
- Make emergency call.

Inputs/outputs

The following graphic shows the sensors and actuators of the MRS 5:



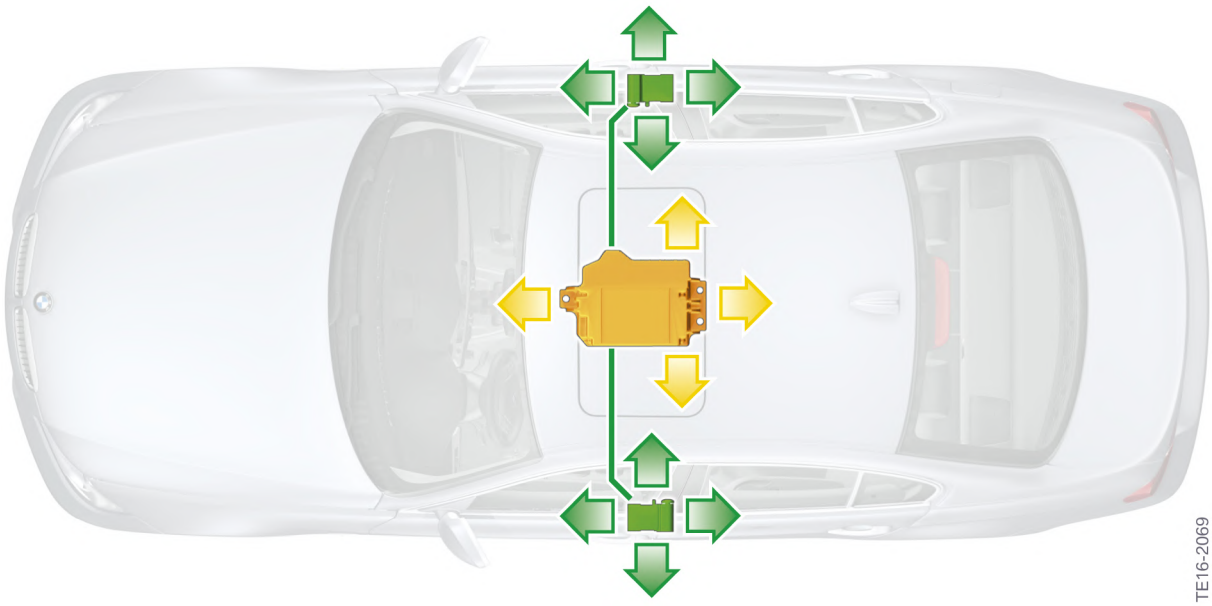
Input/Output MRS 5

SAFETY SYSTEMS.
2. CONTROL UNITS.

Index	Explanation
1	Multiple restraint system 5
2	Driver's airbag
3	Front passenger airbag
4	Head airbag
5	Side airbag
6	Belt tensioner
7	Safety battery terminal SBK
8	Emergency call
9	Seat occupancy mat
10	Seat belt buckle switch
11	B-pillar airbag sensor

Installation position and communication

This graphic shows the installation position and communication between the sensors and the MRS 5:



Installation position and communication between sensors and MRS 5

Zusammenfassung

The 5th generation MRS is used in these BMW models:

Generation	vehicles
MRS 5	• E81, E82, E87 • E84 • E90, E91, E92

The 5th generation MRS is used in these MINI models:

Generation	vehicles
MRS 5	• R50 • R52 • R53

2.2.2. Multiple restraint system 6

The 6th generation multiple restraint system consists of the following basic components:

- MRS 6 control unit
- Sensors and switches
- Actuators.

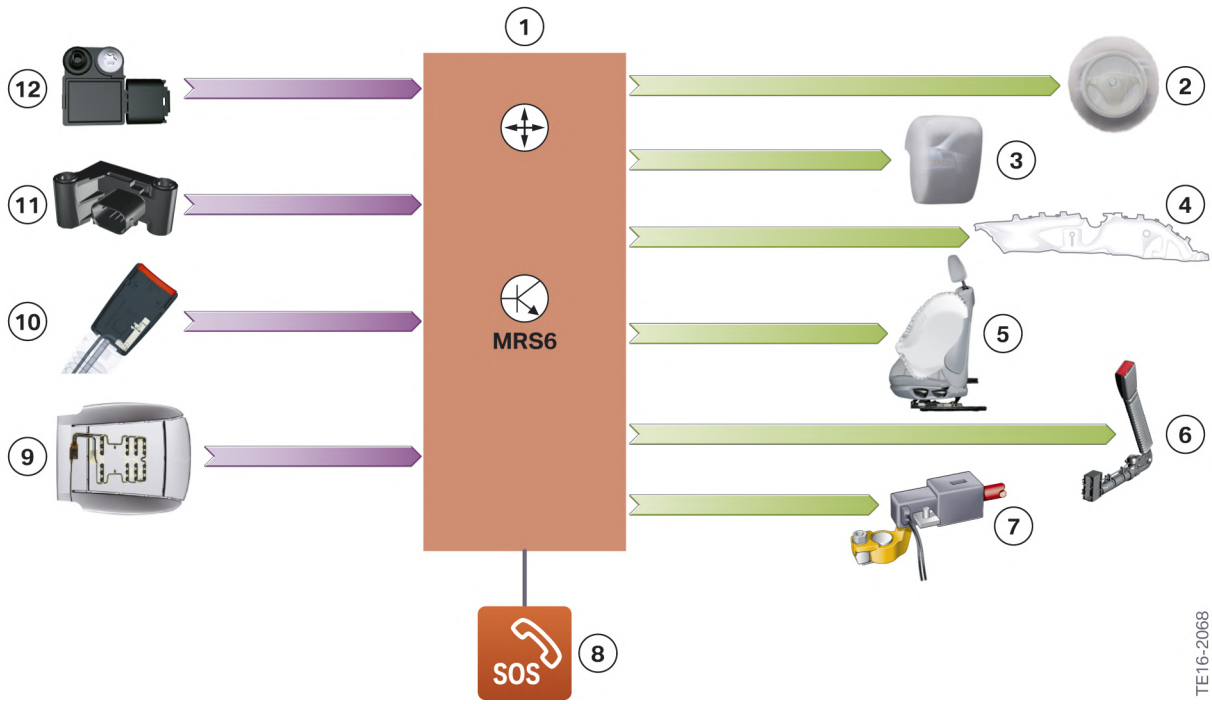
The MRS 6 is a further development of the MRS 5. New features include the acoustic seat belt warning function for the rear seats as well as the pressure sensors in the front doors.

The MRS 6 control unit is located in the centre of the vehicle under the parking brake. It consists of a die-cast housing with integrated connector cover.

Inputs/outputs

The following graphic shows the sensors and actuators of the MRS 6:

SAFETY SYSTEMS.
2. CONTROL UNITS.



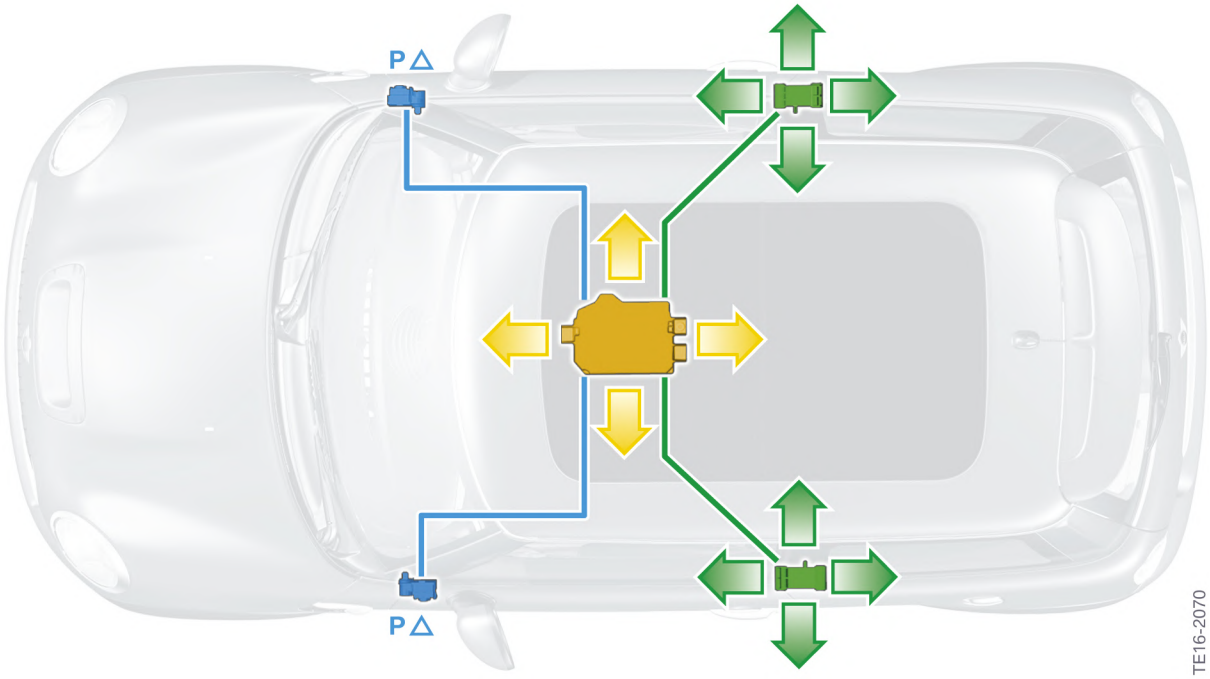
Input/Output MRS 6

Index	Explanation
1	Multiple restraint system 6
2	Driver's airbag
3	Front passenger airbag
4	Head airbag
5	Side airbag
6	Belt tensioner
7	Safety battery terminal SBK
8	Emergency call
9	Seat occupancy mat
10	Seat belt buckle switch
11	B-pillar airbag sensor
12	Pressure sensor (door)

Installation position and communication

This graphic shows the installation position and communication between the sensors and the MRS 6:

SAFETY SYSTEMS.
2. CONTROL UNITS.



Installation position and communication between sensors and MRS 6

Zusammenfassung

The 6th generation MRS is used in these MINI models:

Generation	vehicles
MRS 6	• R55 • R56 • R57 • R58, R59 • R60 • R61

The 7th generation multiple restraint system MRS is a continuous further development of the multiple restraint systems in BMW vehicles.

The MRS 8 additionally monitors the active pedestrian protection and can be found in all LCI versions of the vehicles E82, E88, E92 and E93.

This table contains an overview of the 7th and 8th generation MRS control units:

Generation	vehicles
MRS 7	• E89
MRS 8	• E82, E88 • E92, E93

2.3. Crash Safety Module

In order to guarantee optimum occupant protection, so-called Crash Safety Modules (ACSM) are used in today's vehicles. In the event of a crash, the individual ignition circuits are activated by a central control unit depending on the type and severity of the accident. This is referred to as the “inner safety zone”. This reduces the risk of injury in the interior of the passenger cell through restraint systems and collision protection measures.

Actuators which can be activated by the control unit include, for example:

- Driver's airbag
- Front passenger airbag
- Side airbag
- Head airbag
- Seat belt tensioner.

The Crash and Safety Management (CSM) comprises the following basic components:

- Crash Safety Module
- Sensors and switches
- Actuators.

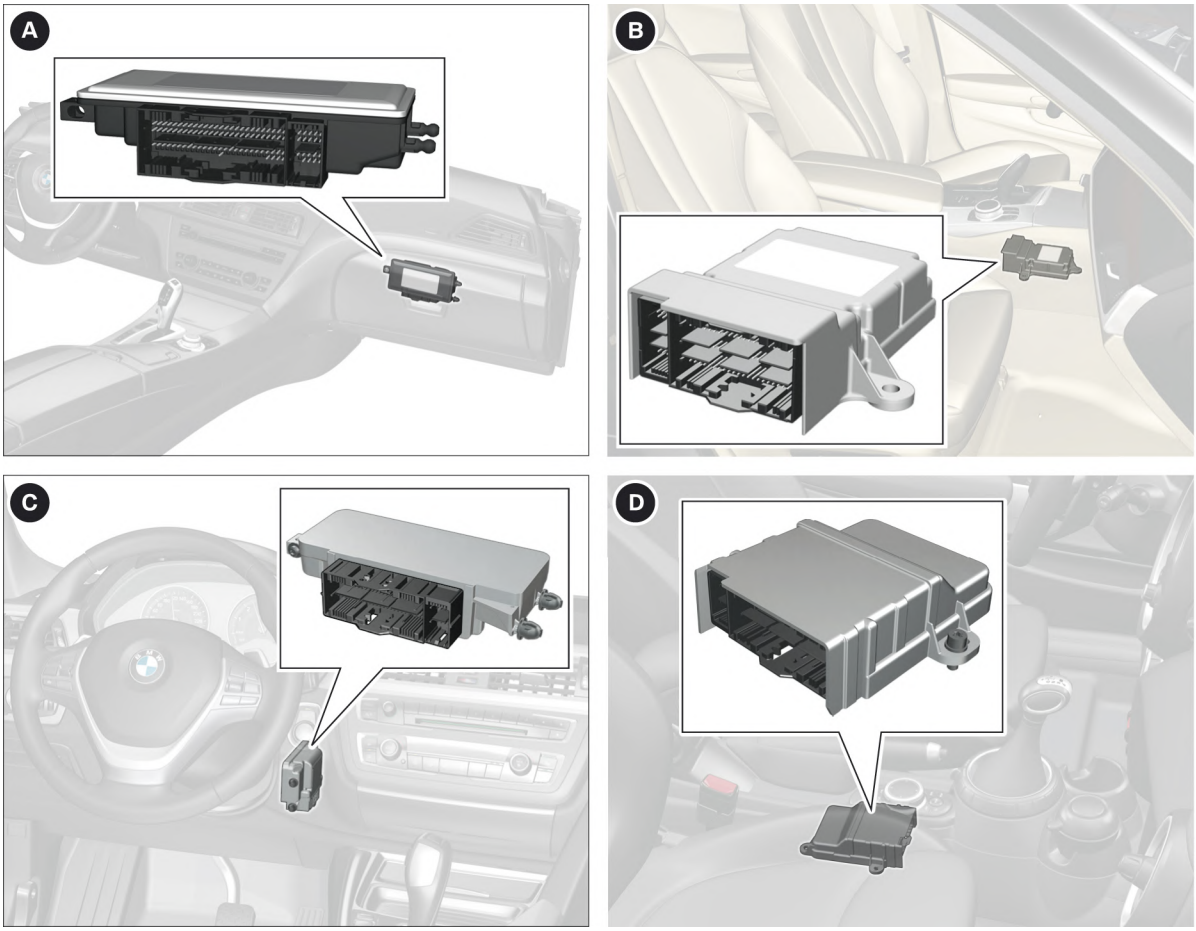
The Crash Safety Module contains 2 acceleration sensors offset at an angle of 90°. The acceleration sensors measure the longitudinal and lateral acceleration of the vehicle.

2.3.1. Installation location

Depending on the vehicle version, the Crash Safety Module is located either centrally in the vehicle on the transmission tunnel or behind the glove box.

This graphic shows the different installation locations in the BMW and MINI models:

SAFETY SYSTEMS.
2. CONTROL UNITS.



Different installation locations of the ACSM control unit

Index	Explanation
A	ACSM behind the glove box (e.g. F01)
B	ACSM on the transmission tunnel under the centre console (e.g. G30)
C	ACSM on the driver's side on the right under the steering wheel (e.g. F20)
D	ACSM on the transmission tunnel under the centre console (e.g. F54)

2.3.2. Tasks

The function of the Advanced Crash Safety Module is to permanently evaluate all sensor signals in order to identify a crash situation. On the basis of the sensor signals and their evaluation, the Crash Safety Module identifies the direction of the crash and the severity of the impact.

The Crash Safety Module evaluates the sensor information and then initiates the relevant measures for selective activation of the necessary restraint systems.

The Crash Safety Module simultaneously monitors the system itself and indicates readiness for operation of the system by switching off the airbag indicator light.



Airbag indicator light in the instrument cluster

If the airbag control unit detects a fault during operation, this is stored in the fault memory of the airbag control unit. The fault memory in the airbag control unit is read out for diagnostic purposes.

If a crash situation is detected, the other control units are notified by means of a crash signal. The relevant control units respond to this signal by executing their own activities according to the severity of the crash.

These activities include:

- Opening the central locking system
- Activating the hazard warning flasher
- Switching on the interior light
- Switching off the fuel pump control electronics
- Making an emergency call.

An additional function of the airbag control unit is the acoustic seat belt warning that reminds the driver and front passenger to fasten their seat belts using visual and acoustic signals. Seat belt buckle switches are used to detect whether the driver and front passenger have fastened their seat belts. For an acoustic signal to sound, the control unit requires information about the occupancy of the front passenger seat.

The functions of the Crash Safety Module generally belong to one of the following areas:

- Crash-relevant functions
- System monitoring functions
- Additional convenience functions.

In the case of convertibles, such as the E88, E93 or the E64, which may also be equipped with an ACSM, there is an additional sensor system for roll-over detection. If an imminent rollover is detected, the rollover protection controller is requested to activate the rollover protection system via the K bus. In newer vehicle models, roll-over detection is integrated in the ICM or directly in the ACSM.

2.3.3. Crash-relevant functions

The Crash Safety Module must fulfil the following crash-relevant functions:

- Evaluating the sensor signals
- Crash detection
- Determining actuators to be activated
- Specifying the trigger time and trigger sequence
- Activation of the ignition circuit output stages

- Sending the crash message to all bus users
- Crash documentation
- Emergency call function.

Evaluating the sensor signals

The sensors have the task of detecting and verifying a frontal crash, side collision and rear-end collision as well as roll-over detection.

The sensors are directly connected to the airbag control unit, .where their signals are evaluated and processed.

Crash detection for European version of vehicles

The European version of the vehicles is equipped with the following sensors:

- One transverse and longitudinal acceleration sensor each in the B-pillars
- One airbag sensor for pressure detection in each of the front doors
- One lateral and one longitudinal acceleration sensor in the ACSM
- A roll rate sensor in the ACSM (in combination with the optional equipment Dynamic Damper Control (OE 223))
- A vertical acceleration sensor in the ACSM.

A side collision is detected with the assistance of the lateral acceleration sensors.

Rollover is detected by the roll rate sensor and vertical acceleration sensor.

A crash whose effective direction does not correspond to the longitudinal or transverse axis of the vehicle is detected by a combined lateral and longitudinal acceleration.

The longitudinal acceleration sensors can measure both positive and negative acceleration. A frontal crash or rear-end collision is detected by means of these signals.

Trigger time and trigger sequence

The airbag control unit detects the crash direction and crash severity on the basis of the values transmitted by the sensors.

In the event of a frontal collision, correspondingly high acceleration forces must be detected by the longitudinal acceleration sensor in the B-pillar and by the longitudinal sensor integrated in the ACSM or ICM (depending on the vehicle type), for example. This information is used to calculate the activation times and sequence of the restraint systems to be activated.

A possible imminent rollover is also detected and the appropriate protection systems such as the rollover protection bar are activated.

Activation of the ignition circuit output stages

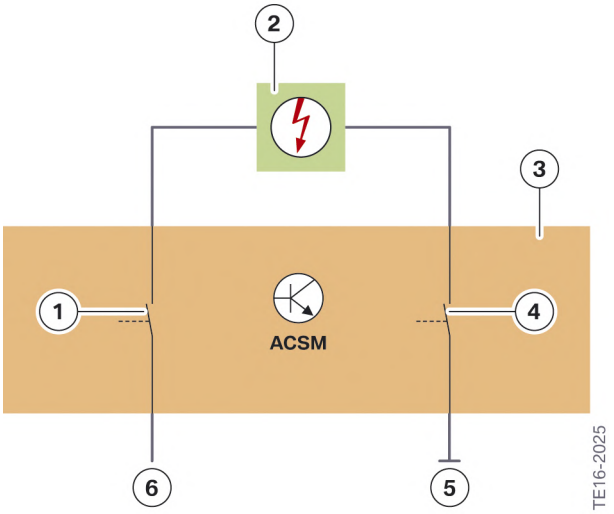
These ignition capacitors make the firing energy available in the event of a crash. If the voltage supply is interrupted during a crash, the ignition capacitors serve briefly as an energy reserve.

SAFETY SYSTEMS.
2. CONTROL UNITS.

The ignition circuit output stages switch the energy required for activation of the ignition squibs. There is one switch for each ignition circuit in the output stages. The first power switch switches the ignition voltage, while the second power switch switches ground. The ignition circuit output stages are controlled by a microcontroller in the airbag module.

Two connections are intentionally switched for the ignition circuits so that short circuits in the lines to the ignition squib do not lead to undesired activation.

The following graphic shows the function of an ignition circuit output stage:



Ignition circuit output stage

Index	Explanation
1	Power switch for ignition voltage
2	Ignition squib
3	Ignition circuit output stage integrated in the ACSM
4	Power switch for ground
5	Ground
6	Ignition capacitor as energy reserve

Sending the crash message

In the event of a collision with activation of the restraint systems, the airbag control unit sends a crash signal to the bus users in the bus network.

The telematics control unit is activated in parallel via a direct single-wire connection.

The telematics control unit serves as a data modem in the vehicle. The emergency GSM aerial has been integrated in the telematics control unit since the Telematic Communication Box 2. This can be used to establish a telephone connection via the connected vehicle microphone even if the vehicle has rolled over. In older vehicles, the emergency GSM aerial was always realised as a separate aerial under storage shelf of the vehicle. Emergency calls are made possible by means of a SIM card.

Depending on the crash severity, the following functions are performed by the respective control units:

- Switch off electric fuel pump
- Switch off the auxiliary heater
- Switch off hazard warning lights
- Switch on interior light
- Make emergency call.

Crash documentation

In the event of a collision where at least one actuator is activated, a crash entry is stored in a non-erasable memory in the ACSM. The airbag indicator light also lights up continuously.



The crash entries are stored during the course of an accident event. Each crash entry is assigned a system time. The control unit remains capable of firing even after the crash entries. The crash entries cannot be erased and serve the purpose of subsequent device diagnosis. Only a certain number of crash entries are stored. The control unit must then be replaced.

2.3.4. System monitoring functions

The Crash Safety Module has the following system monitoring functions:

- System self test
- Display of system operability
- Cyclic monitoring
- Fault display and fault storage
- Output of faults (diagnosis)
- Acoustic and visual seat belt warning
- Deactivation of the front passenger airbag and side airbag on the front passenger side with front passenger airbag deactivation (OE 5DA).

System self test

The Crash Safety Module performs a system self test from terminal 15. The airbag indicator light is activated for approx. 5 seconds during the system self-test.

Once the system self-test is complete and no faults have been identified, the airbag indicator light goes out and the system is operational.

Display of system operability

The airbag indicator light in the instrument panel goes out to indicate that the Crash Safety Module is ready for operation.

Cyclic monitoring

Once the system self-test has been successfully concluded and the system is ready for operation, a cyclic monitoring procedure is performed for fault monitoring purposes. This cyclical monitoring serves the internal diagnosis of the control unit and overall airbag system. Cyclical monitoring takes place continuously from terminal 15. This also continues when logical terminal R is reached after the engine is switched off.

Fault display and fault storage

The Crash Safety Module has a non-volatile fault memory. The airbag indicator light lights up to indicate a fault code entry.

Events such as activation of an airbag or seat belt buckle pretensioner are also stored in the fault memory.



Entry of an activated restraint system in the fault memory does not mean that the restraint system was faulty in the event of a crash. It means that the activated restraint system is not available for further activation.

Output of faults (diagnosis)

The fault memory is read out via the diagnostic interface using the workshop information system. After rectifying the faults or after renewing activated components, the fault memory is cleared with the diagnostic command "Clear fault memory".

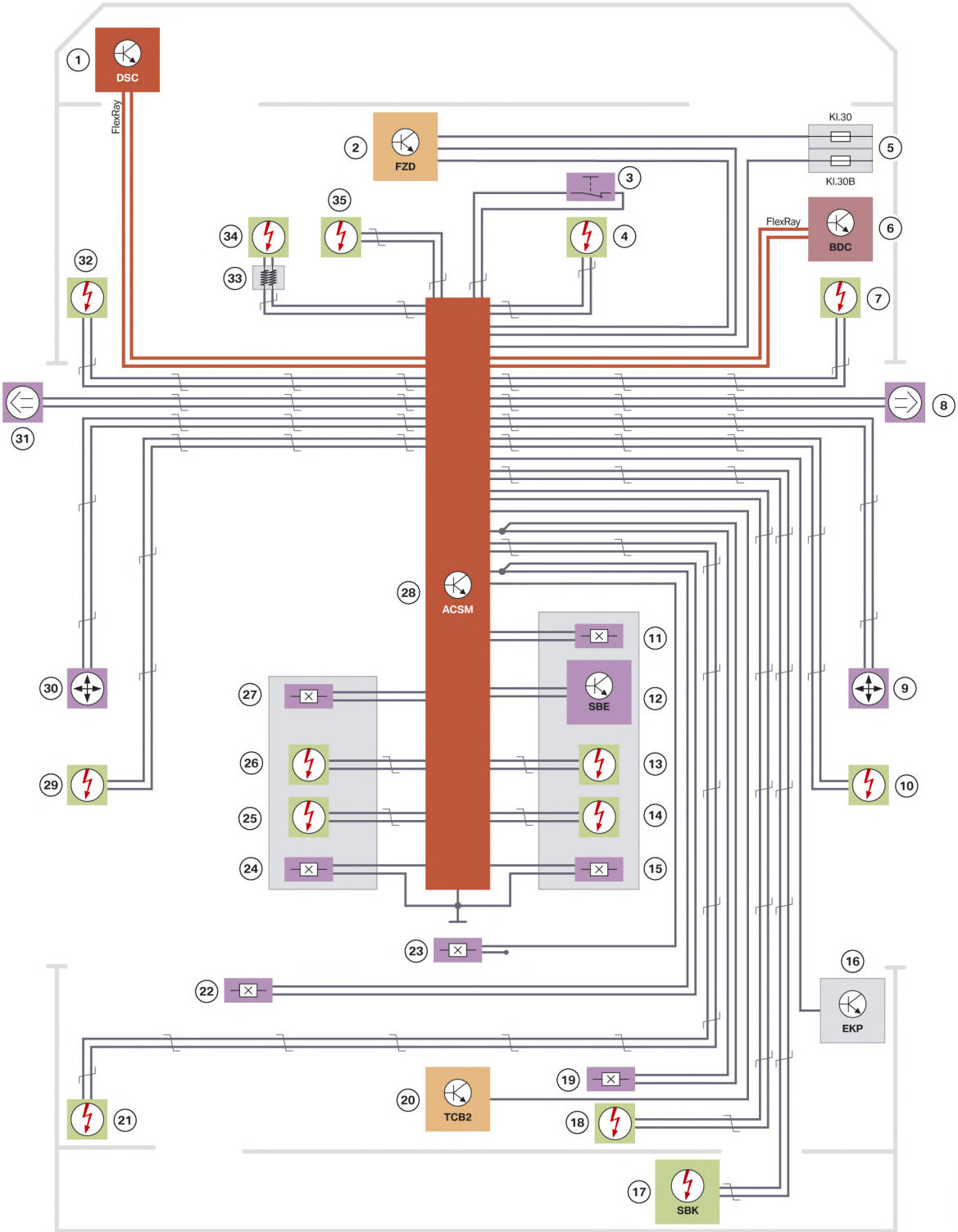
Acoustic and visual seat belt warning

An acoustic and visual seat belt warning is a standard feature of all vehicles equipped with the Crash Safety Module. The Crash Safety Module detects whether or not the driver and front passenger have fastened their seat belts. If they have not, an acoustic and visual warning is output to remind them to fasten their seat belts. Both seat belt buckle switches are monitored separately.

2.3.5. System wiring diagram

The following graphic shows a system wiring diagram of a G30.

SAFETY SYSTEMS.
2. CONTROL UNITS.



G30 system wiring diagram in European version with 5th generation ACSM

SAFETY SYSTEMS.
2. CONTROL UNITS.

Index	Explanation
1	Dynamic Stability Control (DSC)
2	Roof function centre (FZD)
3	Switch for front passenger airbag deactivation
4	Front passenger airbag
5	Fuses in power distribution box, front right
6	Body Domain Controller (BDC)
7	Head airbag, right
8	Airbag sensor, door, right (pressure)
9	Acceleration sensor, B-pillar on right
10	Reel pretensioner, front passenger
11	Seat-position sensor (on vehicles without seat module)
12	Seat occupancy mat
13	Crash-active headrest, front passenger
14	Side airbag, front passenger
15	Seat belt buckle switch, front passenger
16	Electronic fuel pump control
17	Safety battery terminal SBK
18	Reel pretensioner, rear right
19	Seat belt buckle switch, rear right
20	Telematic Communication Box 2 (TCB2)
21	Reel pretensioner, rear left
22	Seat belt buckle switch, rear left
23	Seat belt buckle switch, rear centre
24	Seat belt buckle switch, driver
25	Side airbag, driver's side
26	Crash-active headrest, driver
27	Seat-position sensor, driver (on vehicles without seat module)
28	Crash Safety Module (ACSM)
29	Reel pretensioner, driver
30	Acceleration sensor, B-pillar on left
31	Airbag sensor, door, left (pressure)
32	Head airbag, left
33	Clock spring
34	Driver's airbag
35	Knee airbag, driver

2.4. Overview of Crash Safety Module generations

Information is provided here on the development of the Crash Safety Module from the 1st generation.

2.4.1. 1st generation

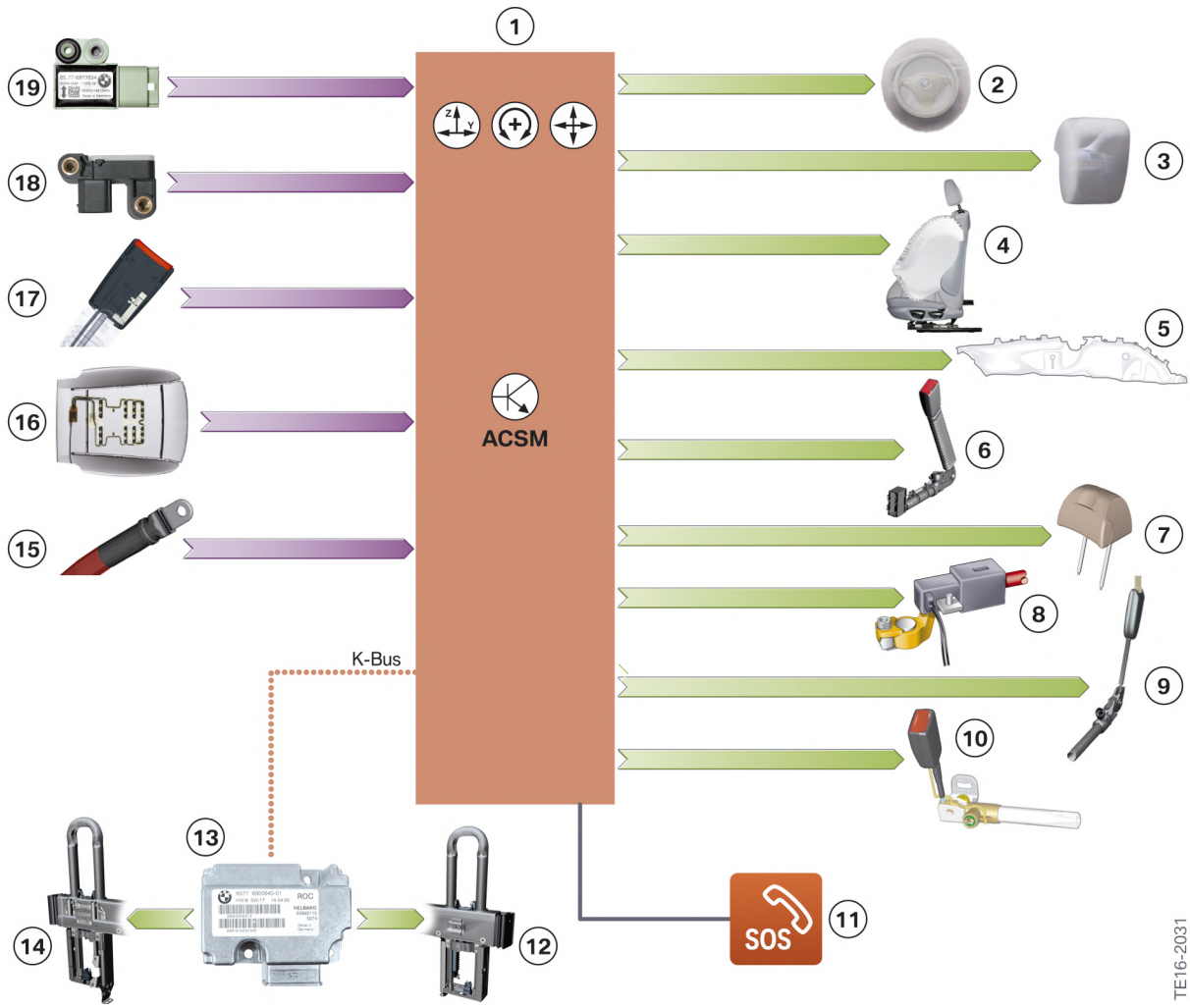
The Advanced Crash Safety Module ACSM is characterised by the fact that it performs the same range of functions as the ASE system. The ASE system consists of 7 individual control units. The ACSM has only one control unit. Use of only one control unit has the following advantages:

- Reduction in the number of control units
- Weight reduction
- Simpler diagnosis
- Easier programming/updates.

Inputs/outputs

The following graphic shows the sensors and actuators of the ACSM 1:

SAFETY SYSTEMS.
2. CONTROL UNITS.



Input/Output ACSM 1

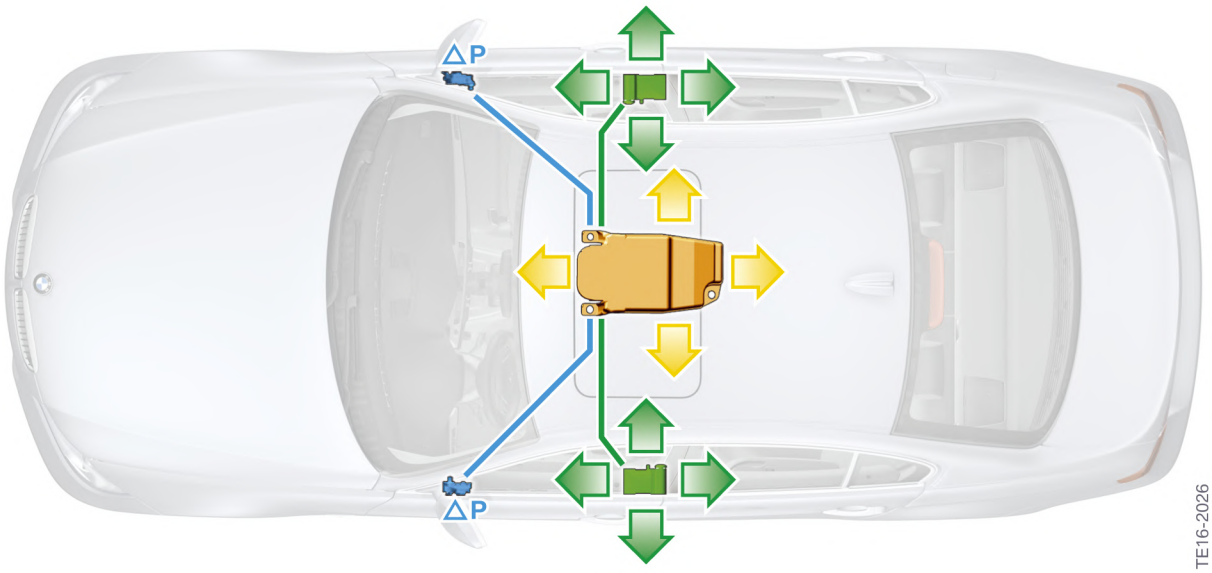
Index	Explanation
1	Crash Safety Module (ACSM)1
2	Driver's airbag
3	Front passenger airbag
4	Side airbag
5	Head airbag
6	Belt tensioner
7	Crash-active headrest
8	Safety battery terminal SBK
9	End fitting pretensioner
10	Seat belt buckle pretensioner, rear passenger compartment
11	Emergency call function

SAFETY SYSTEMS.
2. CONTROL UNITS.

Index	Explanation
12	Rollover protection actuator, left
13	Rollover protection controller
14	Rollover protection actuator, right
15	Intelligent battery sensor
16	Seat occupancy detection
17	Seat belt buckle switch
18	Lateral and longitudinal acceleration sensor
19	Pressure sensor (door)

Installation position and communication

This graphic shows the installation position and communication between the sensors and the ACSM 1:



Installation position and communication between sensors and ACSM 1

Zusammenfassung

The 1st generation ACSM is used in these BMW models:

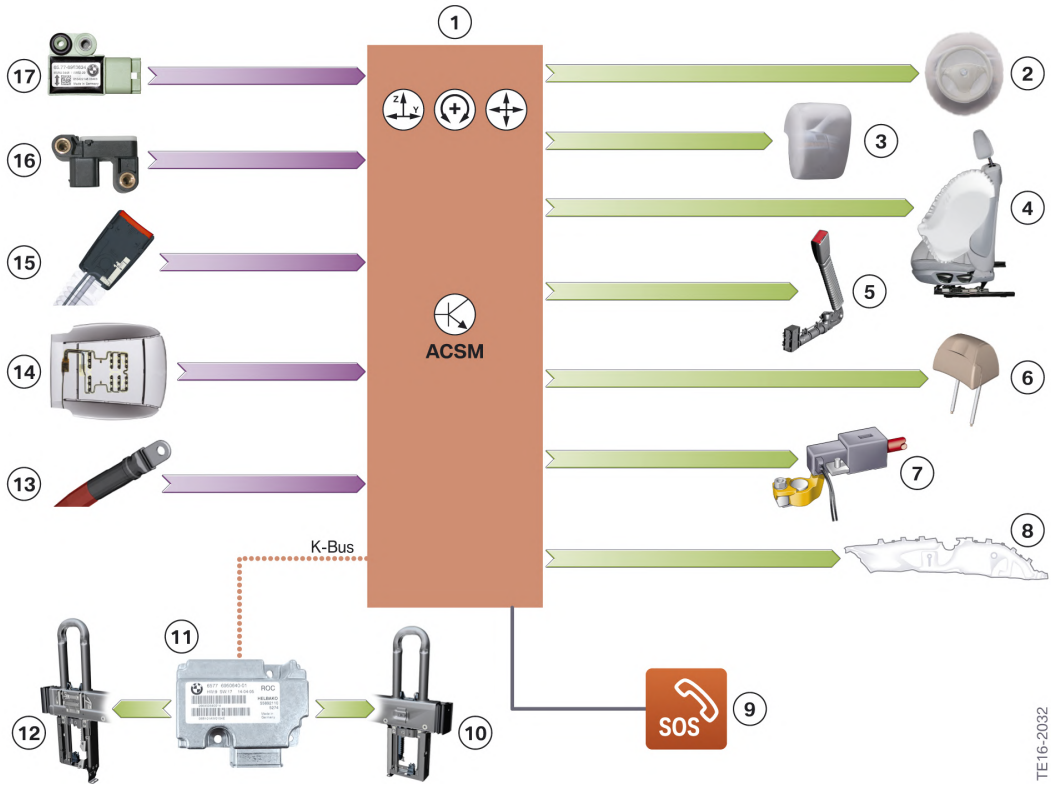
Generation	vehicles
ACSM 1	• E60, E61, E63, E64 • E85, E86

2.4.2. 2nd generation

- The ACSM 2 was extended by additional interfaces for future functions.

Inputs/outputs

The following graphic shows the sensors and actuators of the ACSM 2:



Input/Output ACSM 2

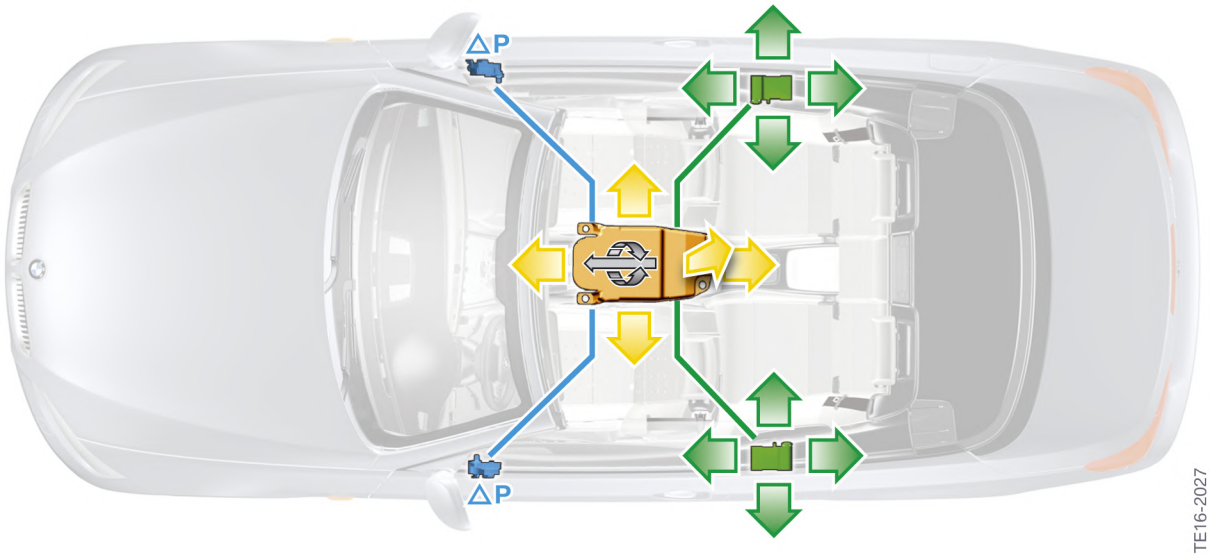
Index	Explanation
1	Crash Safety Module (ACSM)2
2	Driver's airbag
3	Front passenger airbag
4	Side airbag
5	Belt tensioner
6	Crash-active headrest
7	Safety battery terminal SBK
8	Head airbag
9	Emergency call function
10	Rollover protection actuator, left
11	Rollover protection controller

SAFETY SYSTEMS.
2. CONTROL UNITS.

Index	Explanation
12	Rollover protection actuator, right
13	Intelligent battery sensor
14	Seat occupancy detection
15	Seat belt buckle switch
16	Lateral and longitudinal acceleration sensor
17	Pressure sensor (door)

Installation position and communication

This graphic shows the installation position and communication between the sensors and the ACSM 2:



Installation position and communication between sensors and ACSM 2

Zusammenfassung

The 2nd generation ACSM is used in these BMW models:

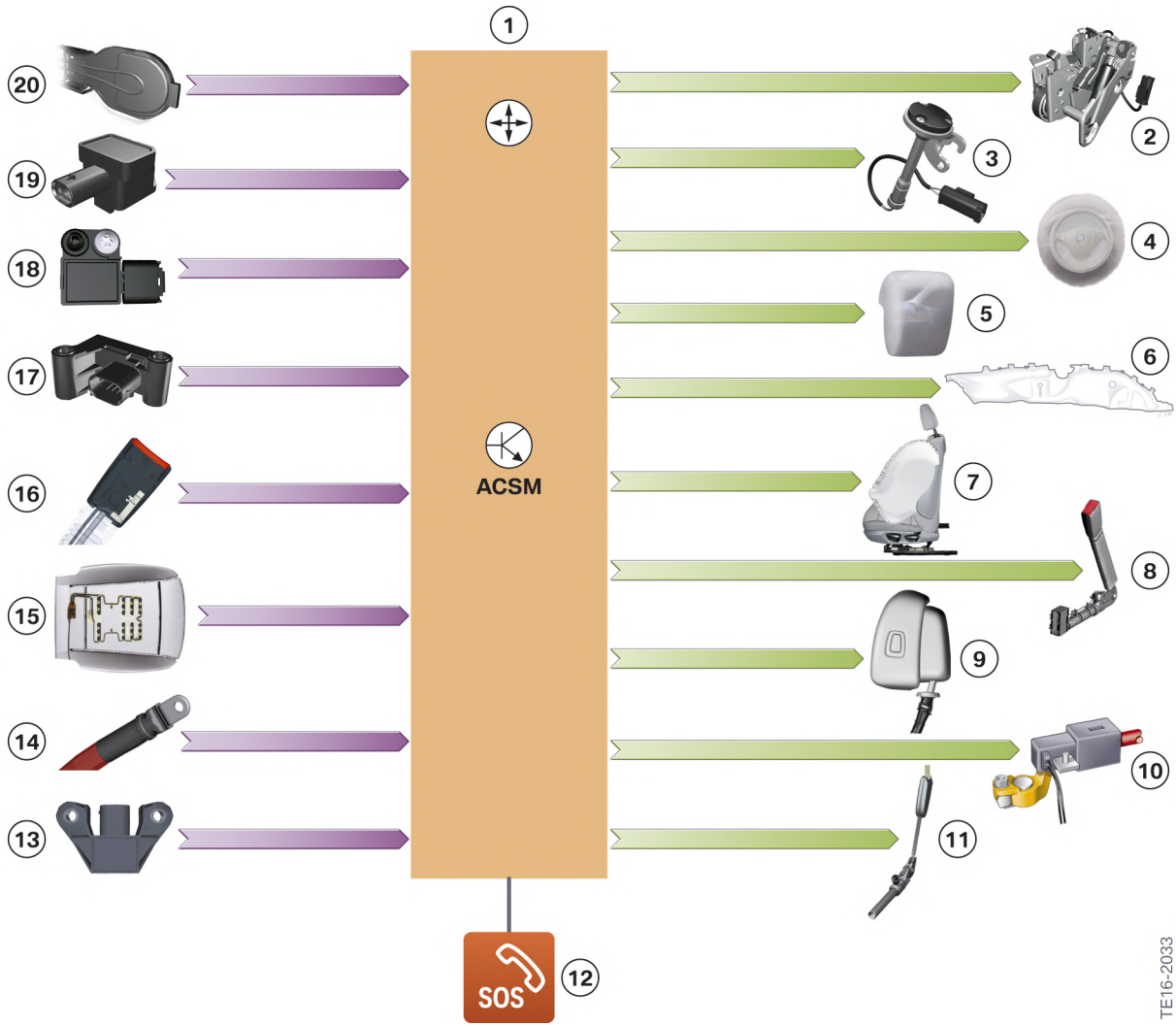
Generation	vehicles
ACSM 2	• E70, E71 • E88 • E93

2.4.3. 3rd generation

- In the 3rd generation, the sensor system integrated in the ACSM was relocated externally to a central sensor.
- The ACSM 3 provides up to 32 ignition outputs.

Inputs/outputs

The following graphic shows the sensors and actuators of the ACSM 3:



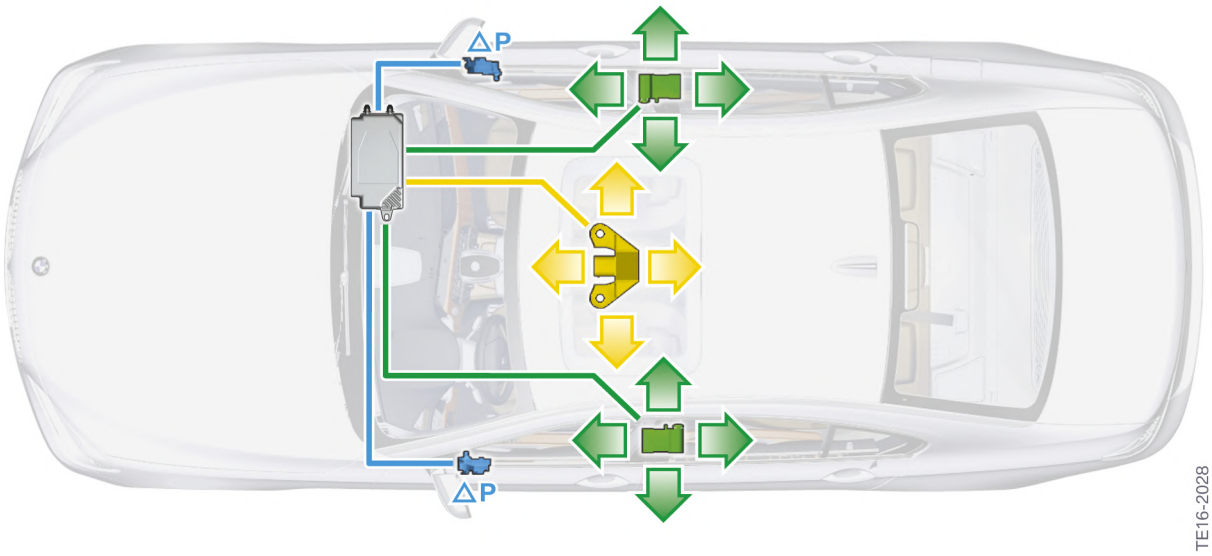
Input/Output ACSM 3

SAFETY SYSTEMS.
2. CONTROL UNITS.

Index	Explanation
1	Crash Safety Module (ACSM)3
2	Bonnet catch actuator
3	Bonnet hinge actuator
4	Driver's airbag
5	Front passenger airbag
6	Head airbag
7	Side airbag
8	Belt tensioner
9	Crash-active headrest
10	Safety battery terminal SBK
11	End fitting pretensioner
12	Emergency call function
13	Central sensor
14	Intelligent battery sensor
15	Seat occupancy detection
16	Seat belt buckle switch
17	Acceleration sensor, B-pillar on left, right
18	Pressure sensor (door)
19	Pedestrian protection sensors
20	Optical fibre with pedestrian protection sensor

Installation position and communication

This graphic shows the installation position and communication between the sensors and the ACSM 3:



Installation position and communication between sensors and ACSM 3

Zusammenfassung

The 3rd generation ACSM is used in these BMW models:

Generation	vehicles
ACSM 3	• F01, F02, F07 • F10, F11

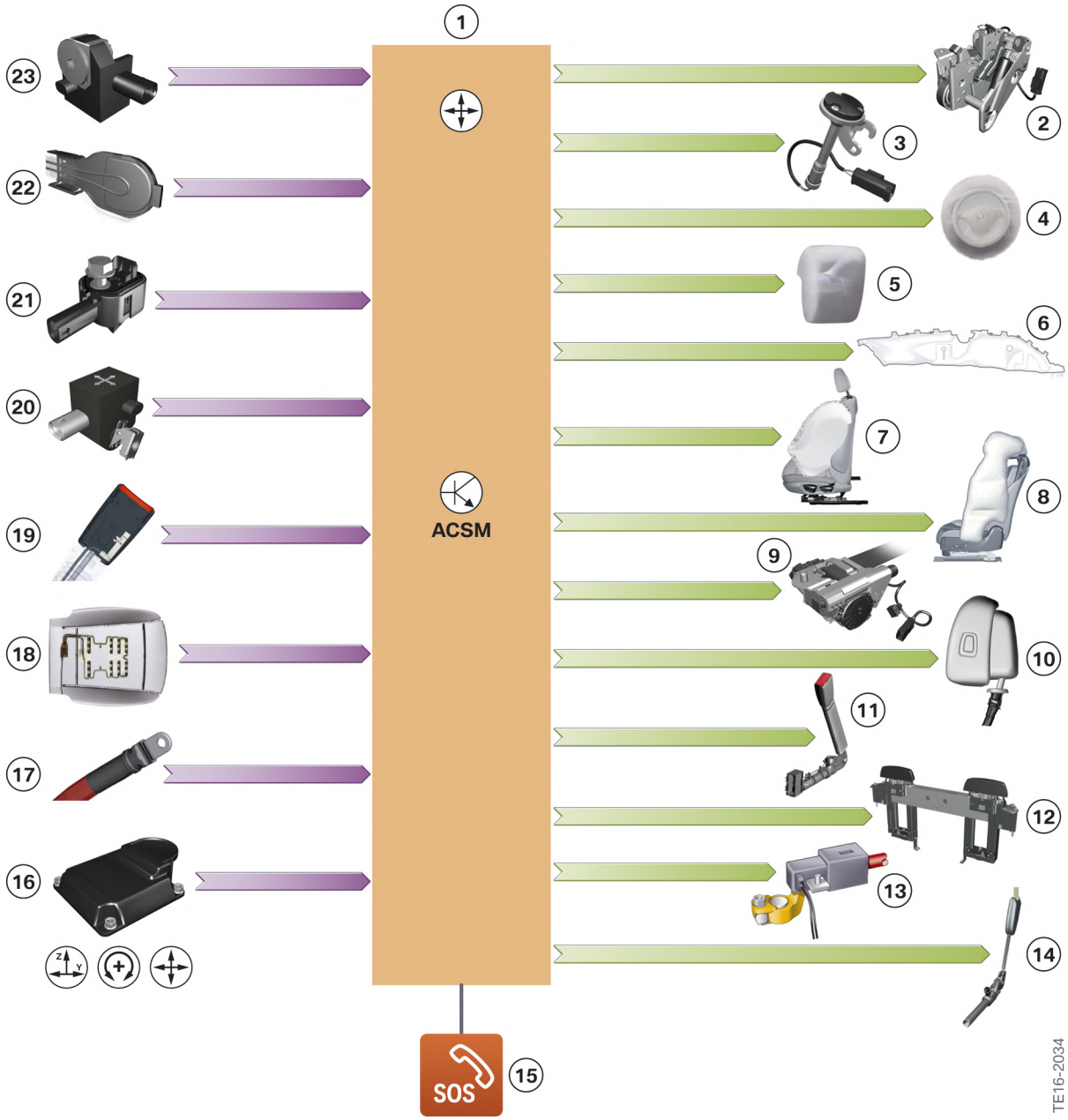
2.4.4. 4th generation

- As in the 3rd generation ACSM, the central sensor system is located outside the control unit, but not in a separate central sensor.
- The ACSM 4 evaluates the sensor data from the ICM.

Inputs/outputs

The following graphic shows the sensors and actuators of the ACSM 4:

SAFETY SYSTEMS.
2. CONTROL UNITS.



System wiring diagram ACSM 4

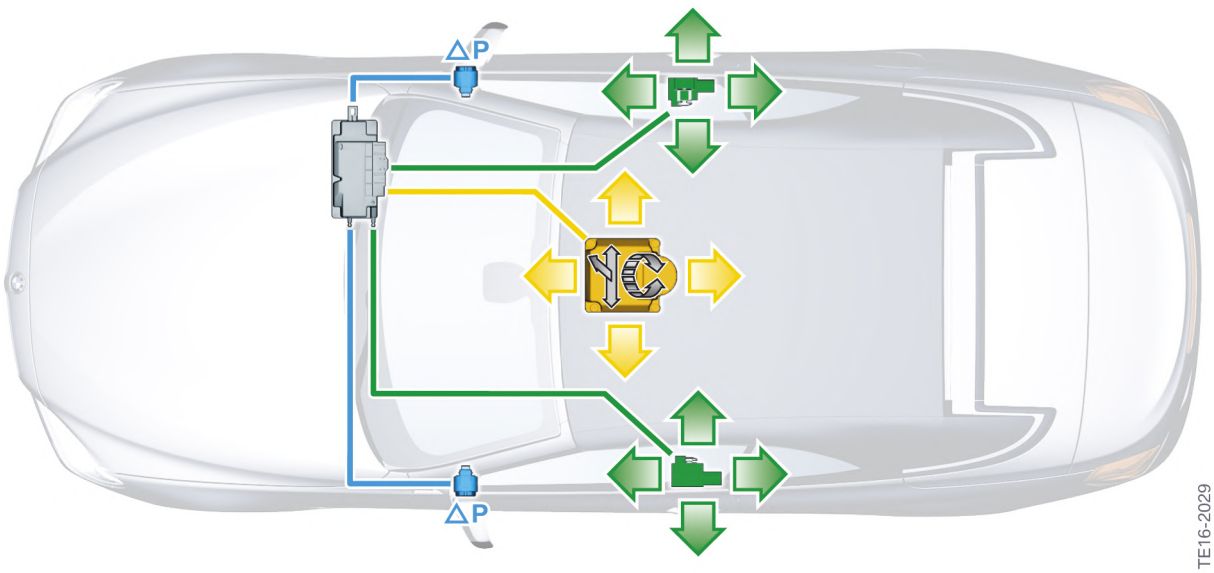
Index	Explanation
1	Crash Safety Module (ACSM)4
2	Bonnet catch actuator
3	Bonnet hinge actuator
4	Driver's airbag
5	Front passenger airbag
6	Head airbag

SAFETY SYSTEMS.
2. CONTROL UNITS.

Index	Explanation
7	Side airbag
8	Side head airbag
9	Automatic tensioner with linear force limiter
10	Crash-active headrest
11	Belt tensioner
12	Rollover protection actuators
13	Safety battery terminal
14	End fitting pretensioner
15	Emergency call function
16	Integrated Chassis Module ICM
17	Intelligent battery sensor
18	Seat occupancy detection, front passenger
19	Seat belt buckle switch
20	Acceleration sensor, B-pillar on left, right
21	Pressure sensor (door)
22	Optical fibre with pedestrian protection sensor
23	Pedestrian protection sensors

Installation position and communication

This graphic shows the installation position and communication between the sensors and the ACSM 4:



Installation position and communication between sensors and ACSM 4

SAFETY SYSTEMS.
2. CONTROL UNITS.

Zusammenfassung

The 4th generation ACSM is used in these BMW models:

Generation	vehicles
ACSM 4	• F06 • F12, F13 • F15, F16 • F20, F21, F22 • F25, F26 • F30, F31, F32, F33, F34, F36 • F45, F46, F48

The 4th generation ACSM is used in these MINI models:

Generation	vehicles
ACSM 4	• F54 • F55 • F56 • F57 • F60

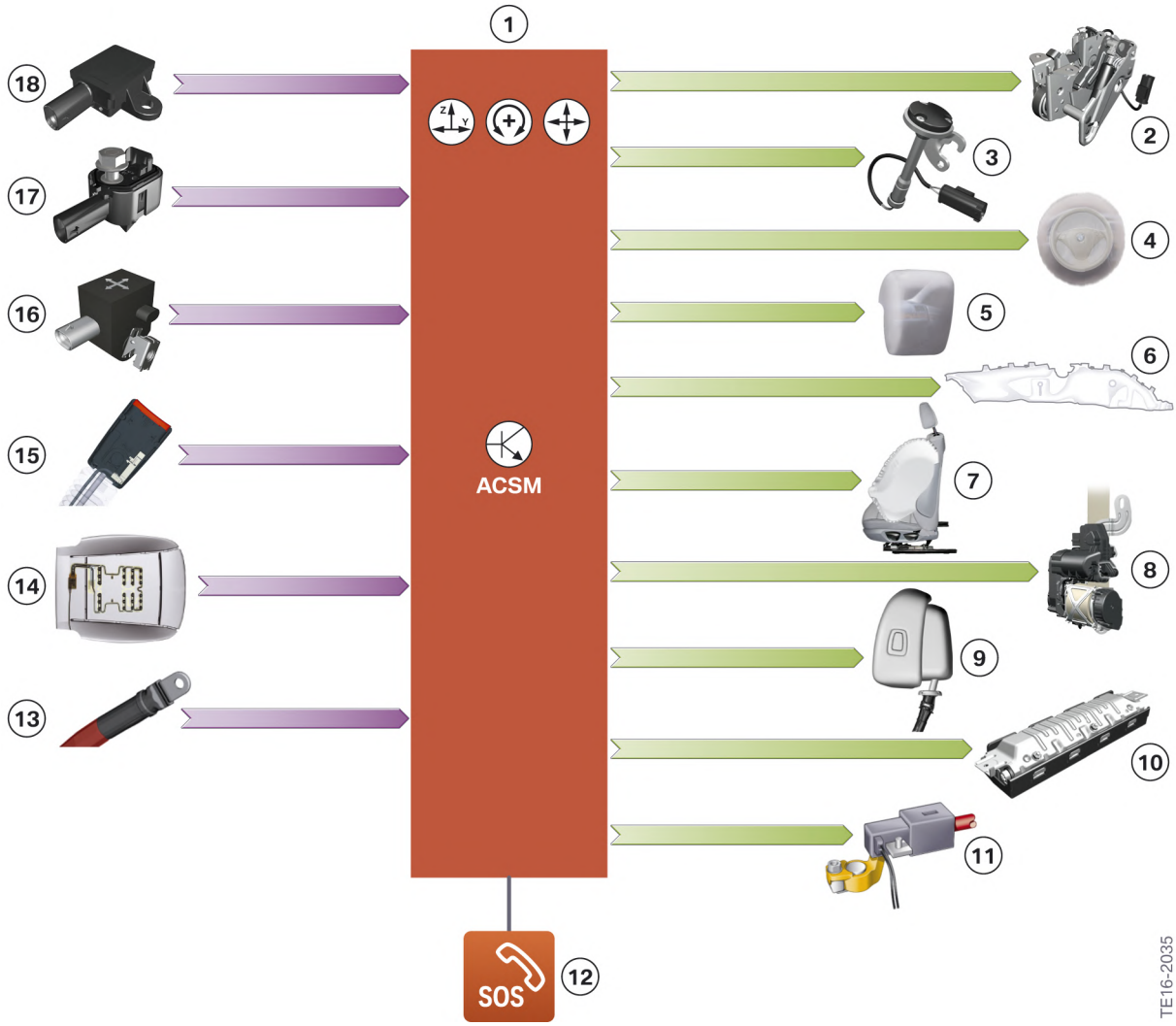
2.4.5. 5. Generation

- With the passive safety system, the central sensor system (sensors for lateral and longitudinal acceleration, roll rate sensor and vertical acceleration sensor) is integrated in the ACSM control unit, which is installed in the centre of vehicle.
- The ACSM is a bus user in the FlexRay.

Inputs/outputs

The following graphic shows the sensors and actuators of the ACSM 5:

SAFETY SYSTEMS.
2. CONTROL UNITS.



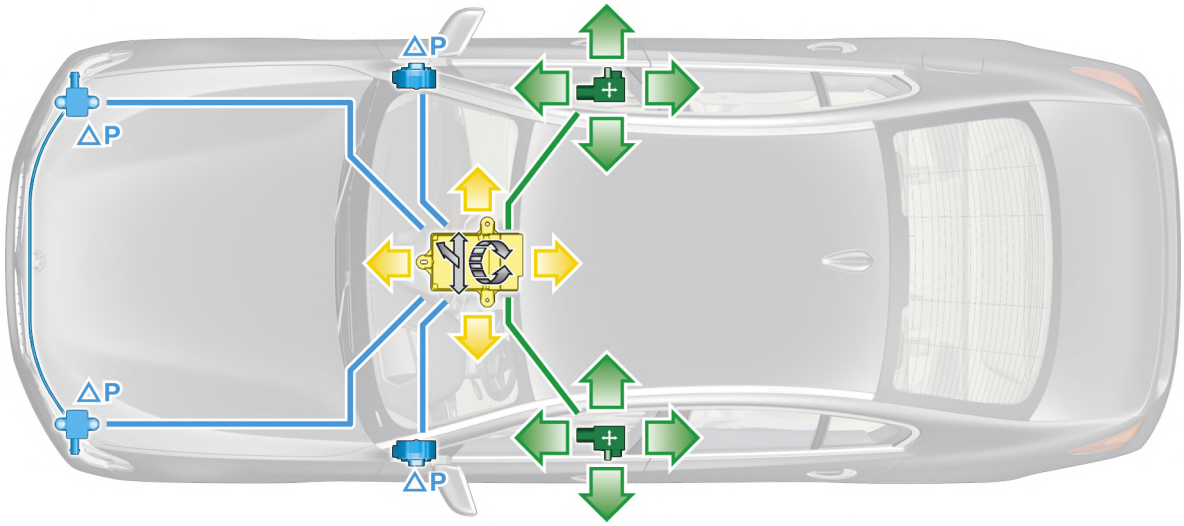
System wiring diagram ACSM 5

Index	Explanation
1	Crash Safety Module (ACSM)5
2	Bonnet catch actuator
3	Bonnet hinge actuator
4	Driver's airbag
5	Front passenger airbag
6	Head airbag
7	Side airbag
8	Automatic tensioner with linear mechanical force limiter
9	Crash-active headrest
10	Knee airbag (standard equipment for G30 driver's side)
11	Safety battery terminal

Index	Explanation
12	Emergency call function
13	Intelligent battery sensor
14	Seat occupancy detection, front passenger
15	Seat belt buckle switch
16	Lateral and longitudinal acceleration sensor, B-pillar
17	Pressure sensor (door)
18	Pressure hose sensor

Installation position and communication

This graphic shows the installation position and communication between the sensors and the ACSM 5:



Installation position and communication between sensors and ACSM 5

Zusammenfassung

The 5th generation ACSM is used in these BMW models:

Generation	vehicles
ACSM 5	• G11, G12 • G30

2.5. Integrated Chassis Management

Sensors are installed in the Integrated Chassis Management ICM (depending on equipment). The ICM control unit calculates the variables which are of importance for the driving dynamics condition of the vehicle using these sensors. The airbag sensors are also integrated in the ICM control unit.

This means that passive and active safety are networked. The ACSM control unit ensures the voltage supply for the sensors. The ICM is located in the centre of the vehicle and was used only in combination with the 4th generation ACSM.

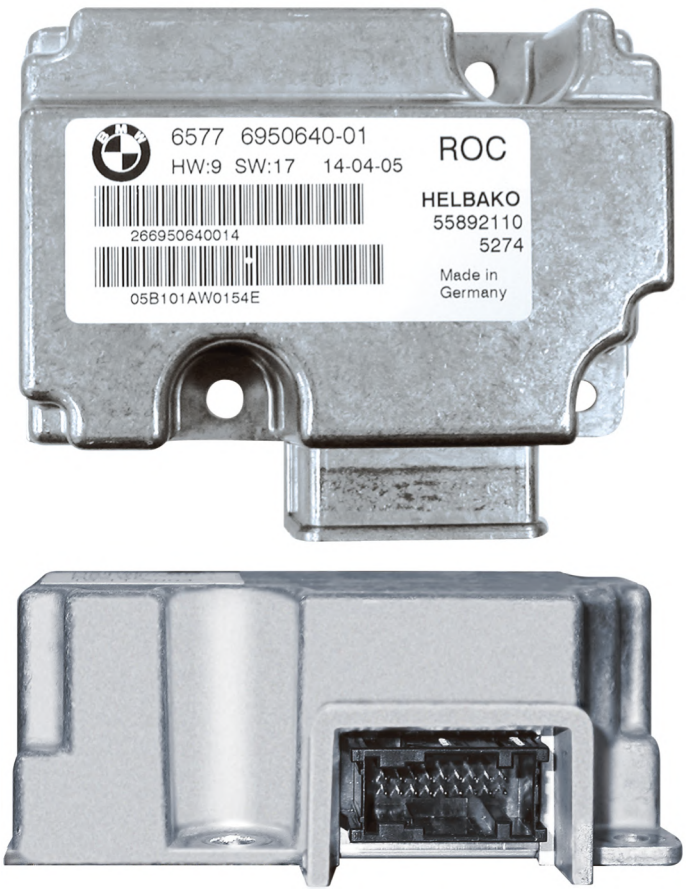
The following sensors and switches are installed, among others:

- Sensors for lateral and longitudinal acceleration
- Roll rate sensor
- Vertical acceleration sensor.

2.6. Rollover protection controller

On vehicles such as the E64 convertible, the Advanced Crash and Safety Management is equipped with an additional rollover protection controller. The control unit is connected with the ACSM via a K bus. The rollover protection controller was used since the power for activating the actuators was not yet sufficient in the control units at the time.

The following graphic shows a rollover protection controller:



TE17-0320

Rollover protection controller

SAFETY SYSTEMS.
2. CONTROL UNITS.

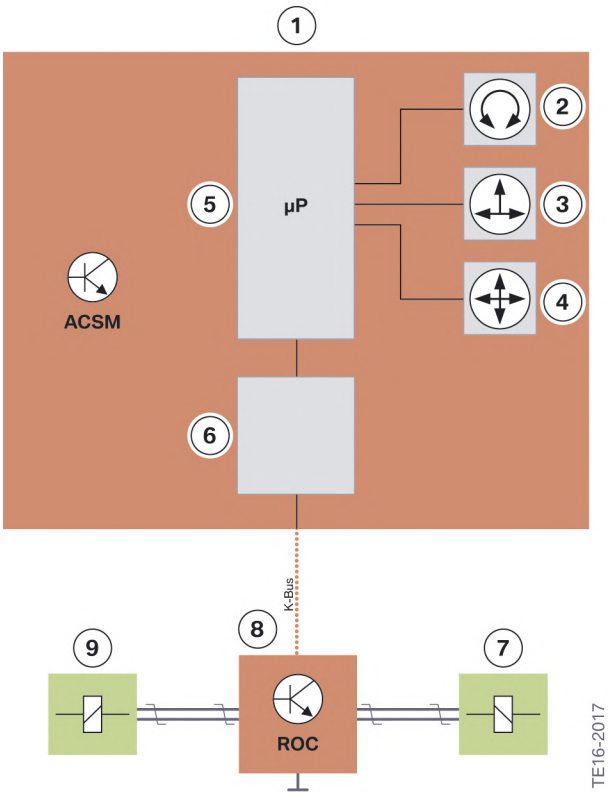
2.6.1. Function

In the event of an imminent rollover, the rollover protection controller has the task of activating the actuators of the rollover protection system. Rollover detection takes place in the Crash Safety Module. 2 telegrams are sent to the rollover protection controller when the threshold values are reached.

The rollover protection controller activates the two rollover protection bars in the event of a rollover.

The E64 convertible has a special Crash Safety Module. A special sensor system for detection of rollovers is integrated in the Crash Safety Module. In addition to the two sensors for longitudinal (X-axis) and lateral acceleration (Y-axis), there is a yaw sensor and a vertical acceleration sensor for the Z-axis and for the Y-axis.

The following graphic shows a block diagram of the rollover sensor system:



Block diagram of the rollover sensor system

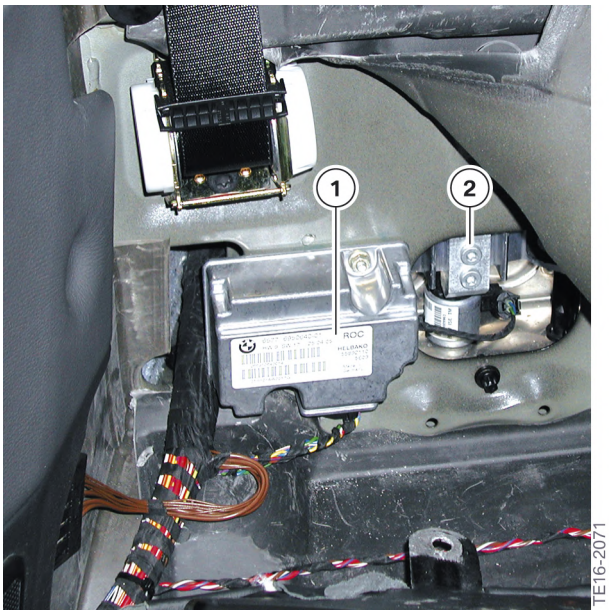
Index	Explanation
1	Crash Safety Module
2	Yaw sensor
3	Vertical acceleration sensors (Z-/Y-axis)
4	Longitudinal and lateral acceleration sensor
5	Microcontroller

Index	Explanation
6	K bus interface
7	Rollover protection actuator, right
8	Rollover protection controller
9	Rollover protection actuator, left

The sensors supply a voltage as the measured variable. The processor evaluates the signals of the longitudinal and lateral acceleration sensor and from the two vertical acceleration sensors. The yaw sensor is also included in the calculation. The results are compared with the stored algorithm. If the processor detects an imminent rollover, it sends 2 messages to the rollover protection controller in a defined time frame with the request to activate the actuators.

2.6.2. Installation location

The rollover protection controller is installed behind the right rear seat on the support structure of the rollover protection system.



Installation location of rollover protection controller

Index	Explanation
1	Rollover protection controller
2	Rollover protection actuator, right

2.6.3. Diagnostics

The rollover protection controller does not have direct diagnostic capability. It monitors itself internally. The two circuits for the actuators are also monitored by the rollover protection controller. In the event of a fault, the rollover protection controller sends the fault to the Crash Safety Module ACSM, where the fault is stored in the fault memory. The ACSM activates the airbag indicator light in the instrument cluster.

2.7. Steering column switch cluster

The clock spring establishes a movable electrical connection between the wiring harness on the steering column and the gas generator in the steering wheel.

The spiral-wound conductor foil is able to follow the steering wheel turns in both directions.

The clock spring has the task of transferring the following electrical signals from and to the multifunction steering wheel:

- Activation of the driver's airbag
- Control buttons
- Multifunction buttons
- Horn.

The following graphic shows an exploded illustration of the steering column switch cluster:

SAFETY SYSTEMS.
2. CONTROL UNITS.



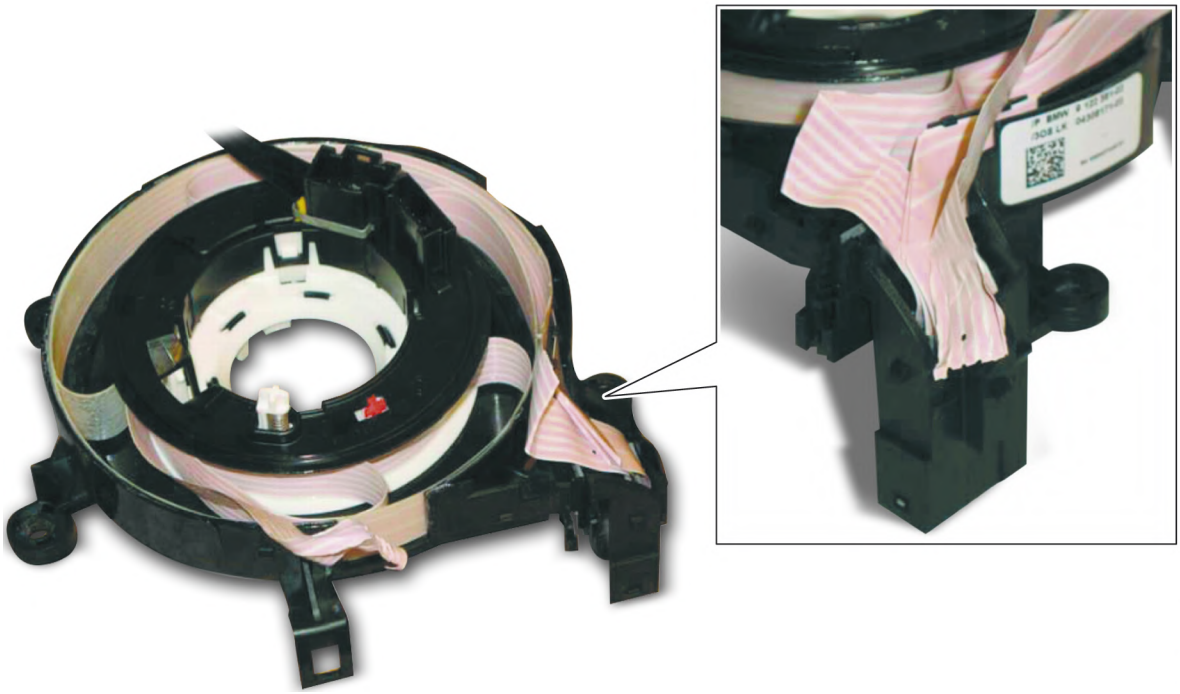
G30 installation location of clock spring

Index	Explanation
1	Bushing contact (to clock spring)
2	Bushing contact (to airbag)
3	Airbag module, driver's side
4	Steering wheel
5	Clock spring
6	Steering column switch cluster

When carrying out installation work, it must be ensured in all cases that the steering wheel is removed with the steering in centre position and the wheels in straight-ahead position. The return ring must be fixed in centre position. Before installing a new clock spring, it must always be ensured that the fixing strips on the housing are undamaged.

The following graphic shows a damaged clock spring:

SAFETY SYSTEMS.
2. CONTROL UNITS.



TE17-0321

Damaged coil spring cassette

The coil spring cassette can be replaced only as a complete unit.

If the steering wheel is not in centre position during installation of the clock spring, there is a risk of the coil spring cassette being damaged.

SAFETY SYSTEMS.
3. ACTIVE PROTECTION.

Active Protection is a safety package and comprises the following 3 functionally independent subsystems:

- PreCrash
- PostCrash
- Alertness assistant.

The system increases the safety of the vehicle occupants through preventative measures in the event of critical driving situations (PreCrash) and through measures after a crash (PostCrash). The alertness assistant monitors the driving style of the driver during a journey and encourages the driver to take breaks if necessary.

3.1. PreCrash

The PreCrash subsystem introduces preventative protective measures for the occupants in known critical driving situations which may lead to an accident.

3.1.1. Identification of critical driving situations

Critical driving situations are identified with help of the inertial sensor system installed in the vehicle. Critical driving situations include:

- Emergency braking initiated
- Strong understeer
- Strong oversteer
- Rollover of the vehicle.

3.1.2. PreCrash functions

In vehicles with the optional equipment Driving Assistant (OE 5AS) or Driving Assistant Plus (OE 5AT), imminent collisions are also taken into account which are detected by means of the radar sensors or KAFAS camera. If both systems are installed, then the collisions identified by the radar sensor are evaluated. For PreCrash detection, separate thresholds to the collision warning thresholds are made available by the KAFAS camera or radar sensor for activating the PreCrash functions.

3.1.3. Measures for occupant protection

If a critical driving situation has been detected, the following measures are introduced to protect the occupants:

- Automatic pre-tensioning of front seat belts
- Automatic closing of side window up to a gap
- Closing of slide/tilt sunroof
- Front passenger backrest is moved to upright position.

If a critical driving situation is detected and the seat belt buckle contacts are closed, the seat belts of the driver and front passenger are preventively tensioned by the reversible electromotive reel REMA and the belt slack reduced. The effectiveness of the belt tensioner is thus increased and the severity of injuries in the event of a crash can be reduced. The seat belt is released again as soon as the critical driving situation no longer exists.

Belt slack is the term used to describe the part of the seat belt which is loose or does not have contact with the body. As a result of the belt slack, the position of the occupant moves forward in an accident, and therefore results in higher deceleration forces and more serious injuries.

Closing the side window and the slide/tilt sunroof prevents objects entering the passenger compartment in the event of a crash. If the vehicle is at a standstill after a crash, the side windows are opened a fraction so that the gases released when the airbags were triggered can escape quicker. The side window and slide/tilt sunroof are controlled by the control unit of the reversible electromotive reel left, or REMALI for short.

In addition, the reversible electromotive reel makes available the comfort tensioning function. This reduces the belt slack of the driver and front passenger each time the belt contact is closed as from a certain speed.

3.2. PostCrash

The PostCrash subfunction includes the automatic braking function after an accident, which brakes the vehicle to a standstill in certain accident situations without intervention from the driver. The risk of another collision can thus be reduced. The driver can override the automatic braking after an accident by pressing the brake pedal or accelerator pedal.

The automatic braking function after an accident is activated if a crash with sufficient severity has been identified in the Crash Safety Module ACSM. The Crash Safety Module ACSM then sends a signal to the electronic fuel pump control module (EKPS) to switch off the electric fuel pump. This signal is evaluated by the DSC control unit. The DSC control unit then brakes the vehicle with help of the DSC pump and brings it to a standstill.

During automatic braking the accelerator pedal and brake pedal are monitored. The automatic braking is interrupted if an acceleration request is identified or if the brake pedal is pressed for a longer period by the driver.

The brake is automatically released approx. 1.5 s after reaching standstill. The parking brake must be applied to prevent the vehicle from rolling away.

If a crash is detected which leads to activation of the safety battery terminal, then automatic braking cannot be activated as the DSC pump is no longer supplied with current.

Automatic braking is not activated under the following conditions:

- DSC failed before crash
- ABS failed before crash
- Brake fluid level too low.

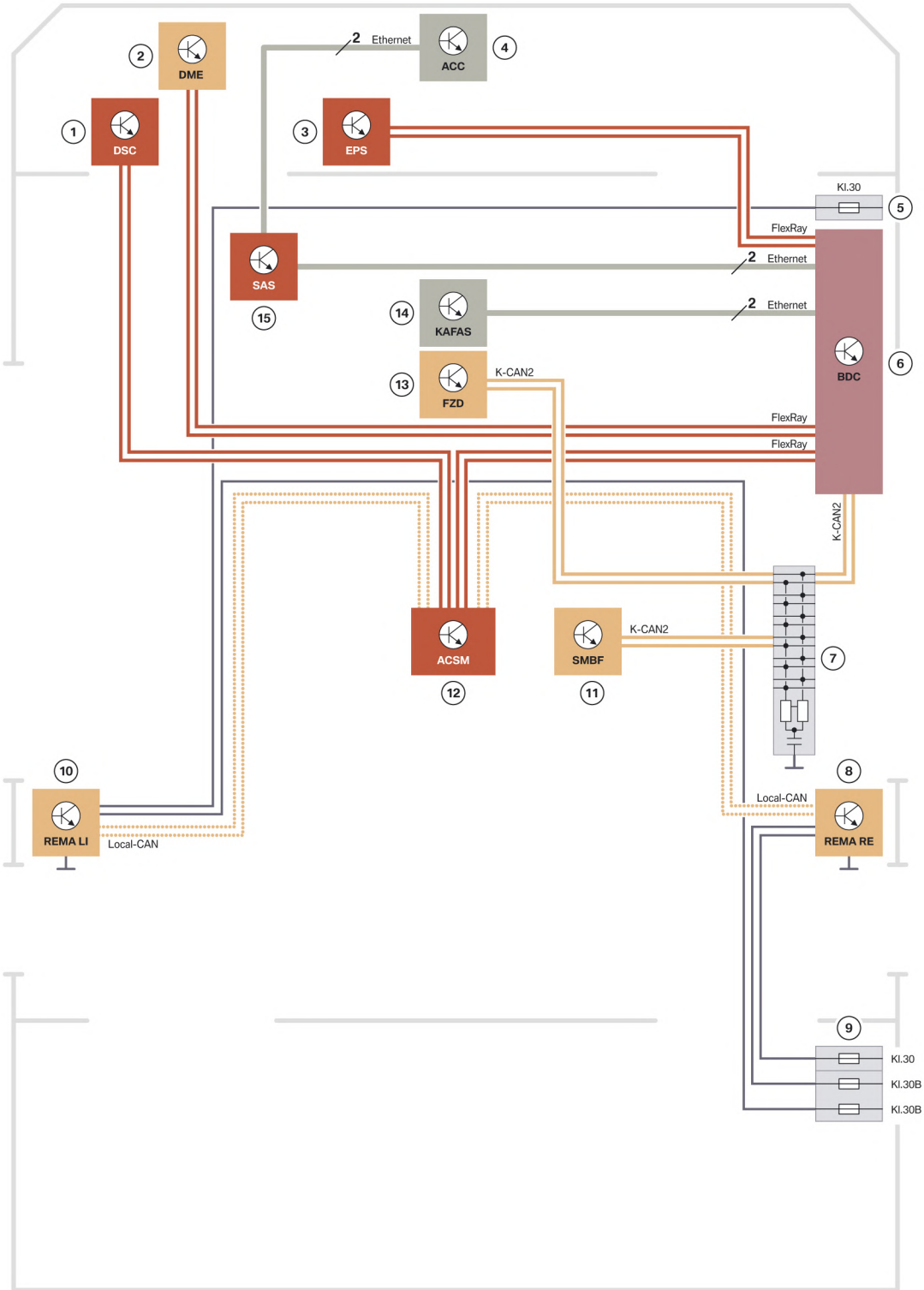
If these faults occur during or after the crash, the control is led to the end as these faults are then interpreted as consequences of the crash.

SAFETY SYSTEMS.
3. ACTIVE PROTECTION.

3.3. System wiring diagram

The following graphic shows the Active Protection system wiring diagram of a G30:

SAFETY SYSTEMS.
3. ACTIVE PROTECTION.



G30 System wiring diagram for Active Protection

SAFETY SYSTEMS.
3. ACTIVE PROTECTION.

Index	Explanation
1	Dynamic Stability Control (DSC)
2	Digital Motor Electronics (DME)/Digital Diesel Electronics (DDE)
3	Electronic Power Steering EPS
4	Radar sensor for active cruise control (ACC)
5	Fuse for front right power distribution box
6	Body Domain Controller (BDC)
7	CAN terminator
8	Reversible electromotive reel right, REMARE
9	Fuses in the power distribution box, rear right
10	Reversible electromotive reel left, REMALI
11	Front passenger seat module, SMBF
12	Crash Safety Module (ACSM)
13	Roof function centre (FZD)
14	Camera-based driver support systems (KAFAS)
15	Control unit for optional equipment system SAS

SAFETY SYSTEMS.
4. AIRBAG SYSTEMS.

4.1. Design

All airbag systems function according to the same principle. When the airbag is activated, it is filled with nitrogen by a gas generator within fractions of a second.

2 different gas generators are used at BMW:

- Gas generator with solid propellant
- Hybrid gas generator with solid propellant and pressurised gas reservoir.

4.1.1. Gas generator with solid propellant

The gas generator with solid propellant is a technology that has proven itself over the course of many years and has been used since the first airbag.

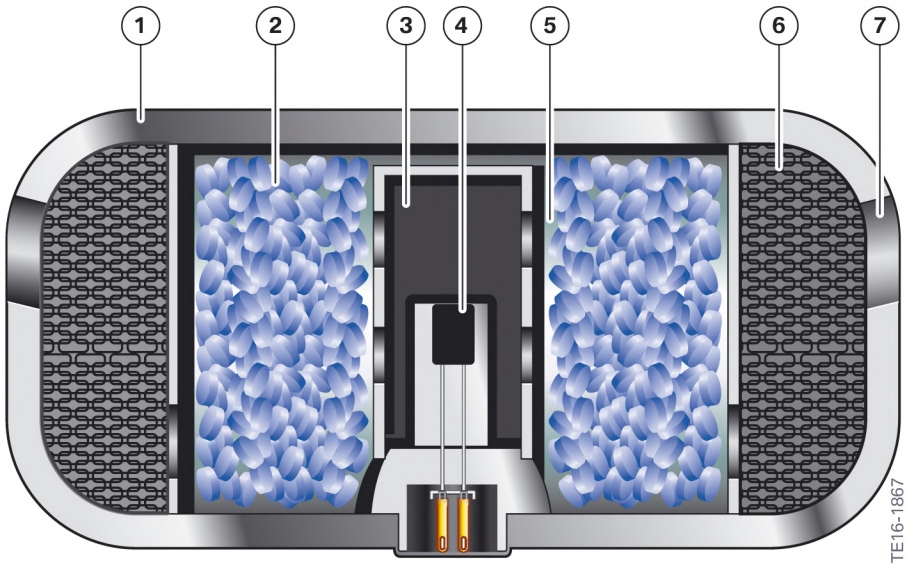
This is used, among other things, for:

- Driver's airbag
- Front passenger airbag.

The solid propellant generator comprises a housing in which a solid propellant charge with igniter is integrated. The design and form of the generator housing are adapted to the installation conditions in each case. As a result, a distinction is made between pot gas generators and tubular gas generators according to their design form.

The solid propellant is used in tablet or ring form. After the solid propellant is ignited, the filling gas comprising mainly nitrogen is produced. This gas is safe for the vehicle occupants.

The following graphic shows the design of a pot gas generator:



Gas generator sectional view

Index	Explanation
1	Housing
2	Propellant in tablet form
3	Priming charge
4	Igniter (bridge igniter)
5	Combustion chamber
6	Filter
7	Discharge hole for filling the airbag

4.1.2. Hybrid gas generator

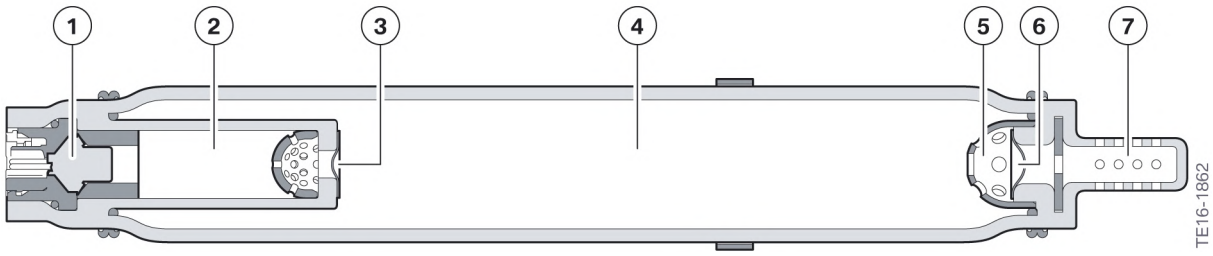
A newer technology is the hybrid gas generator. This is also referred to as a cold gas generator, since hardly any heat it produced when the propellant is released.

In the hybrid gas generator, the propellant required for deployment of the airbag is located in a pressure reservoir as a compressed gas with a pressure of up to 350 bar. Ignition takes place via a bridge igniter and a small quantity of solid propellant. This is the reason for the name hybrid gas generator. When solid propellant combustion takes place, a pressure is generated which opens the diaphragm of the pressure reservoir and thus allows the propellant to escape into the airbag. Since the propellant cools down significantly when it is discharged from the pressure reservoir, the small quantity of solid propellant is used to heat it. This prevents the reservoir opening from freezing. The propellant flows into the airbag via defined openings. The airbag cover breaks at predetermined breaking points and releases the airbag.

Hybrid gas generators are used, among other things, for:

- Front passenger airbag (Z4 only)
- Head airbag
- Seat-integrated side airbag.

Design



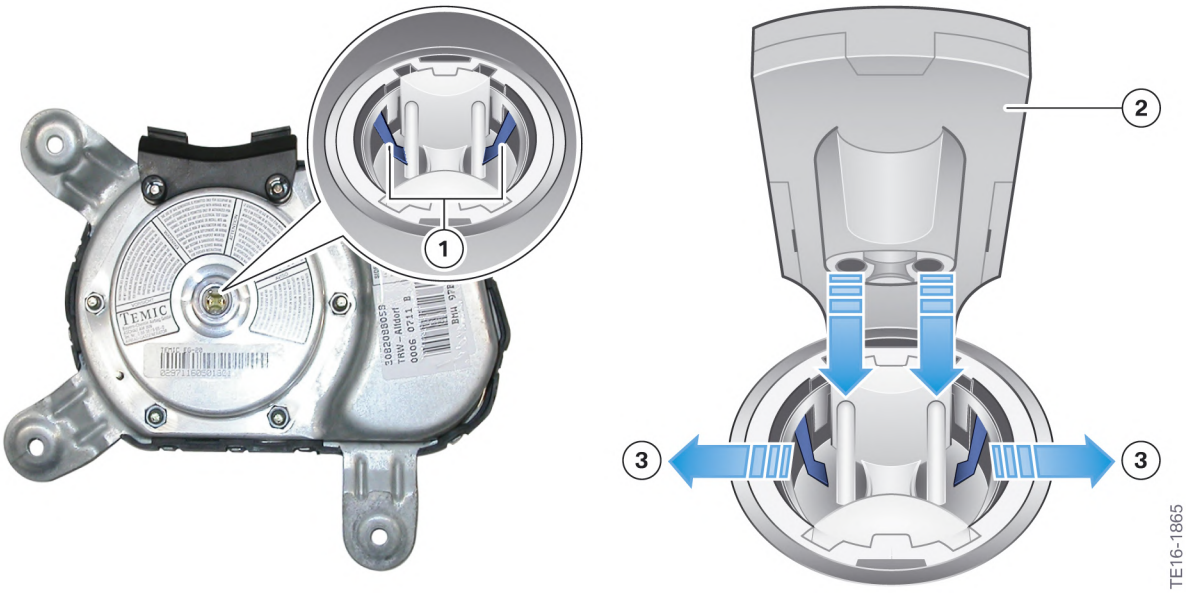
Sectional view of single-stage hybrid gas generator

Index	Explanation
1	Igniter
2	Combustion chamber with solid propellant
3	Burst disc 1
4	Pressurised gas reservoir with gas filling (up to approx. 350 bar)
5	Metal filter
6	Burst disc 2
7	Discharge hole

4.1.3. Plug connections on pyrotechnic components

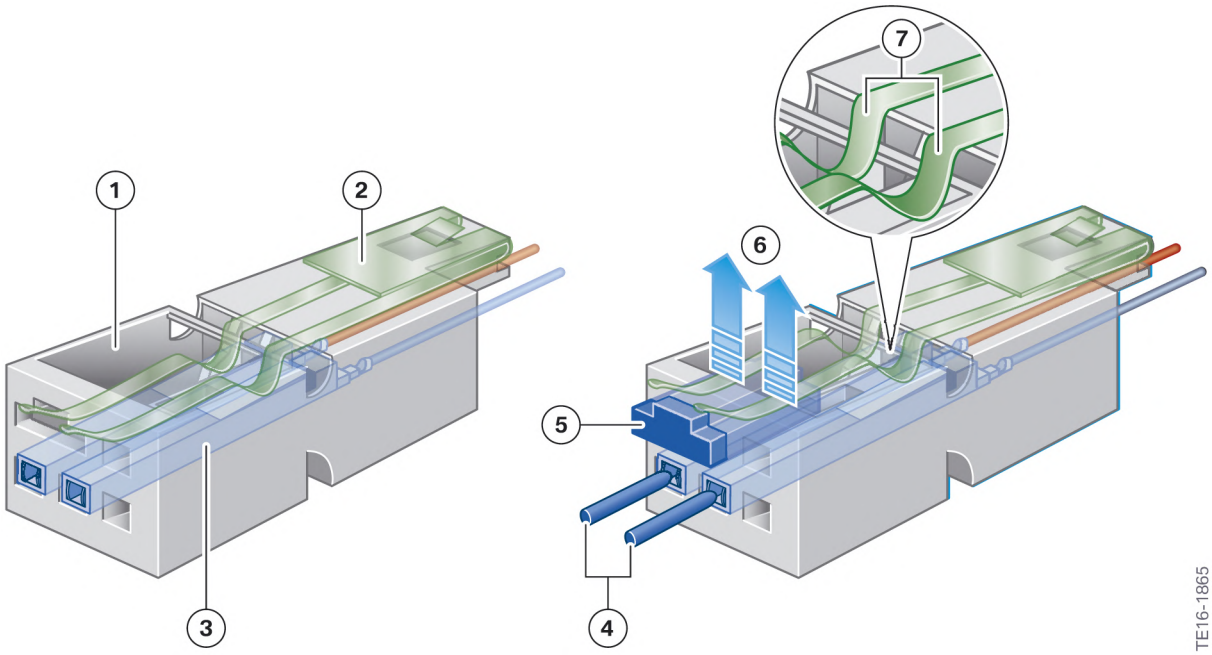
All pyrotechnic components have a short-circuit bridge on all plug connections. Without this short-circuit bridge, different voltage potentials could cause unintentional activation of the pyrotechnic component. This may be the case, for example, if a cable is damaged or as a result of contact with a live cable. The short-circuit bridge ensures that the lines of the pyrotechnic component are always short-circuited when the plug connection is disconnected.

The following graphic shows a short-circuit bridge using the example of a side airbag:



Short-circuit bridge on side airbag

Index	Explanation
1	Springs of short-circuit bridge (lines short-circuited)
2	Connector
3	Short-circuit bridge not functional (the springs are pressed to the outside by the connector)



Plug connection short-circuit bridge

Index	Explanation
1	Socket (not plugged)
2	Short-circuit bridge rests on the socket contacts
3	Socket contacts
4	Connector pins
5	Plastic pin of connector
6	Short-circuit bridge is lifted by the plastic pin of the connector
7	Short-circuit bridge no longer rests on the socket contacts

When the plug connection is made, the short circuit is cancelled and operational readiness is therefore ensured again.

4.2. Actuators

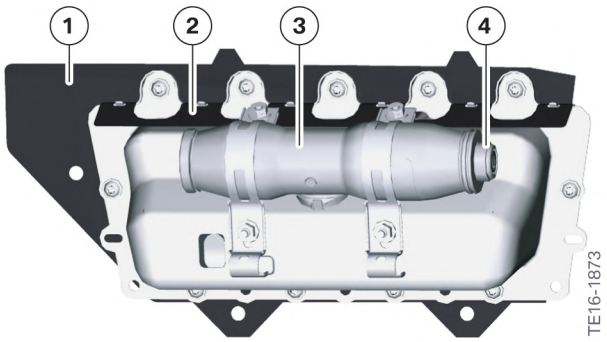
Installation position	Component
 TE16-1882	**Driver's airbag** The purpose of the driver's airbag is to reduce the risk of injury to the driver in combination with the seat belt when a front-end crash occurs. The driver's airbag is located in the steering wheel impact plate. The driver's airbag is equipped with a gas generator.
 TE16-2019	**Front passenger airbag** The front passenger airbag is located in the instrument panel. When the front passenger airbag expands, the instrument panel tears open at defined points. It is equipped with a single-stage gas generator.
 TE16-2020	**Knee airbag** Due to new requirements, vehicles with European version have a knee airbag installed on the driver's side. This is located behind a cover under steering column. The knee airbag is designed as a single-stage airbag with inflator assembly.

Installation position	Airbag system
 TE16-2021	**side airbag** The side airbags have the task of reducing the risk of injuries to the pelvis and torso areas of the body in the event of a side collision. The side airbag at the front ignites out of the front seat backrest in all current BMW models. The side airbags and gas generators are accommodated in a plastic housing, the airbag module. This is installed in the front seat backrest and is concealed by the seat cover. The cover and seat cover have a predetermined breaking point. When triggered, the side airbag emerges outwards from the backrest frame and spreads between the side structure and occupant. The air cushion between the door and occupants provides controlled collision damping and therefore reduces the forces acting on the occupants.
 TE16-1880	**Side head airbag** Like the side airbag, the side head airbag ignites from the front seat backrest. Since vehicles such as the F12 have a convertible top, the head airbags are omitted in this case. The side airbags help to protect the thorax and head of the vehicle occupants and have an increased volume.

Make sure that you fit only seat covers that have been approved by BMW. If covers are used incorrectly, there is a risk that the function of the side airbags may be considerably impaired or even prevented.

SAFETY SYSTEMS.
4. AIRBAG SYSTEMS.

4.2.1. Design of front passenger airbag



F12 Front passenger airbag in European version of vehicles

Index	Explanation
1	Cover
2	Housing, airbag
3	Gas generator
4	Ignition squib

4.2.2. Head airbag

In the event of a side collision, there are fewer deformable metal structures in comparison with a front-end collision. The enormous speed difference between the inwardly moving door and the occupant determines the injury potential. Head airbags are installed to protection the heads of occupants in side collisions. These extend from the A-pillar to the C-pillar and cover the entire area of the side windows. The head airbag is deployed between the occupants and the side structure.

System features:

- Extended covered area across all front and rear side windows
- Protection of occupants from glass splinters
- Covered area for different-sized occupants optimised.

Head airbags have been installed since the E83 and can be found in all vehicles up to the present day.

Head airbags are deployed between the vehicle occupants, side window and trim panels and are activated via a hybrid gas generator.

The head airbag is filled via 2 valves in order to achieve uniform filling.

System features:

- Large area coverage for the front and rear side windows
- Reduction of the risk of glass splinters and penetrating objects
- Optimised area coverage for occupants of different sizes.

SAFETY SYSTEMS.
4. AIRBAG SYSTEMS.

Installation location

The head airbag extends from the A-pillar to the C-pillar and covers the complete side area.

Depending on the vehicle model, the gas generators may be installed in the front roof area under the sun visors, in the area of the sun visors, in the area of the B-pillar, between the C- and D-pillars and also in the rear roof area.



G30 Head airbag

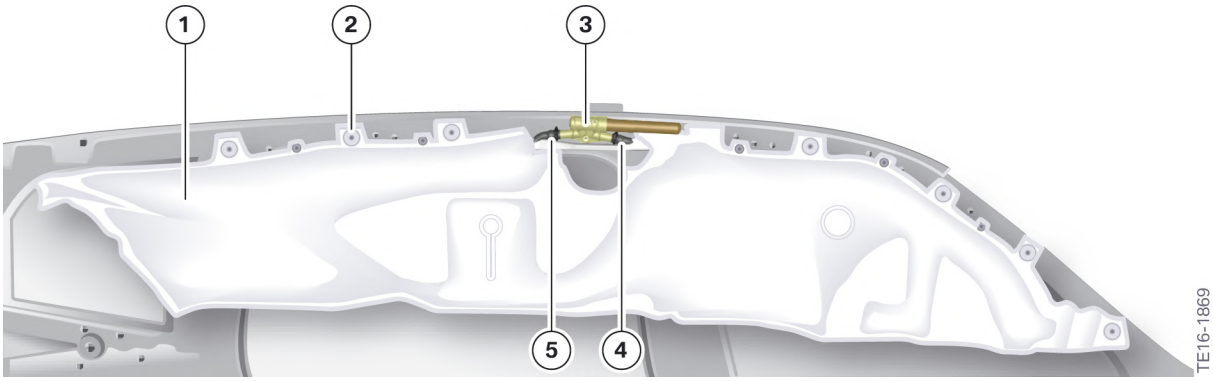
Design

The head airbag is accommodated folded up in the roof frame area. It consists of the gas generator, the two gas lances and the air bag.

Function

In the event of a side collision, the generator is ignited and gas flows through the two gas lances into the air bag. More uniform filling is achieved through simultaneous filling of the air bag at front and rear. The head airbag is brought into position by the mounting on the A-pillar and C-pillar. It then extends between the side window, trim panels and occupants.

The closed system ensures that the structural rigidity and stability are maintained for several seconds.



Head airbag deployed

Index	Explanation
1	Air bag
2	Airbag mountings
3	Gas generator
4	Front gas lance
5	Rear gas lance

4.3. Safety battery terminal

If the Crash Safety Module ACSM detects a crash of sufficient severity, the safety battery terminal is triggered at different thresholds. The connecting line from the positive battery terminal to the starter motor/alternator is then interrupted.

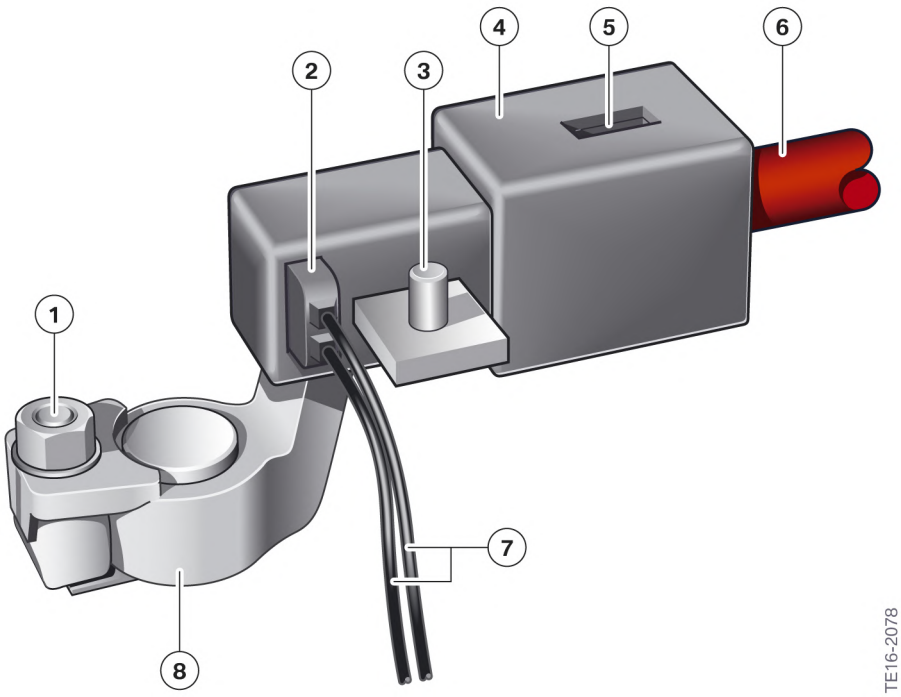
When the safety battery terminal is triggered, the voltage supply to the safety-relevant consumers (hazard warning lights, airbag, emergency call etc.) is still guaranteed.

The safety battery terminal is normally located directly at or close to the positive terminal of the vehicle battery.

4.3.1. Design

It consists of a conventional terminal secured on the terminal from above and a socket contact connected with the clamping surface. The socket contact contains a pyrotechnic ignition squib. Due to the low quantity of propellant, the safety battery terminal is not subject to any danger classification in accordance with the German law relating to the use of explosives.

The housing contains 2 retaining springs which guarantee reliable mechanical separation of the connection after the triggering process and prevent renewed contacting. For this reason, the safety battery terminal must be renewed by Service after triggering.



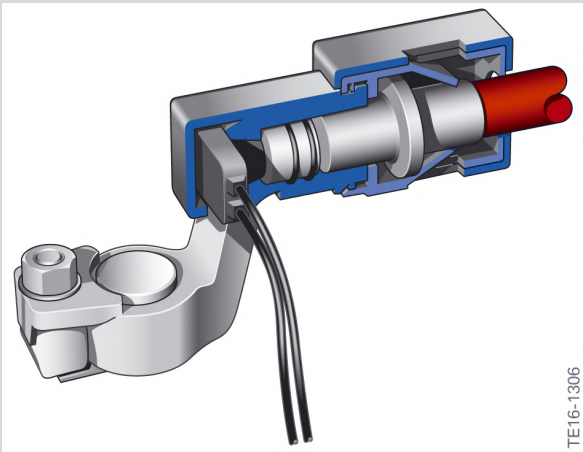
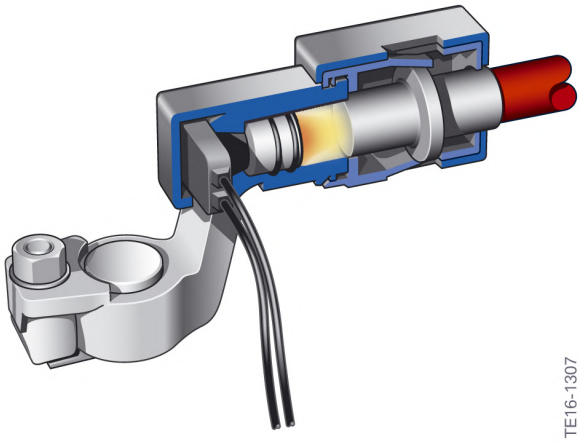
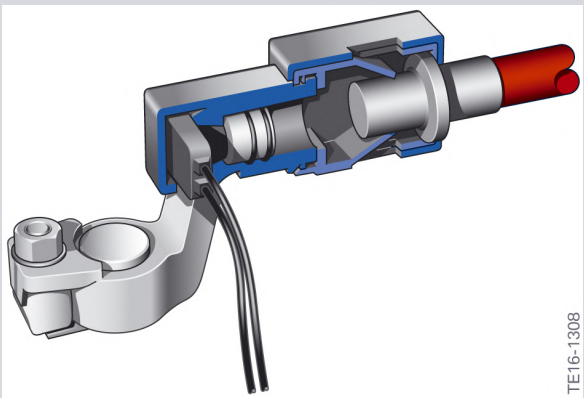
Safety battery terminal (SBK)

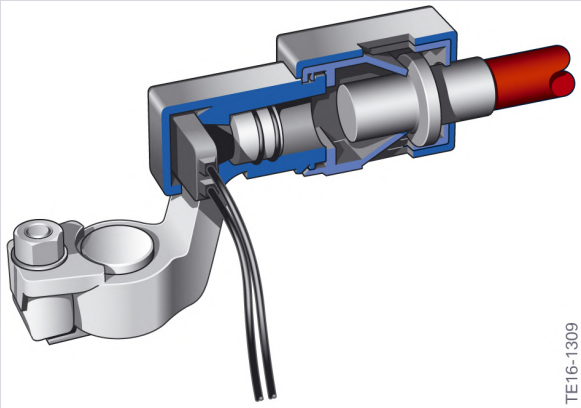
Index	Explanation
1	Battery positive terminal screw connection M6
2	Ignition squib
3	Vehicle electrical supply connection
4	Plastic housing
5	Retaining spring
6	B+ cable
7	Ignition squib connections, two-pin
8	Battery terminal clamp

4.3.2. Triggering process

The following table shows the triggering process of a safety battery terminal:

SAFETY SYSTEMS.
4. AIRBAG SYSTEMS.

Safety battery terminal SBK	Explanation
 TE16-1306	**Initial status** The safety battery terminal SBK is triggered by the ACSM or an airbag sensor as from a certain crash severity. The process of cable separation is shown in the following illustrations. In the normal operating condition, the battery cable is connected with the battery terminal clamp via the contact bush.
 TE16-1307	**Initiated isolation process** After triggering, the B+ cable is pressed out of the contact bush. In this way, the line is reliably mechanically separated and also electrically isolated.
 TE16-1308	**Completed isolation process** When isolation of the B+ cable has been completed, the line is held in position by the two retaining springs. This prevents renewed contact with the positive terminal of the vehicle battery.

Safety battery terminal (SBK)	Explanation
	**Retained safety battery terminal** The protective function achieved in this way includes the complete line routing from the battery via the vehicle interior, engine compartment and alternator. The alternator is also switched off simultaneously.

As a result of the targeted separation, all of the rest of the vehicle electrical system remains functional when the safety battery terminal is tripped. A prerequisite for this is that none of the main fuses disconnects a circuit due to a short circuit. This ensures continued operation of all important functions such as lights, hazard warning lights, telephone, airbag etc.

Triggering of the safety battery terminal in the event of a collision from any direction is ensured by linking the different restraint system triggering thresholds with the safety battery terminal.

In the event of a front-end collision, the safety battery terminal is triggered parallel to the front airbags.

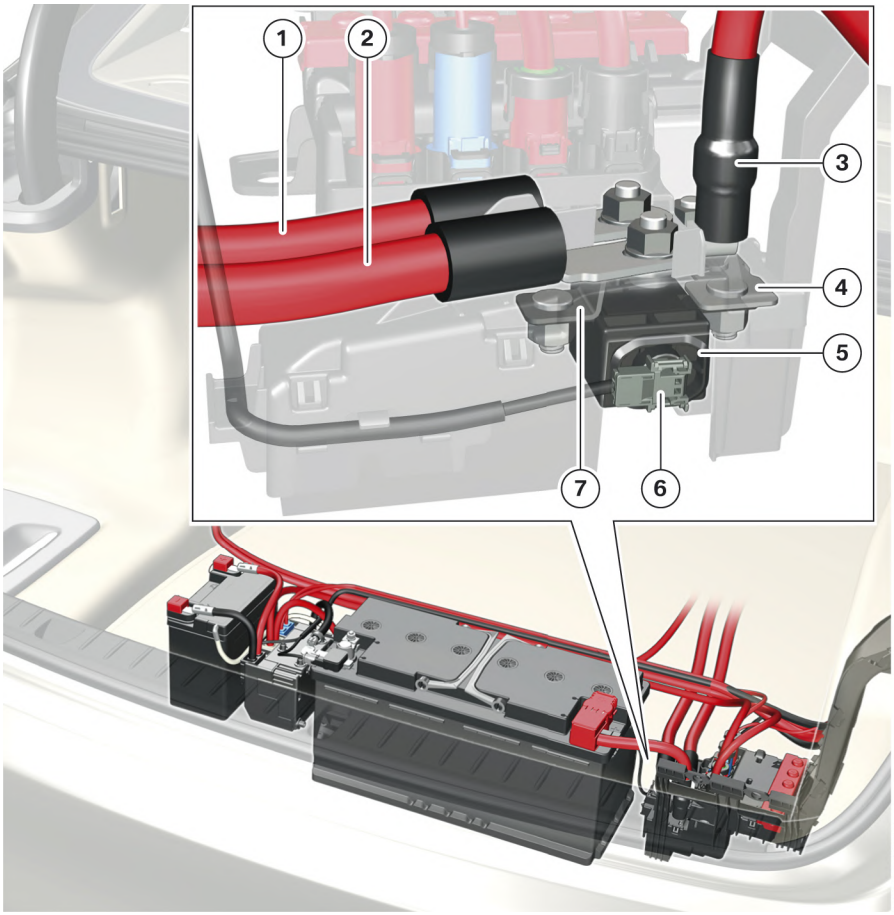
In the case of side collision detection, the triggering threshold of the safety battery terminal is higher than that of the side and head airbags. This is because the vehicle might still be operational after a side collision and it would then no longer be possible to start the vehicle.

The safety battery terminal must be replaced if the B+ cable has been disconnected from the positive battery terminal by the safety battery terminal.

If the metal conical clamp connector is visible, the power supply from the starter battery to the starter motor is disconnected. The safety battery terminal has visibly tripped and interrupted the voltage supply. It is no longer possible for customers to start a BMW vehicle when the safety battery terminal has been triggered.

Safety battery terminal G11/G12/G30

The safety battery terminal in the G11/G12/G30 is not installed on the positive terminal of the battery, but is located in the power distribution box next to the battery.



Detailed view of safety battery terminal in G11/G12/G30

Index	Explanation
1	B+ cable to the starter motor and alternator
2	B+ cable to the jump start terminal point
3	Positive wire in the power distribution box
4	Busbar infeed
5	Safety battery terminal
6	Safety battery terminal actuation
7	Power distribution box busbar

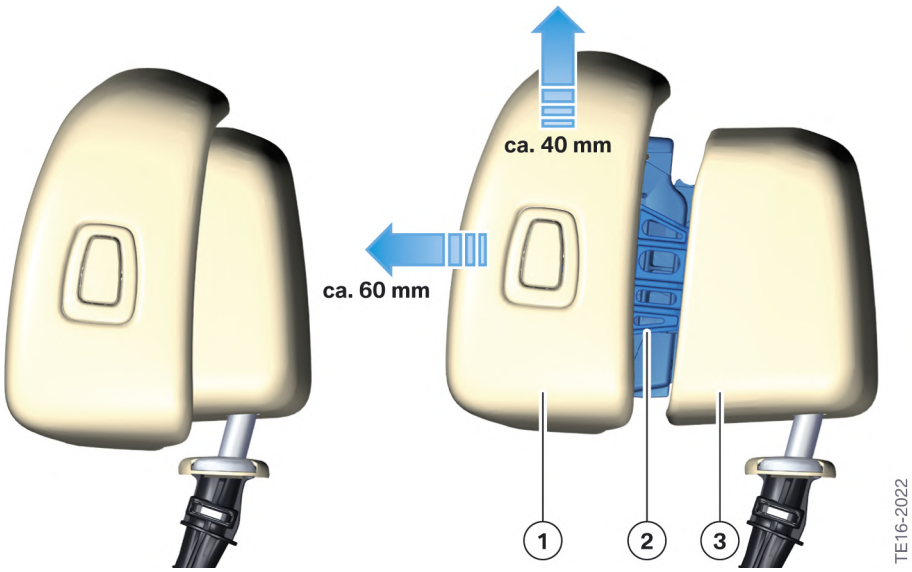
4.4. Crash-active headrest

Since September 2007, a crash-active head restraint has been installed on the driver and passenger's sides depending on the optional equipment. This is a head restraint with a pyrotechnic actuator that optimises the distance and height of the headrest in relation to the head if a rear-end collision of sufficient force occurs. This reduces the load on the cervical vertebrae in the event of a rear-end collision. The head restraint is activated early, even prior to backwards displacement of the occupant's head.

4.4.1. Activation in the event of rear-end collision

The safety electronics of the vehicle detects by means of sensors whether a rear-end collision has taken place. The crash-active head restraint will be activated if the hit vehicle reaches a certain speed delta within a defined period. This can occur even for a slight rear-end collision.

If the safety electronics detects a massive rear-end collision, other safety components can also be additionally activated, e.g. seat belt tensioner and triggering of the safety battery terminal.

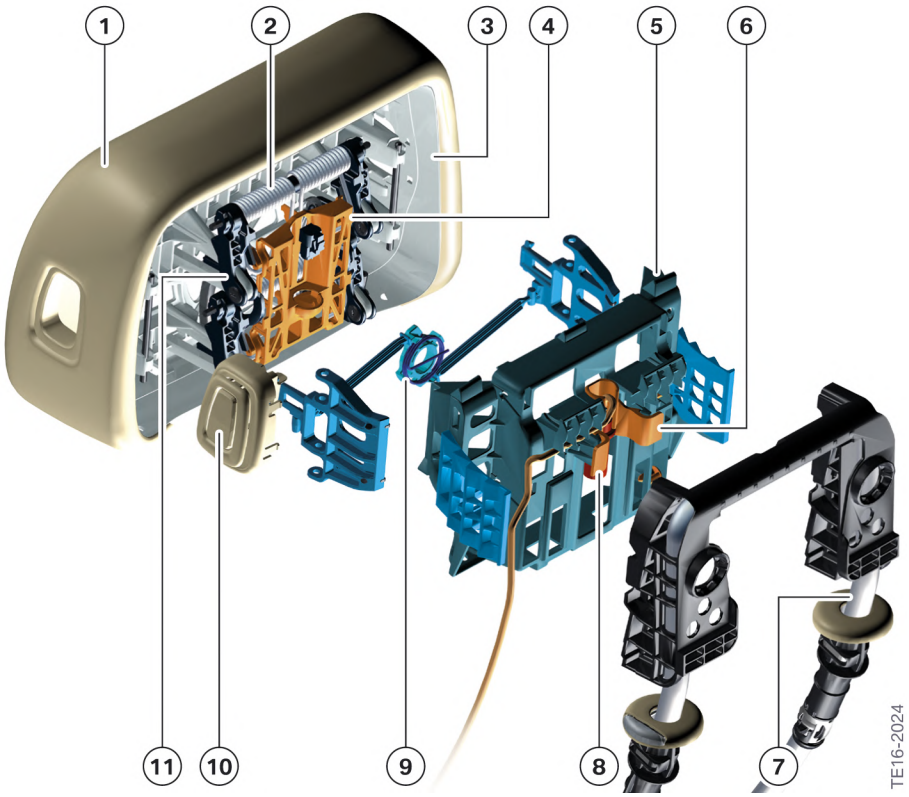


Left: Head restraint in E60 in initial condition, right: Head restraint in E60 after activation

Index	Explanation
1	Head restraint front section
2	Head restraint drive
3	Head restraint support

SAFETY SYSTEMS.
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4.4.2. 1st generation

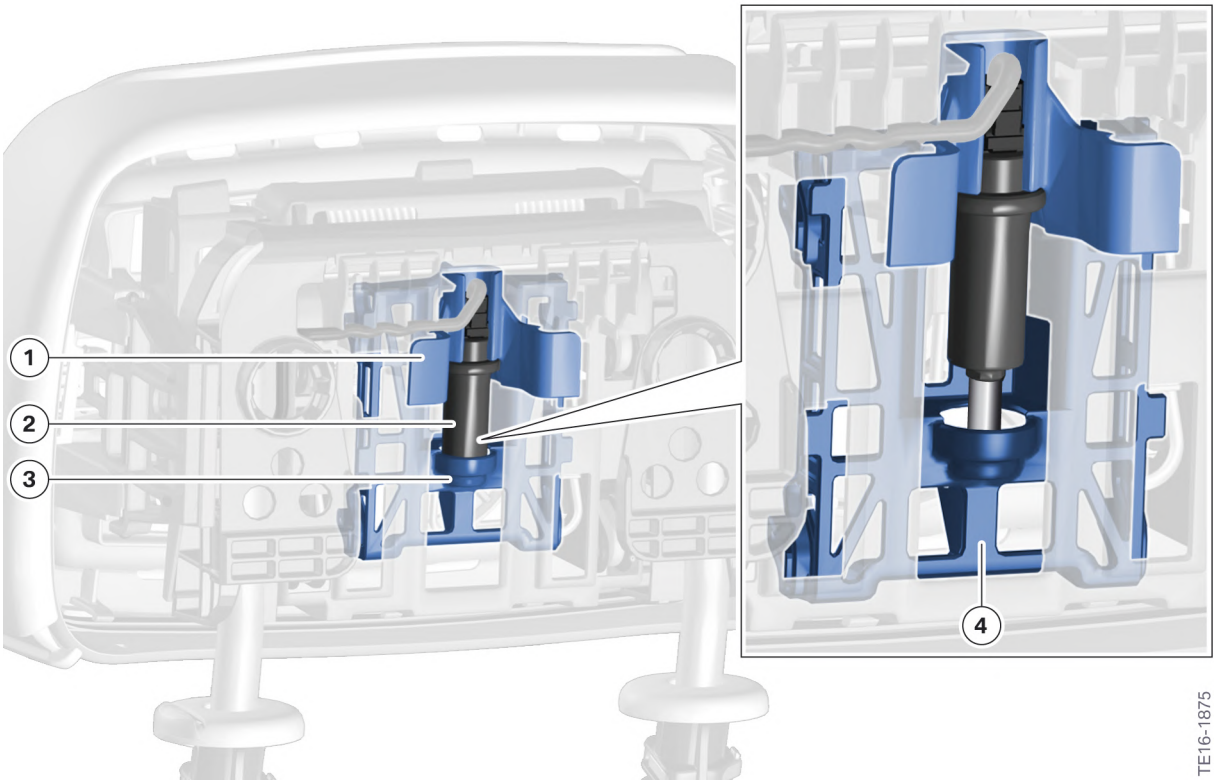


Exploded diagram of crash-active head restraint

Index	Explanation
1	Head restraint cushion
2	Spring
3	Impact plate of the head restraint
4	Trigger plate
5	Head restraint support
6	Actuator holder
7	Head restraint pin
8	Actuator
9	Mechanism for head restraint movement
10	Button for head restraint adjustment
11	Mechanism for head restraint movement

The airbag control unit initiates activation of the crash-active head restraint. The actuator releases the head restraint by actuating the trigger plate. As a result, the front section of the head restraint is moved forwards by a spring. The maximum travel of the head restraint cushion in forward direction is 60 mm. The cushion is raised by up to 40 mm at the same time.

The actuator must be replaced in the workshop after a crash-active head restraint has been triggered. The fault memory must be deleted after replacing the actuator.



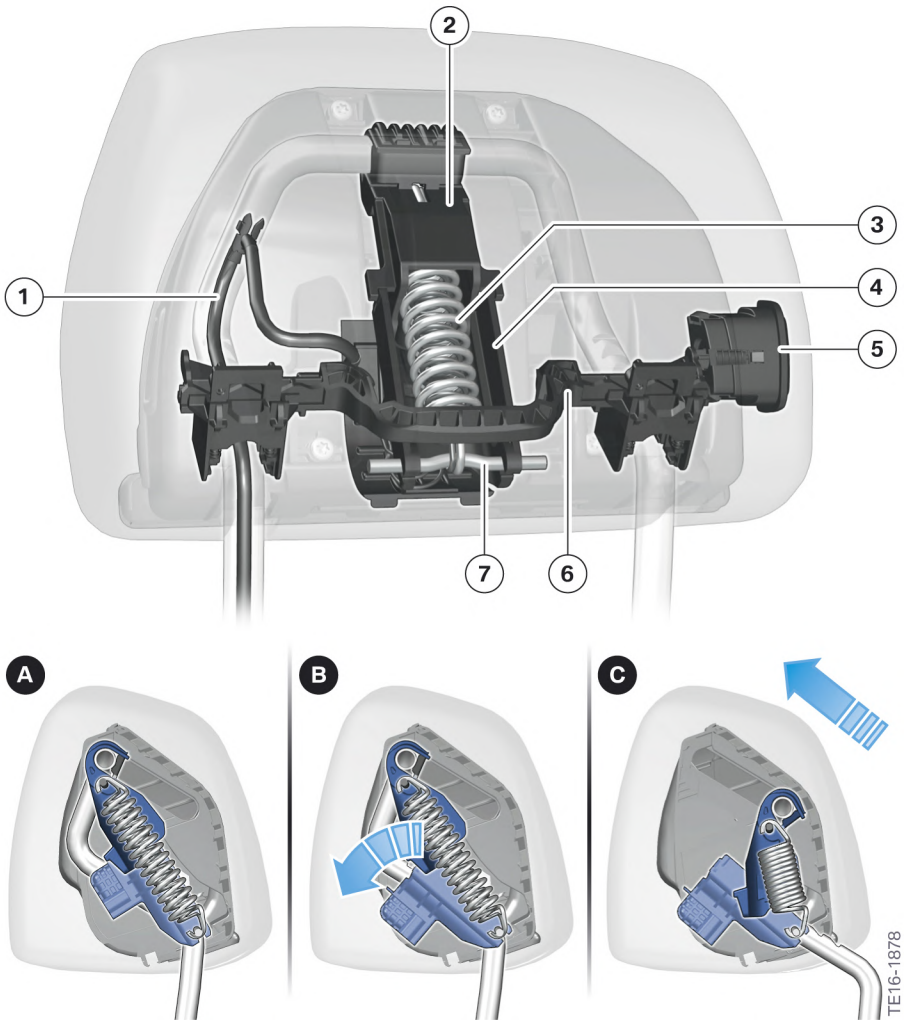
Crash-active head restraint of a E60

Index	Explanation
1	Actuator holder
2	Trigger plate
3	Actuator
4	Pin activated

4.4.3. 2nd generation

SAFETY SYSTEMS.
4. AIRBAG SYSTEMS.

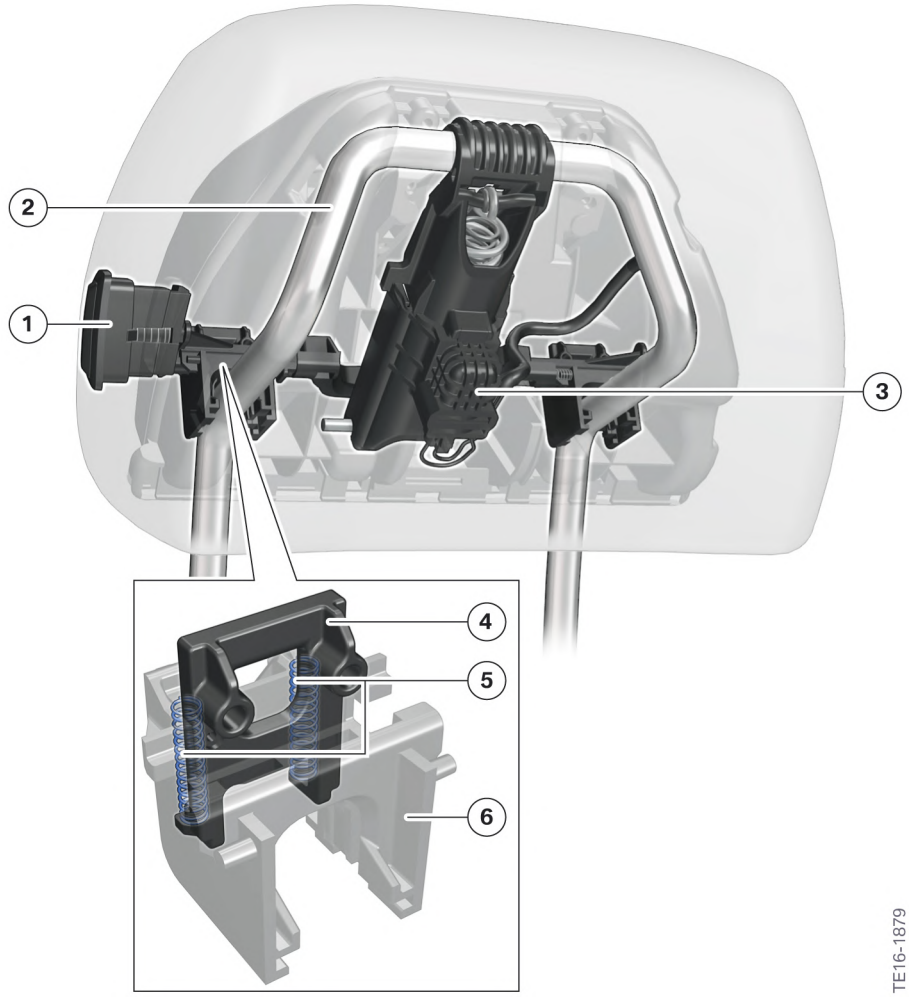
Front view



F12 Design and activation of a crash-active head restraint

Index	Explanation
A	Driving position
B	Support activated
C	Collision position
1	Line
2	Support, upper part
3	Tension spring
4	Support, bottom section
5	Actuating button
6	Cross slide
7	Spring attachment

Rear view



F12 Design and rear view of a crash-active head restraint

Index	Explanation
1	Actuating button
2	Head restraint pin
3	Lower section with actuator
4	Lock
5	Compression springs
6	Pin guide

4.4.4. Service

The crash-active head restraint is retained at the end stop and cannot be pulled out upwards because it is connected to the Crash Safety Module by a two-pin line.

SAFETY SYSTEMS.
4. AIRBAG SYSTEMS.

Check control message

A Check Control message is output on the instrument cluster after the crash-active head restraint has been triggered. Additional information is displayed in the central information system CID:

Check control message	Meaning	Information in central information display
	Restraint system fault	Restraint system. Function of airbags, belt tensioners and belt force limiters fault. Ensure your seat belt is fastened despite the fault. Have the problem checked by the nearest BMW Service Partner.



If the crash-active headrests have been triggered, the pyrotechnic actuators must be replaced. The head restraint can be moved manually back to its initial position.

Seat or head restraint covers and/or accessories that could impair the protective effect must not be mounted on the head restraints.

SAFETY SYSTEMS.
5. PEDESTRIAN PROTECTION.

5.1. Introduction

The pedestrian protection system has the task of detecting an accident situation between a pedestrian and a passenger car in the area of the bonnet and selectively activating the system depending on certain marginal conditions.

The pedestrian protection system is a complete system consisting of the following components:

- Control unit (Crash Safety Module)
- Pedestrian protection sensors
- Pedestrian protection actuators.

The airbag control unit evaluates the signals of the sensors. In the event of a collision with a pedestrian, the airbag control unit triggers the corresponding actuators for pedestrian protection.

The pedestrian protection is triggered only within a speed range between approx. 20 and 55 km/h. The function is blocked in the airbag control unit outside this speed range.

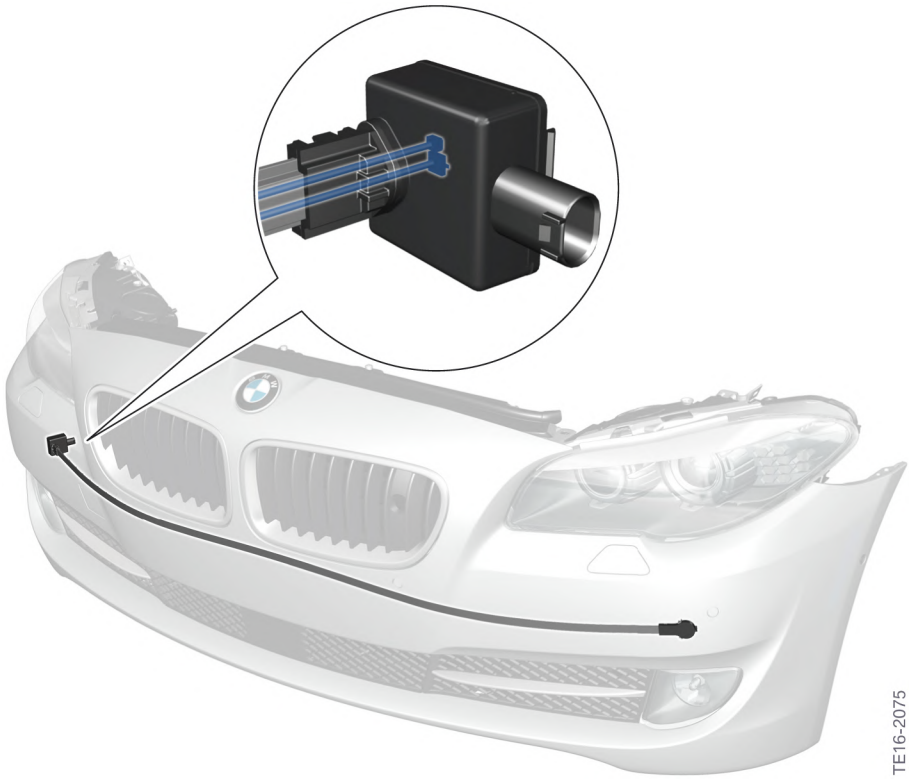
5.2. Pedestrian protection sensors

The various pedestrian protection sensors are listed below.

5.2.1. Pedestrian protection sensor with optical fibre

An optical fibre is installed behind the bumper. The optical fibre is part of the pedestrian protection sensor. The light in the optical fibre is transmitted by the sensor on one side of the bumper and then routed back to the sensor via a loop at the other end of the bumper. The optical fibre is deformed between the interference structures when a force acts on it. This dampens the light signal in the optical fibre.

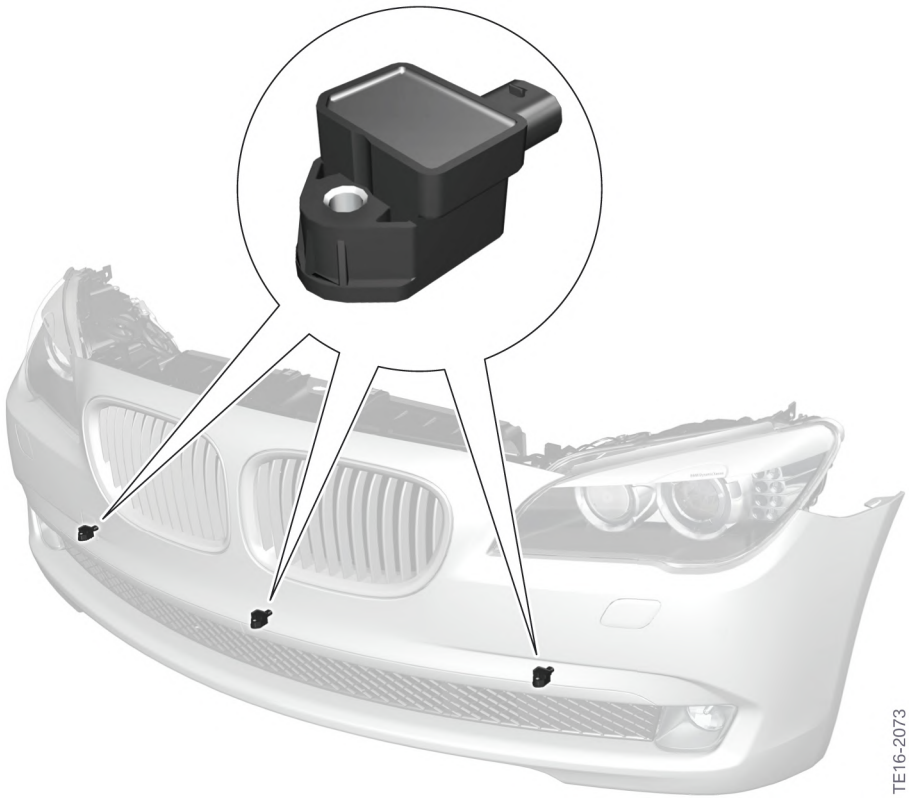
SAFETY SYSTEMS.
5. PEDESTRIAN PROTECTION.



Sensor band in a F10

5.2.2. 3 acceleration sensors

In this version, there are 3 acceleration sensors directly behind the bumper. If a collision is detected, the recorded acceleration values of the acceleration sensor are converted into digital signals and transferred to the airbag control unit.

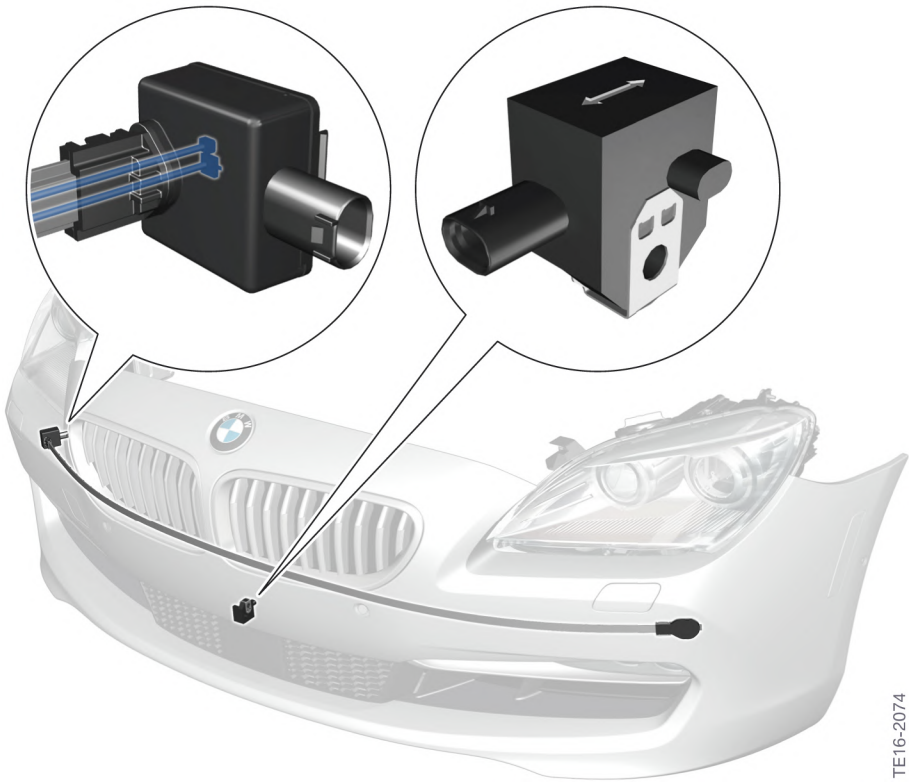


3 acceleration sensors in a F01

5.2.3. Optical fibre with acceleration sensor

An optical fibre and an acceleration sensor are installed behind the bumper. The acceleration sensor is installed centrally on the bumper. Both sensors must detect a crash simultaneously in order for the active bonnet to be triggered.

SAFETY SYSTEMS.
5. PEDESTRIAN PROTECTION.

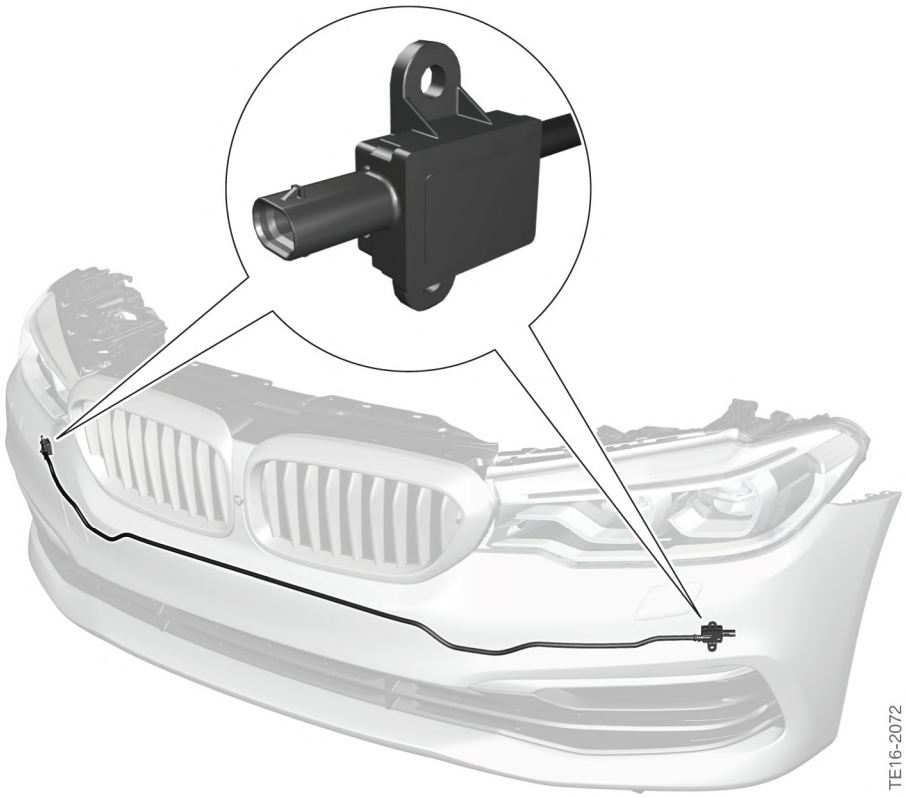


Optical fibres with an acceleration sensor in a F12

5.2.4. Pressure hose sensor

There are 2 pressure sensors (left pedestrian protection sensor and right pedestrian protection sensor) installed which are connected with each other via a pressure hose. The pressure hose is part of the pedestrian protection sensor. Any force that acts on the pressure hose deforms its structure. The displaced air volume leads to a pressure increase at both pressure sensors.

SAFETY SYSTEMS.
5. PEDESTRIAN PROTECTION.



Pressure hose sensor in a G30

5.3. Active engine compartment lid

If a pedestrian collides frontally with the vehicle in the speed range between 20-55 km/h, this is first detected by the sensors. The signals are transmitted via data lines to the Crash Safety Module ACSM. The Crash Safety Module (ACSM) uses the data to determine whether the threshold values for detection of a crash with a pedestrian have been reached or exceeded. This process determines whether triggering of the pyrotechnic actuators on the bonnet takes place.

For safety reasons, the system may also occasionally be activated if it cannot exclude the possibility that an impact with a pedestrian has occurred.

The following situations are possible:

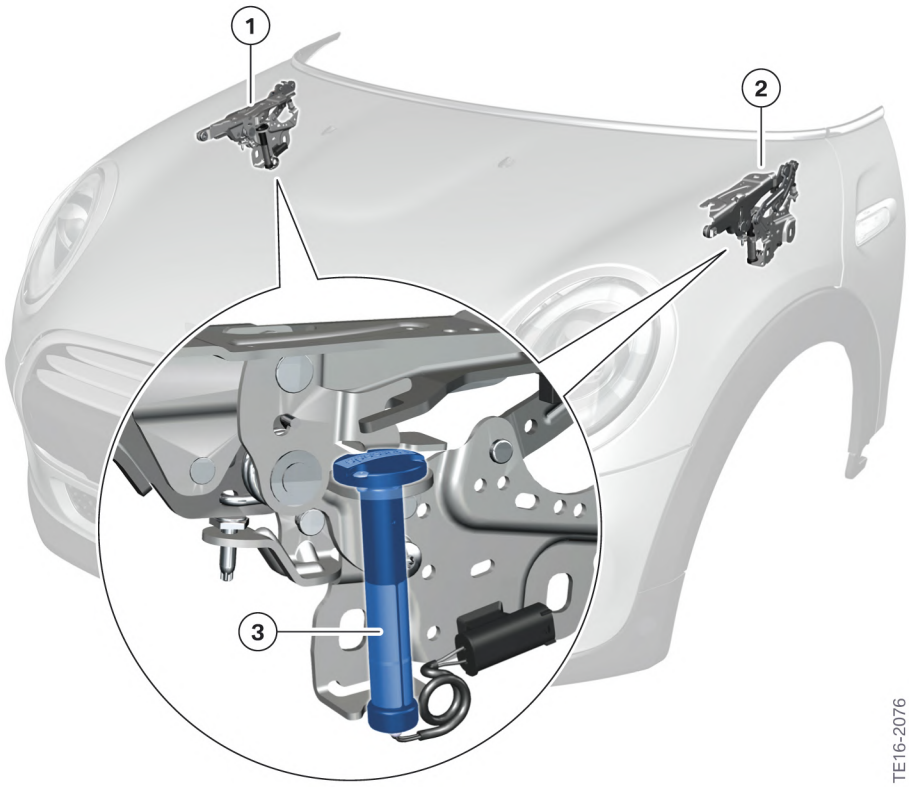
- if the vehicle crashes into a container or a road marker post
- if the vehicle collides with an animal
- in the event of stone chipping
- if the vehicle is driven into a snowdrift

SAFETY SYSTEMS.
5. PEDESTRIAN PROTECTION.

5.3.1. Semi-active bonnet.

The semi-active bonnet has been installed in MINI models since the F56. The semi-active bonnet is also used on BMW models such as the E93 life cycle impulse. If the vehicle collides with a pedestrian, the bonnet is raised by the bonnet hinges. The hinge locks are unlocked by pyrotechnic actuators. The bonnet is raised via the two gas pressure springs on the bonnet.

The increased distance between the bonnet and units in the engine compartment creates an additional deformation space. This means that the risk of injury to pedestrians can be reduced in certain accident situations.



F57 Semi-active bonnet

Index	Explanation
1	Right bonnet hinge with actuator
2	Left bonnet hinge with actuator
3	Actuator

Observe the repair instructions when working on the bonnet catches and bonnet hinges. There is a danger of injury as a result of unintentional triggering of the system.

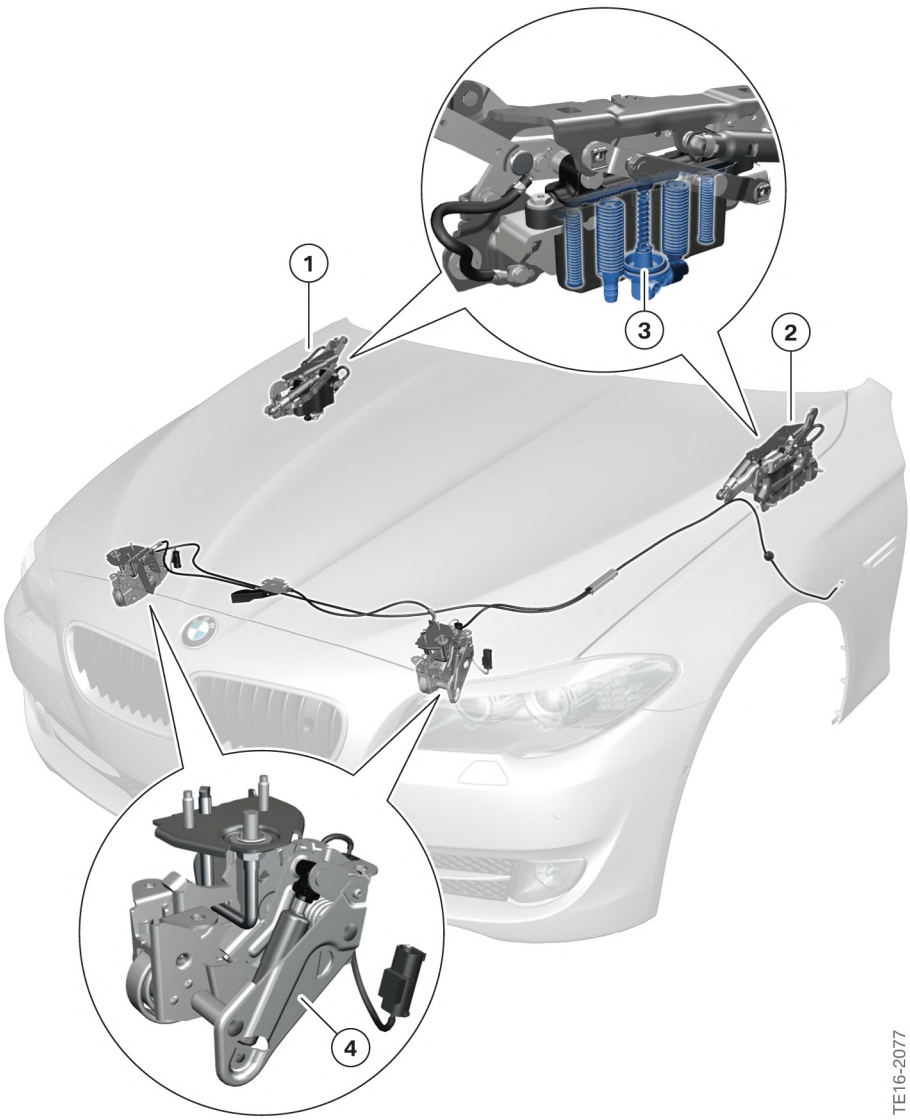
The **expiry date** of the gas pressure springs must be checked regularly. This is otherwise part of the scope of maintenance work.

5.3.2. Fully active bonnet

An active bonnet is installed depending on national-market version and model. In the event of a collision with a pedestrian, the bonnet is lifted by up to 30 mm by the bonnet catches. At the same time, the bonnet hinges are raised by up to 60 mm. The bonnet hinge locks and catches are unlocked by pyrotechnic actuators. The bonnet is then lifted by spring force.

The increased distance between the bonnet and units in the engine compartment creates an additional deformation space. This means that the risk of injury to pedestrians can be reduced in certain accident situations.

If the active bonnet is triggered, a red warning light is displayed in the instrument cluster and a Check Control Message is displayed in the central information display. The engine compartment lid remains raised, thus impairing visibility. There is no safety risk as a result of the activated engine compartment lid. It is possible to drive carefully to the nearest workshop. The active pedestrian protection only becomes available again, however, after the component has been replaced.



F10 Fully active bonnet

SAFETY SYSTEMS.
5. PEDESTRIAN PROTECTION.

Index	Explanation
1	Right bonnet hinge
2	Left bonnet hinge
3	Actuator
4	Bonnet catch on left and right with actuator

On vehicles such as the F45, it is possible to reset the active bonnet to the closed initial position after triggering.

On vehicles such as the G11/G12, temporary engagement of the bonnet hinges like on the F45 is not possible. In spite of this, it is still possible to drive to the next BMW Service after the system has been activated.



Observe the repair instructions when working on the bonnet catches and bonnet hinges. There is a danger of injury as a result of unintentional triggering of the system.

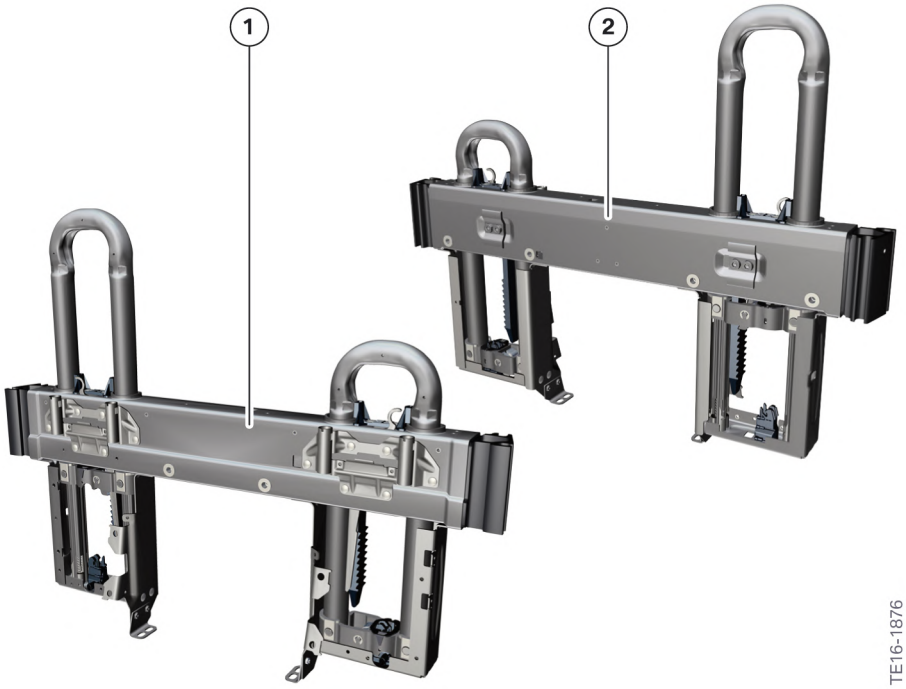
The **expiry date** of the gas pressure springs must be checked regularly. This is otherwise part of the scope of maintenance work.

SAFETY SYSTEMS.
6. ROLLOVER PROTECTION.

6.1. Introduction

The rollover protection makes a significant contribution to the passive safety of convertibles.

The rollover protection comprises the reinforced windscreen frame and one or more rollover protection bars. They can be designed as fixed or retractable elements. All current BMW models have retractable rollover protection bars for the driver's side and the passenger's side.



E88 Rollover protection system

Index	Explanation
1	Front view
2	Rear view

The rollover protection bars are generally located in a common module. In the event of a vehicle rollover, the rollover protection bars extend by the spring force and lock into place with a positive connection. This applies to both vehicles with a softtop and a retractable hardtop, both in open and closed state. Together with the reinforced windscreen frame, this ensures the survival space for the occupants.

The actuators release the spring-biased rollover bars. The two rollover bars extend and lock in their end position by means of pawl and toothed strip.



When the rollover protection bars are extended, the softtop or retractable hardtop must not be operated, as this may cause damage.

SAFETY SYSTEMS.
6. ROLLOVER PROTECTION.

6.2. Special features

- Increased passive safety
- Structure-orientated modular construction
- Reduced weight compared with rollover protection system with fixed-bodyshell partition wall
- Load-bearing parts made from lightweight, high-strength aluminium
- Supported vehicle rigidity.

6.3. Triggering criteria

The decision about triggering the rollover protection system is made by the airbag control unit. The ACSM calculates the probability of an imminent rollover using the available sensor signals. The airbag control unit is therefore able to already activate the rollover protection bars before the actual rollover. A front-end, rear or side collision of corresponding severity can also lead to triggering of the rollover protection system.

The key sensor signals for calculating an imminent rollover are:

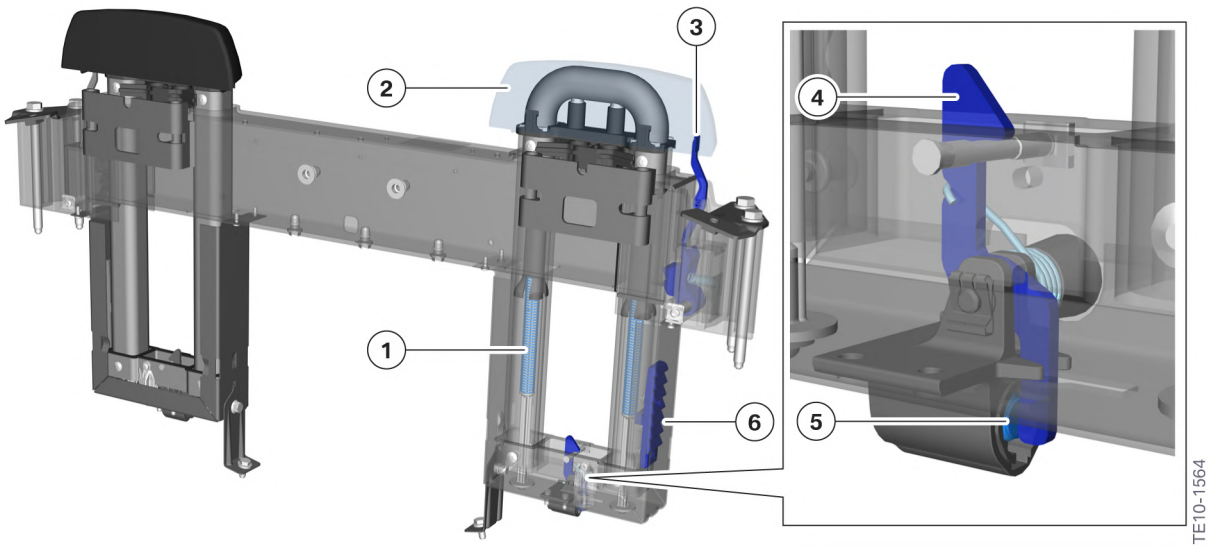
- Longitudinal acceleration
- Lateral acceleration
- Yaw rate.

The airbag control unit has special acceleration sensors in order to be able to detect an imminent rollover, also at very slow tilting speeds. They are also able to detect minimal accelerations.

6.4. Activation systems

2 different systems are used for triggering the rollover protection bars at BMW. However, actual extension is performed purely mechanically by means of coil springs.

6.4.1. Pyrotechnic actuators



F12 Pyrotechnic actuator

Index	Explanation
1	Spring
2	Cover
3	Release lever
4	Lock
5	Ignition squib
6	Serrated strip

With this system, the lock of the rollover protection bars is opened by two pyrotechnic actuators.

If the rollover protection system has been activated, the rollover protection bars can be reset only after a mechanical lock is released. For this purpose, the two rollover protection bars must be pressed down manually against the spring force. In order to guarantee mechanical functioning of the rollover protection system, it is possible to manually trigger the rollover protection system in Service.

6.4.2. Electromagnetic actuators



F33 Electromagnetic actuator

Index	Explanation
1	Serrated strip
2	Interlock pawl
3	Electromagnetic actuator

With this system, the lock of the rollover protection bars is opened by two electromagnetic actuators.

The rollover protection system can be manually activated in Service. This is done either with help of a service function of the diagnosis system or manually by applying voltage to the electromagnetic actuators.

Rollover protection controller

Some vehicles require an additional control unit, the rollover protection controller, for activating the rollover protection system. This contains power electronics for activating the actuators.

When the rollover protection controller detects an imminent rollover, it sends 2 electric messages within a defined time frame:

- **Firing readiness**
  The first message requests the rollover protection controller to prepare for firing. Capacitors within the rollover protection controller are charged.
- **Firing command**
  The second message contains the firing command. Now the rollover protection controller discharges the ignition capacitors and the two actuators are supplied with voltage.

6.5. Service

If the rollover protection system is removed due to a repair, it must be mechanically triggered using the following method before removal:

- Open soft top
- Open tailgate
- Take off the covers on the left and right-hand side of the luggage compartment trim panel
- Using a hexagon socket wrench, turn against the compression force until the bar travels upwards.



Service triggering of rollover protection system on a F12

Index	Explanation
1	Covers

Reset the roll-over protection system following service activation

SAFETY SYSTEMS.
6. ROLLOVER PROTECTION.

- Pull the release lever out and hold in this position
- Push rollover bar downwards
- Let go of the release lever just before the rollover bar engages
- Push the rollover bar right down until it audibly engages.



Danger of injury!

Before activating the system, make sure that nobody is in the immediate vicinity of the rollover bar.

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.

7.1. Lateral and longitudinal acceleration sensor, B-pillar

The B-pillar airbag sensor consists of a longitudinal acceleration sensor and a lateral acceleration sensor. The acceleration sensors measure both the acceleration and the deceleration in the X and Y directions. The result from the X and Y signals is decisive in identifying the direction of the crash. The airbag sensors in the B-pillar assist in detection of front-end, side and rear-end crashes. The airbag sensors on the left and right are of identical design and are fixed by way of mechanical encoding upon installation.



G30 Lateral and longitudinal acceleration sensor, B-pillar

7.2. Airbag front sensor

Some models additionally have airbag front sensors installed in the front area of the engine supports. The measured values are forwarded to the Advanced Crash Safety Module (ACSM) where they are evaluated.

The airbag front sensors assist in detection of a front-end collision.

The sensors deliver additional information to the Advanced Crash Safety Module (ACSM) on the characteristics and severity of the collision. Each sensor contains an acceleration sensor for recording the deceleration, signal processing technology and electronics for data transfer. The measured values are sent in the form of a message to the Crash Safety Module ACSM and are included in the calculation of the algorithm.

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.



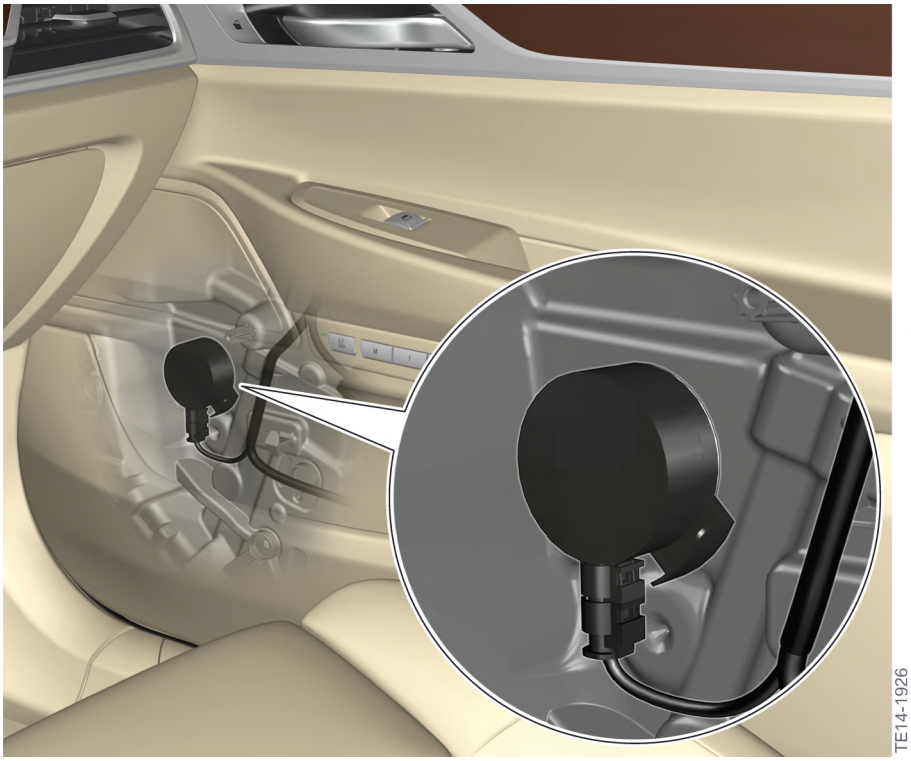
G11/G12 Airbag front sensor

7.3. Door airbag sensor (pressure)

On vehicles such as the F01, pressure sensors are installed in the driver's and front passenger door. A new feature is that the pressure sensors in the doors are no longer secured by means of a screw as in the past. They are now screwed into the holder (inner door panel). The electrical connections or plug connections can be connected only after installing the respective sensors.

The airbag sensors in the doors assist in side impact detection. In addition to the lateral acceleration values, the pressure in the door cavity also increases in the event of a side collision.

The airbag sensors in the doors are used to verify the plausibility of the acceleration signals from the B-pillar airbag sensors and the Crash Safety Module when a side impact is detected. The airbag sensors are situated in the inner panel of the doors and measure the pressure increase in the event of a side-on collision. In the event of a side-on collision at the door the outer skin is pushed inwards. The door interior is reduced in size and the inner pressure increases. This pressure change is measured by the airbag sensors. The airbag sensor also includes electronics, in addition to the pressure sensor, that digitise the pressure readings and transmit them cyclically to the Advanced Crash Safety Module (ACSM). The data transfer is effected analogue to the airbag sensors in the B-pillars. The pressure readings are evaluated in the Advanced Crash Safety Module (ACSM).



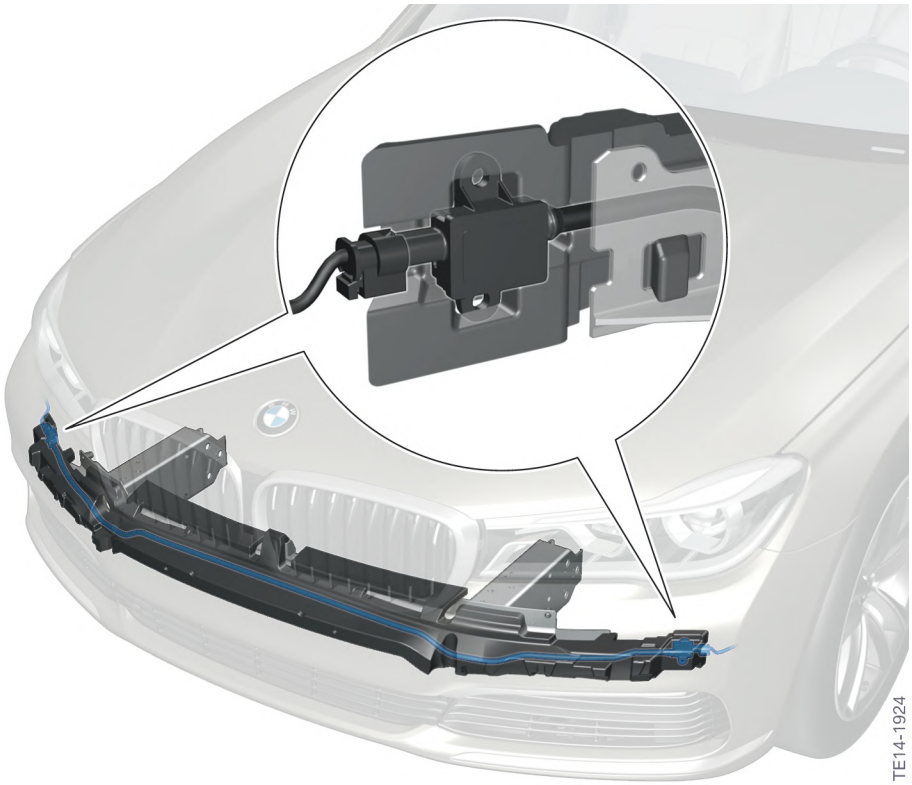
G11/G12 Door airbag sensor (pressure) connected

7.4. Pressure hose with sensors

Any force that acts on the pressure hose deforms its structure. The displaced air volume leads to a pressure increase at both pressure sensors (left pedestrian protection sensor and right pedestrian protection sensor).

Each pressure sensor transmits a pressure signal dependent on the mass and rigidity of the object that collides with the vehicle to the Crash Safety Module (ACSM) via a data line. The characteristic of the signal is analysed in the Crash Safety Module (ACSM) and a triggering decision taken if appropriate.

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.

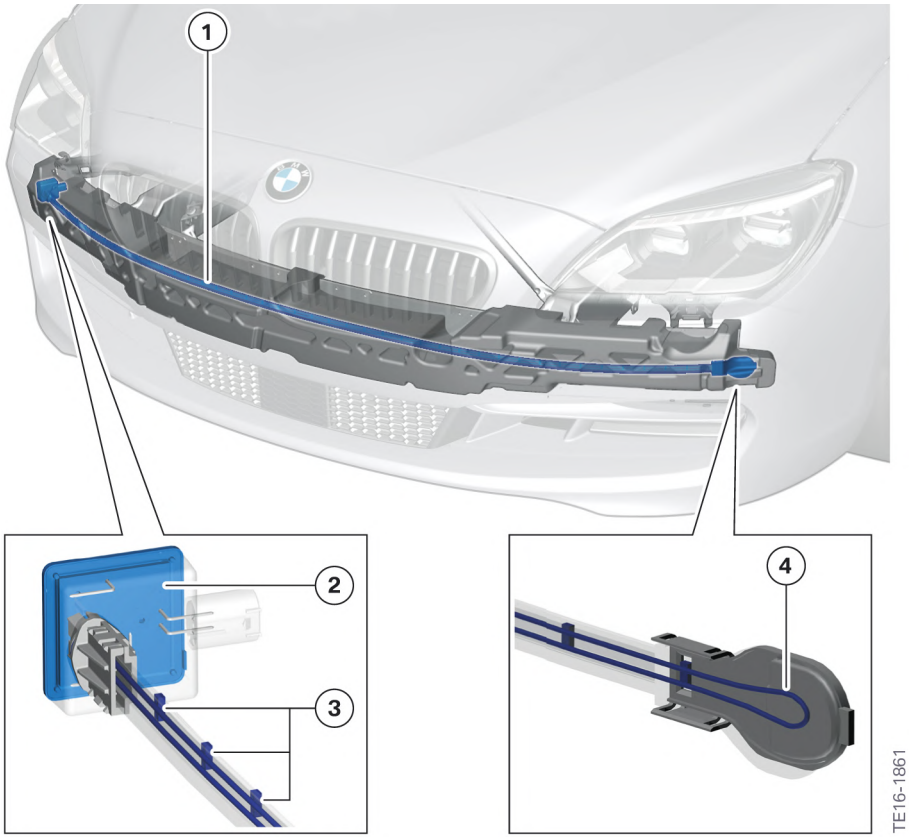


G30 Pressure hose with sensors

7.5. Optical fibre

The optical fibre is deformed between the interference structures when a force acts on it. This dampens the light signal in the optical fibre. The degree of damping depends on the force applied. As the damping of the light signal varies depending on the mass and rigidity of the colliding object, a characteristic signal is generated.

This signal is measured by the sensor and forwarded via a data line to the Crash Safety Module ACSM. On the basis of this data and the data from the central acceleration sensor in the bumper, the Crash Safety Module ACSM determines whether the threshold values for detection of a collision with a pedestrian have been reached or exceeded and thus takes the triggering decision for the actuators on the bonnet.



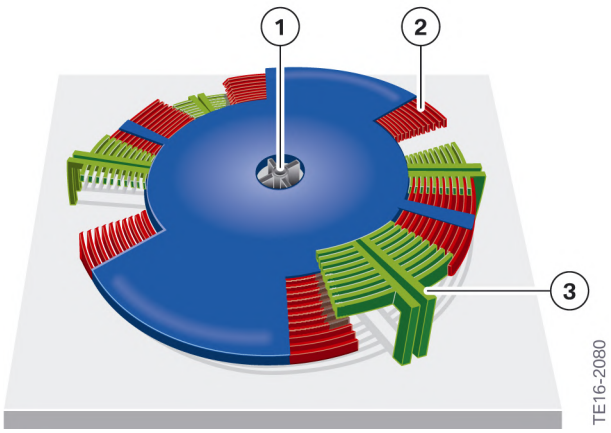
F12/F13 Optical fibre

Index	Explanation
1	Optical fibre
2	Sensor
3	Interference structure
4	Loop

7.6. Yaw sensor

A yaw sensor is used for detection of the rollover. The yaw sensor is designed with a comb structure. It comprises the fixed comb structure and the torsional oscillator with the moving comb structure. The torsional oscillator is mounted centrally on a spindle so that it is capable of movement.

If a rollover takes place, this results in a change in capacitance. The change in capacitance generates a change in voltage in the mV range which is dependent on the deflection of the torsional oscillator. This change in voltage is evaluated as a measured variable.



Yaw sensor

Index	Explanation
1	Rotational axis
2	Torsional oscillator
3	Fixed comb structure

7.7. Sensors in the Integrated Chassis Management

In addition to the driving dynamics control sensors, the ICM also has sensors for crash and rollover detection depending on vehicle. The fact that the airbag sensors are integrated in the ICM control unit means that passive and active safety are networked. The ACSM control unit ensures the voltage supply for the sensors.

The ICM control unit (Integrated Chassis Management) is located centrally on the transmission tunnel.

The following sensors may be installed:

- Longitudinal acceleration sensor (max. 3)
- Lateral acceleration sensor (max. 3)
- Vertical acceleration sensor
- Yaw sensor.

The ICM is connected directly via 4 lines to the 4th generation ACSM for transmission of the sensor data. The ACSM evaluates the data provided by the sensors.

The sensor data of the ICM assists the 4th generation ACSM in detection of side or front-end collisions and in rollover detection.

The following graphic shows an ICM control unit:



Integrated Chassis Management

The ICM control unit must be removed or installed only when the battery is disconnected. The airbag control unit uses the sensor system in the ICM control unit for airbag deployment. Otherwise there is a risk of unintentional airbag deployment!

7.8. Central sensor

The central sensor is an acceleration sensor and a lateral acceleration sensor and is installed centrally in the vehicle on the transmission tunnel in combination with the 3rd generation ACSM. The central sensor detects both acceleration and deceleration (deceleration indicates a front-end collision and acceleration a rear-end collision). The side on which the impact occurs is detected by means of the polarity of the sensor signals.

The following graphic shows a central sensor:



Central sensor

The central sensor is supplied with voltage by the Crash Safety Module (ACSM). The data is transmitted via a two-wire connection.

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.

7.9. Switch for front passenger airbag deactivation

In vehicles with the front passenger airbag deactivation optional equipment (OE 5DA) the switch for front passenger airbag deactivation is used to deactivate the front and side airbags on the passenger's side manually.

The switch for front passenger airbag deactivation is operated manually using the mechanical key. The switch position is determined via a hall effect sensor. The Crash Safety Module (ACSM) supplies voltage to the sensor and performs the evaluation.

The following graphic shows a switch for front passenger airbag deactivation:



G11/G12 switch for front passenger airbag deactivation

7.10. Battery cable monitoring

On some models, the battery cable for the starter motor and alternator is routed from the luggage compartment to the engine compartment on the underbody.

If the battery cable is routed parallel to the fuel line on the underbody, it is necessary to monitor the battery cable on vehicles with petrol engine. Monitoring can be omitted on diesel vehicles due to the fact that the fuel is not easily flammable. In the event of damage to the insulation of the battery cable in an accident or if the vehicle slides over an obstacle (e.g. crash barrier), the battery cable is disconnected from the battery and the alternator switched off. This prevents a short circuit and possible sparking.

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.

7.11. Warning lights

7.11.1. Airbag indicator light

The airbag indicator light is located in the instrument cluster. The system readiness of the Crash Safety Module ACSM is indicated by the airbag indicator lamp lighting up and then going out.
The airbag indicator light is controlled via a message from the Crash Safety Module ACSM to the instrument cluster. The instrument cluster receives a message cyclically. The airbag indicator light is activated if the message is not received.

The following graphic shows the airbag indicator light in the instrument cluster:



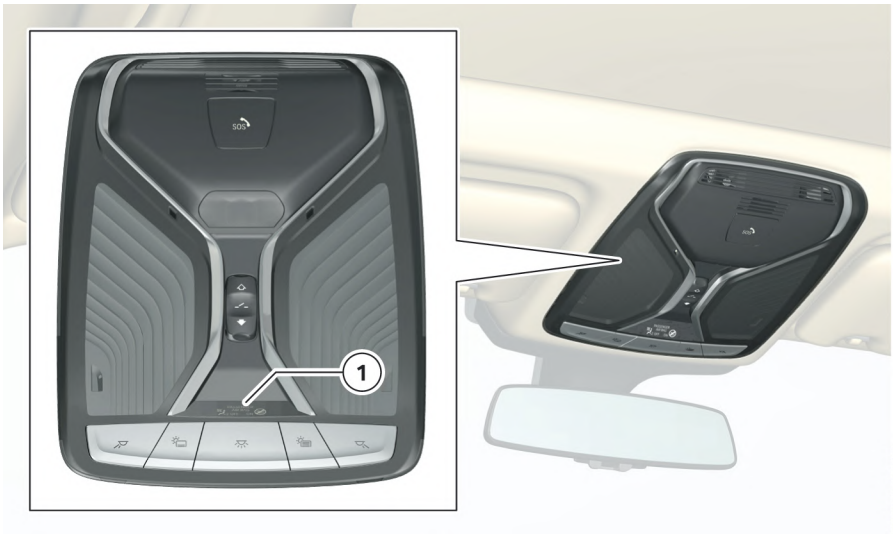
G11G12 Airbag indicator light

7.11.2. Indicator lamp for front passenger airbag deactivation

On vehicles such as the G11/G12, the indicator light for front passenger airbag deactivation is located in the roof function centre FZD. The indicator lamp for front passenger airbag deactivation is activated and lights up yellow if the front passenger airbag and the side airbag on the front passenger side have been deactivated by the switch for front passenger airbag deactivation.

This figures shows the indicator lamp for front passenger airbag deactivation in a G11/G12:

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.



G11/G12 indicator lamp for front passenger airbag deactivation

Index	Explanation
1	Indicator lamp for front passenger airbag deactivation

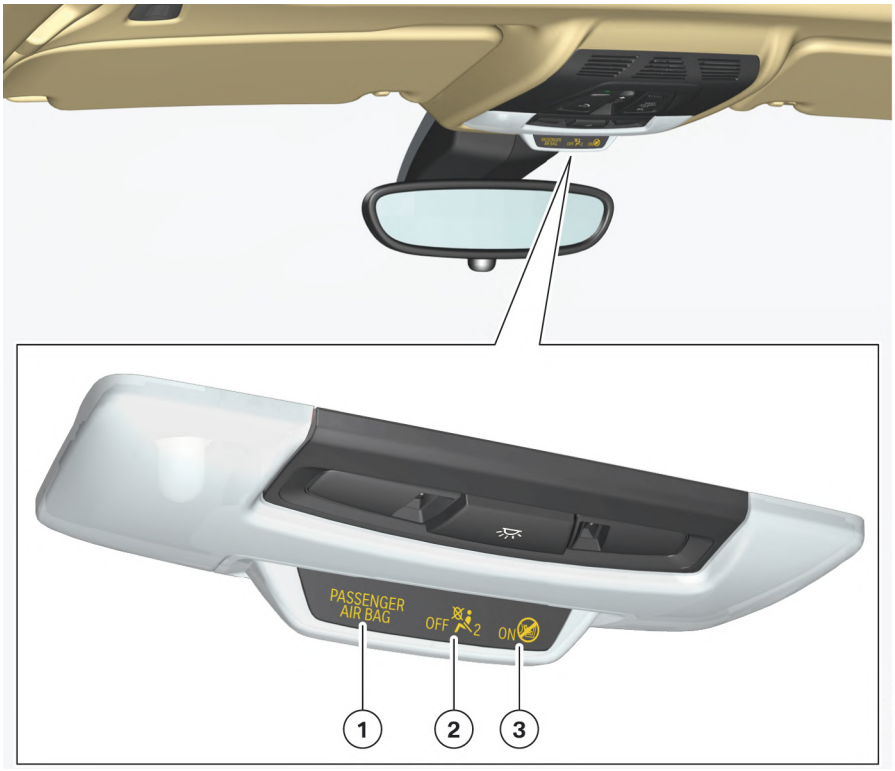
In vehicles such as the F45, the indicator lamp for front passenger airbag deactivation is no longer located in the roof function centre, but in a specially installed light in front of the roof function centre. With this installation location, the visibility of the status of the front passenger airbag for the occupants is improved and satisfies the current requirements of the European New Car Assessment Programme. In vehicles which have the optional equipment European version (SA230), the redesigned indicator light is installed in conjunction with at least one of the following optional equipment items:

- Front passenger airbag deactivation (SA5DA)
- ISOFIX child seat mountings (SA470)
- Front passenger backrest, folding (SA4FK).

For each engine start, the occupant is shown whether the front passenger airbag is activated or deactivated.

This figures shows the indicator lamp for front passenger airbag deactivation in a F45:

SAFETY SYSTEMS.
7. SENSORS AND SWITCHES.



F45 roof function centre with indicator lamp for front passenger airbag deactivation in European version

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.

8.1. Introduction

Three-point seat belts are installed in all current BMW vehicles. The high protective effect of the three-point seat belt has been proven by a large number of statistical accident analyses.

The seat belt is a restraint system which is secured to the vehicle body and which holds the occupants in their seats in the event of abrupt vehicle deceleration (e.g. due braking manoeuvres and accidents) and therefore prevents them from being thrown around or out of the passenger compartment.

The following graphic shows a driver's seat with a three-point seat belt:



Three-point seat belt, driver's seat

Index	Explanation
1	Seat belt buckle
2	Retractor tensioner

A seat belt which has become unusable must be disposed of after removal so that renewed use is not possible.

Three-point seat belts must be renewed in the following situations:

- Creasing
- Pinching damage
- Detached fibres
- Tear and cut damage
- Melt marks.

8.2. Belt tensioner

8.2.1. Introduction

A high level of wearing comfort must be achieved in order to ensure high acceptance for wearing seat belts among customers. The wearing comfort is determined to a significant extent by the retraction force in the inertia reel. A certain belt slack is therefore allowed. However, the belt slack has a negative influence on the forward movement of the occupants in the event of an accident. For this reason, the belt slack must be removed from the seat belt before the restraining effect can act. This task is performed by the seat belt tensioner systems.

8.2.2. Types of seat belt tensioner systems

4 types of seat belt tensioner systems are used at BMW:

- Retractor tensioner
- Seat belt buckle tensioner
- End fitting pretensioner
- Electromechanical automatic reel.

All systems have the goal of reducing the belt slack and the forces acting on the body in the event of an accident.

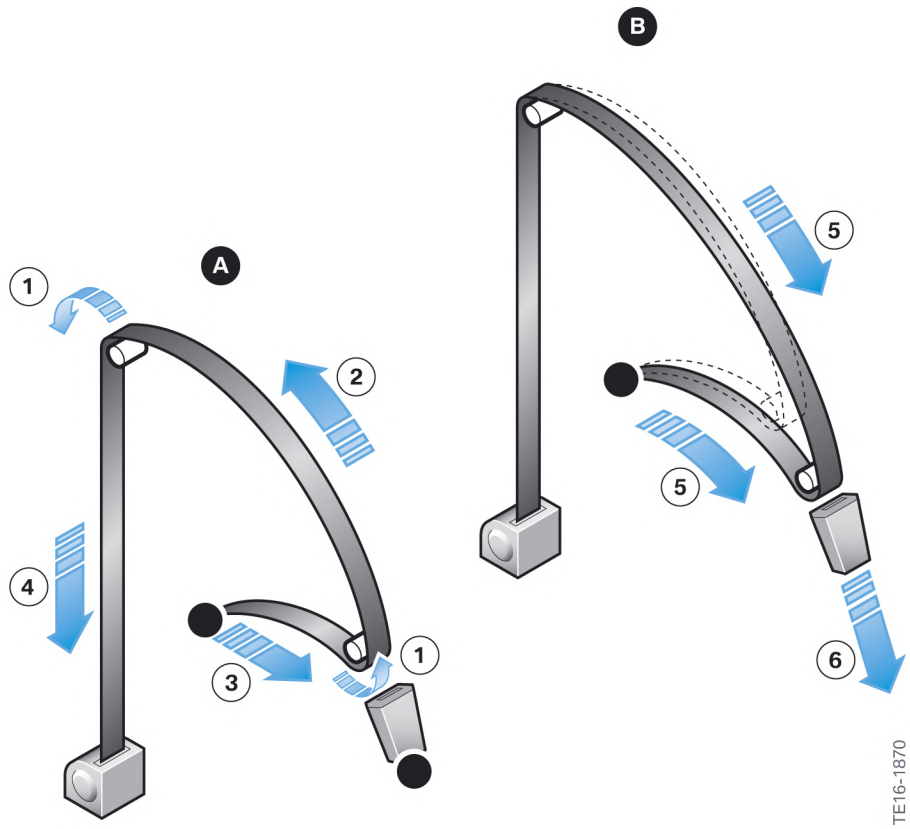
Seat belt buckle tensioner

BMW was the first manufacturer to install a mechanical seat belt buckle pretensioner. The goal was to reduced the belt slack not just in the thoracic region but also in the area of the pelvis.

Simultaneous tensioning of the shoulder and lap belts by the seat belt buckle pretensioner has a greater influence on the restraining effect than is achieved with a pyrotechnic reel pretensioner. The shoulder and pelvis slack are eliminated simultaneously by retraction of the seat belt buckle. The different influences in relation to the tensioning effect can be seen in a direct comparison between a reel pretensioner and a seat belt buckle pretensioner.

This graphic shows a comparison between the reel pretensioner and the seat belt buckle pretensioner:

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.



Tensioner comparison

Index	Explanation
A	Function of reel pretensioner
B	Function of seat belt buckle pretensioner
1	Deflection point
2	100 % travel of the seat belt strap for 30 % of the force generated by the **reel pretensioner**
3	No travel of the seat belt strap for 10 % of the force generated by the **reel pretensioner**
4	100 % travel of the seat belt strap for full force generated by the **reel pretensioner**
5	100 % travel of the seat belt strap for 50 % of the force generated by the **seat belt buckle pretensioner**
6	100 % travel of the seat belt strap for full force generated by the **seat belt buckle pretensioner**

Seat belt tensioning at the seat belt buckle results in the shoulder and lap belts each being tensions by approximately the complete amount of the belt buckle travel. In conventional systems, only the single tensioning travel is available for the entire seat belt strap. BMW uses pyrotechnic seat belt buckle pretensioner technology in all current models. The pyrotechnic seat belt buckle pretensioners replace the reel pretensioners and mechanical seat belt buckle pretensioners.

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.

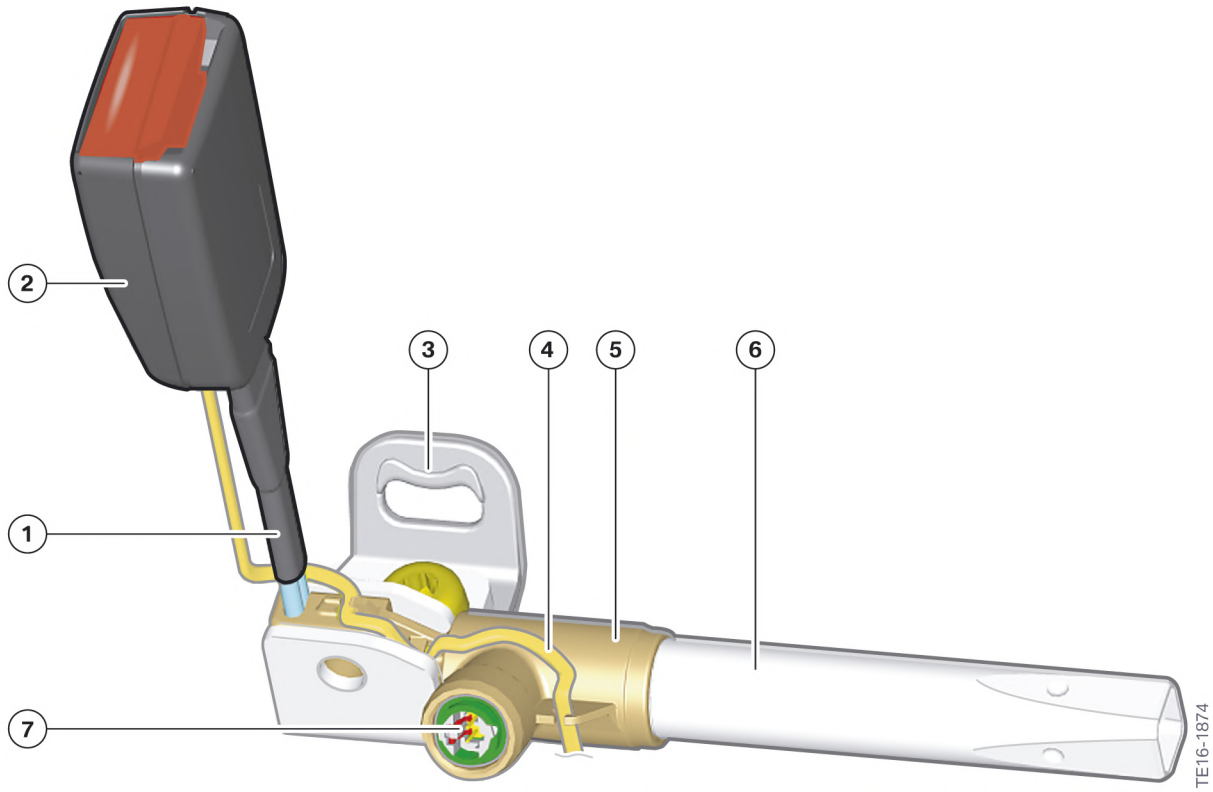
Pyrotechnic seat belt buckle pretensioner

The task of the pyrotechnic seat belt buckle tensioner is to minimise the belt slack in the pelvis and shoulder regions in the event of a crash, thereby improving the restraining effect.

The seat belt buckle pretensioner is locate on the driver's and front passenger seat.

The seat belt buckle is connected to the piston in the tensioning tube by means of a steel cable. If the ignition squib is triggered, gas pressure is created, which moves the piston in the tensioning tube. This causes the cable to pull the seat belt buckle down and the seat belt is tensioned.

A reduction in the forces acting on the thoracic region of the occupants is achieved in combination with the mechanical force limiter in the inertia reel.



Seat belt buckle tensioner

Index	Explanation
1	Buckle anchorage
2	Seat belt buckle with seat belt buckle switch
3	Mounting bracket
4	Seat belt buckle switch connection
5	Gas generator
6	Tensioning tube
7	Connection of ignition squib

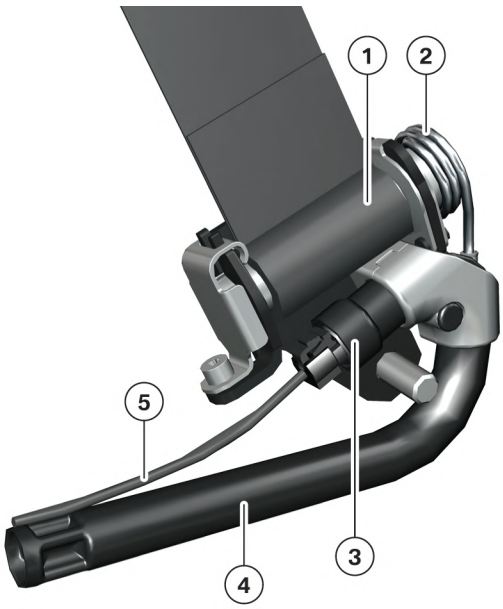
SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.

The seat belt buckle switch is integrated in the seat belt buckle. The seat occupancy is detected and evaluated by means of the seat belt buckle switches.

The airbag control unit installed in the vehicle is responsible for triggering the pyrotechnic seat belt buckle pretensioner. The trigger threshold value of the seat belt buckle pretensioner is below the airbag trigger threshold value. In other words, the seat belt buckle pretensioner may be triggered in an accident where no airbag is deployed.

End fitting pretensioner, front

The end fitting pretensioner acts to reduce the forward travel of the driver's pelvis in the event of a front- or rear-end collision and is installed on the driver's seat in addition to the seat belt tensioner in many vehicles. The end fitting pretensioner functions according to the same principle as the seat belt buckle pretensioner.



End fitting pretensioner, driver's seat

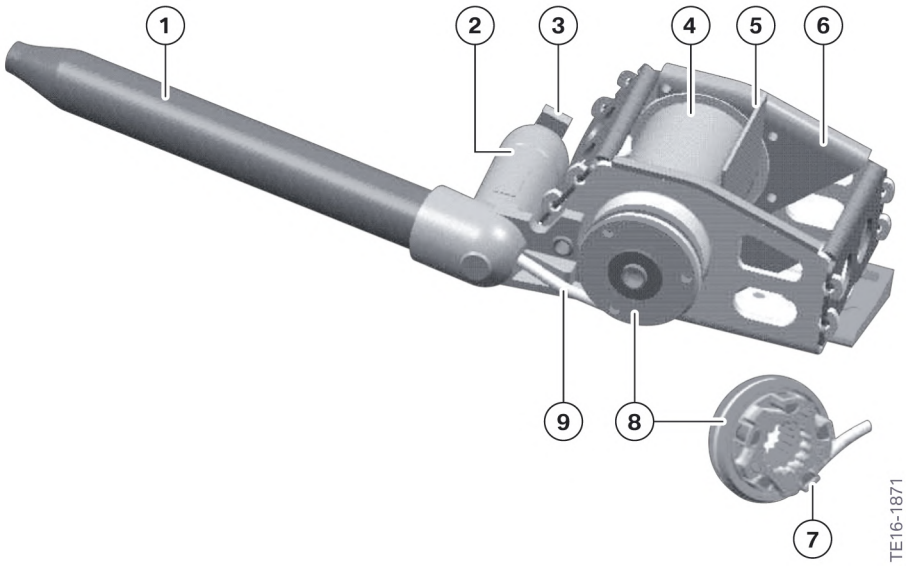
Index	Explanation
1	Belt reel
2	Steel cable
3	Ignition squib
4	Tensioning tube
5	Line

The end fitting pretensioner has a lower triggering threshold than the front airbags. Depending on the accident severity, this means that only the end fitting pretensioner may also be tripped by the Crash Safety Module ACSM.

End fitting pretensioner in the rear passenger compartment

This graphic shows an end fitting pretensioner in the rear passenger compartment:

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.



End fitting pretensioner in the rear passenger compartment

Index	Explanation
1	Tensioning tube
2	Gas generator
3	Connection for seat belt strap ignition squib
4	Belt reel
5	Seat belt webbing
6	Housing
7	Roller coupling
8	Pulley
9	Cable

Combination of end fitting pretensioner and seat belt buckle pretensioner

The combination of a end fitting pretensioner and a seat belt buckle pretensioner prevents a collision with the steering wheel above all for smaller persons who are sitting close to the steering wheel. In addition, the deployment times of the airbag can be adapted in order to reduce the load forces acting on the occupants.

The following graphic shows a seat belt system which includes an end fitting pretensioner as well as a seat belt buckle pretensioner:

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.



End fitting pretensioner and seat belt buckle pretensioner

Index	Explanation
1	Seat belt buckle tensioner
2	Seat belt
3	End fitting pretensioner

Seat-integrated belt system

Three-point seat belts integrated into the seat are used as the seat belt systems in the front seats of convertibles and the F12. The entire seat belt system is integrated into the seat. If the seat backrest is not locked, a Check Control message is displayed when the vehicle is in motion as the seat belts cannot provide a restraining effect if the backrest moves forward in a crash situation.

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.



F12 Seat belt

Index	Explanation
1	Seat belt buckle tensioner
2	Side head airbag
3	Automatic tensioner with linear force limiter

In the F12, the automatic reel is located under the front seat pan, thereby reducing the weight of the backrest. The front seats are equipped with belt tensioners, and the front passenger seat is additionally equipped with a seat occupancy detector.

8.3. Belt force limiter

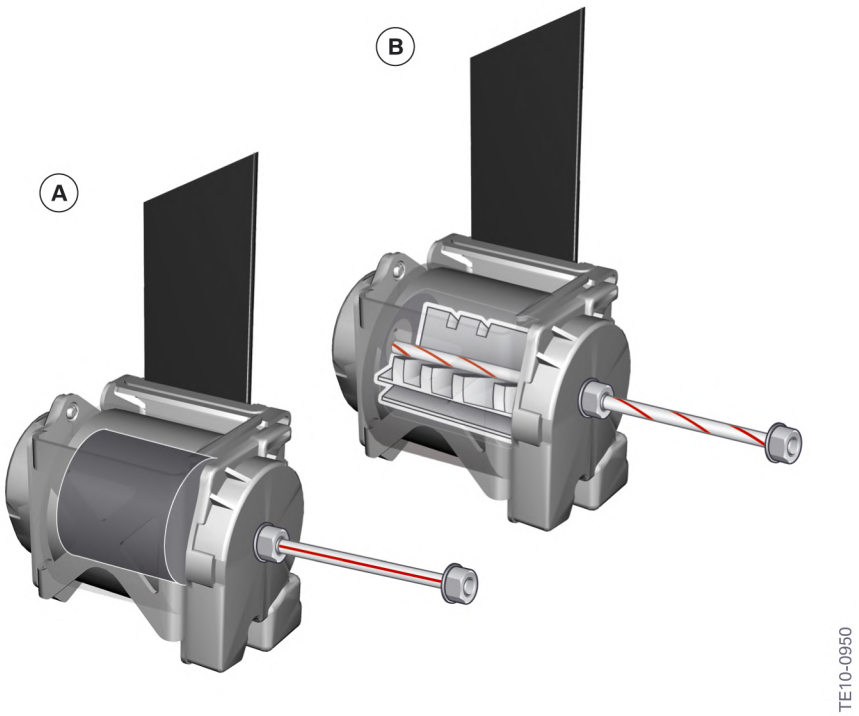
The seat belts are the primary restraint system for all occupants. The seat belts are equipped as standard with belt force limiters in order to minimise the load forces acting in the thoracic region of occupants in a severe front-end collision. The mechanical belt force limiters guarantee that the seat belt strap yields in a defined way as from a certain load. This reduces the danger of injury due to the belt forces acting on the body.

8.3.1. Automatic reel with linear force limiter

In the event of a front-end or rear-end crash or rollover, the automatic reel blocks the extension of the seat belt in order to restrain the occupants. The belt force limiter limits the load exerted by the belt on the occupants during the accident and the airbags provide the remaining restraining effect.

Above the belt force limit, the airbag takes on the forces that occur, in addition to the seat belt. For this reason, the airbag and seat belt are precisely matched to one another.

With the linear system, the belt force is limited after the locking pins have been sheared off by the twisting motion of a torsion bar spring in the seat belt inertia reel.



F30 Automatic reel with force limiter

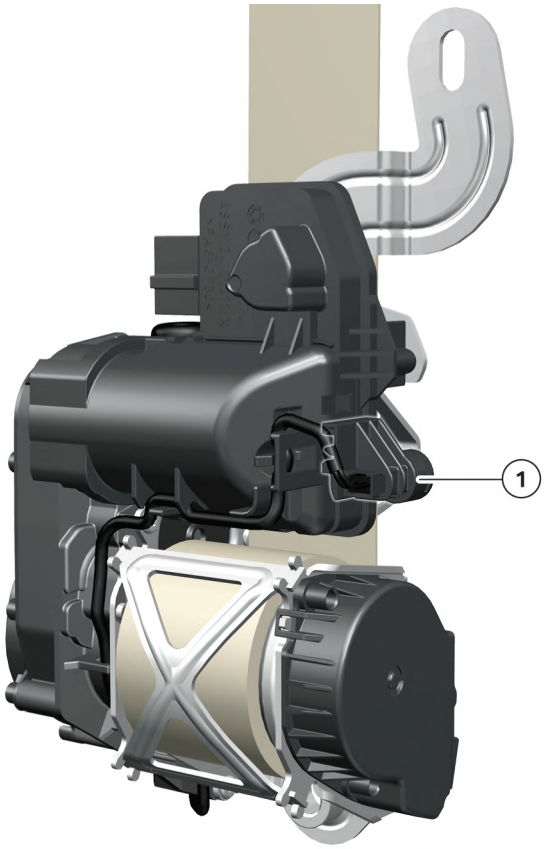
Index	Explanation
A	Torsion bar spring unstressed
B	Torsion bar spring twisted

8.3.2. Automatic tensioner with linear mechanical force limiter

Depending on vehicle, reel pretensioners with linear mechanical force limiters may be installed in the front of the vehicle and in the rear passenger compartment on the left and right.

In the event of a front-end or rear-end crash or rollover, the automatic reel blocks the extension of the seat belt in order to restrain the occupants. The belt force limiter limits the load exerted by the belt on the occupants during the accident and the airbags provide the remaining restraining effect.

With the linear system, the belt force is limited after the locking pins have been sheared off by the twisting motion of a torsion bar in the inertia reel.



G30 Belt tensioner with linear mechanical force limiter

Index	Explanation
1	Connection of ignition squib, belt tensioner

8.3.3. Automatic tensioner with adaptive force limiter

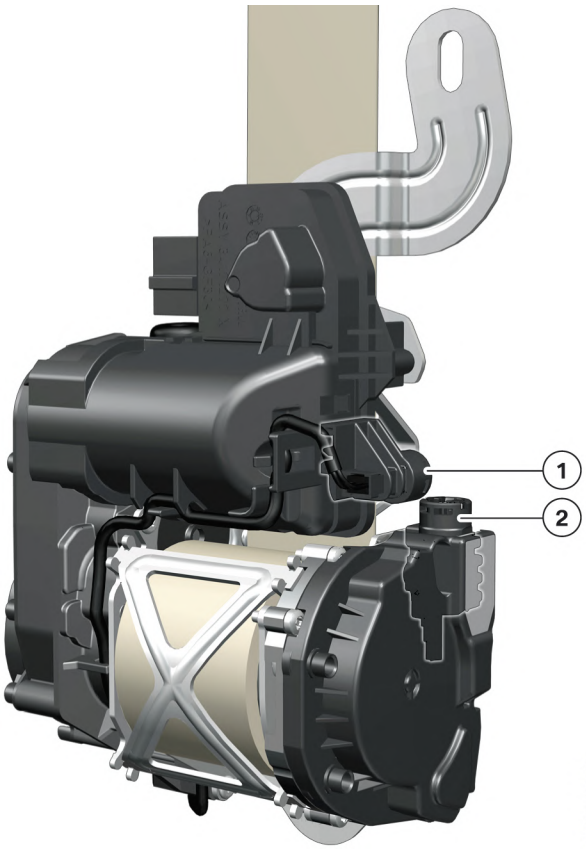
Depending on the vehicle equipment, an automatic reel with adaptive force limitation may be installed for the driver and front passenger. With the aid of a gas generator, there is a changeover from a high to a low power level during the impact, in order to reduce the seat belt restraining forces.

On vehicles with Active Protection (OE 5AL), a reel pretensioner with adaptive force limitation is installed for the driver and front passenger. With the help of a gas generator, the reel pretensioner ensures that the seat belt strap is already retracted in order to reduce the belt slack before the start of the forward movement of the occupants. Since the reel pretensioner with adaptive force limiter is installed only in combination with the optional equipment Active Protection (OE 5AL), it is also equipped with a reversible electromotive reel.

With the assistance of a gas generator, the adaptive force limiter switches from a high level force to a lower level force depending on the seat position. This reduces the restraint force of the seat belt during the crash, also taking into account the size of the occupants. The further forward the seat is, the earlier the changeover takes place.

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.

When optimally harmonised with the airbag, it ensures the kinetic energy acting on the occupants is more evenly dissipated for the duration of the head-on collision. Thus lower occupant stress values are achieved.



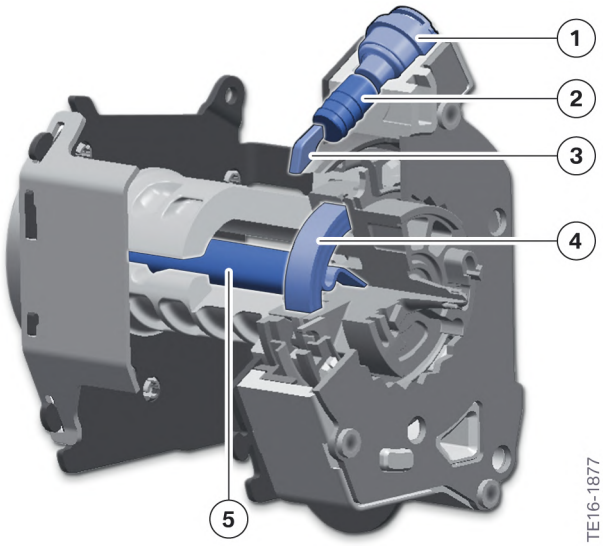
G30 Belt tensioner with adaptive force limiter

Index	Explanation
1	Connection of ignition squib, belt tensioner
2	Connection of ignition squib, adaptive force limiter

Design

The adaptive force limitation is based on a two-step torsion bar (step shaft). The torsion bar consists of the two head ends at left and right, the step and the central head.

Function



Design of an adaptive belt force limiter

Index	Explanation
1	Gas generator
2	Piston
3	Slide
4	Flat spiral spring
5	Torsion shaft

Adaptive force limitation acts in 2 stages. The kinetic energy is first decreased by the flat spiral spring and torsion shaft in the inertia reel. The gas generator is then ignited, which results in the piston being moved.

The piston then presses on the slide, which performs a defined rotational movement as a result. A retaining ring is pressed away by this movement and the flat spiral spring decoupled.

In addition, the slide wedges itself in the housing of the inertia reel and the kinetic energy is subsequently reduced via the torsion shaft. Thanks to optimum coordination with the airbag, the kinetic energy of the occupants is reduced more uniformly over the duration of the impact. Thus lower occupant stress values are achieved.

8.4. Electromotive reel

The electromotive reel EMA was used for the first time in the F01/F02.

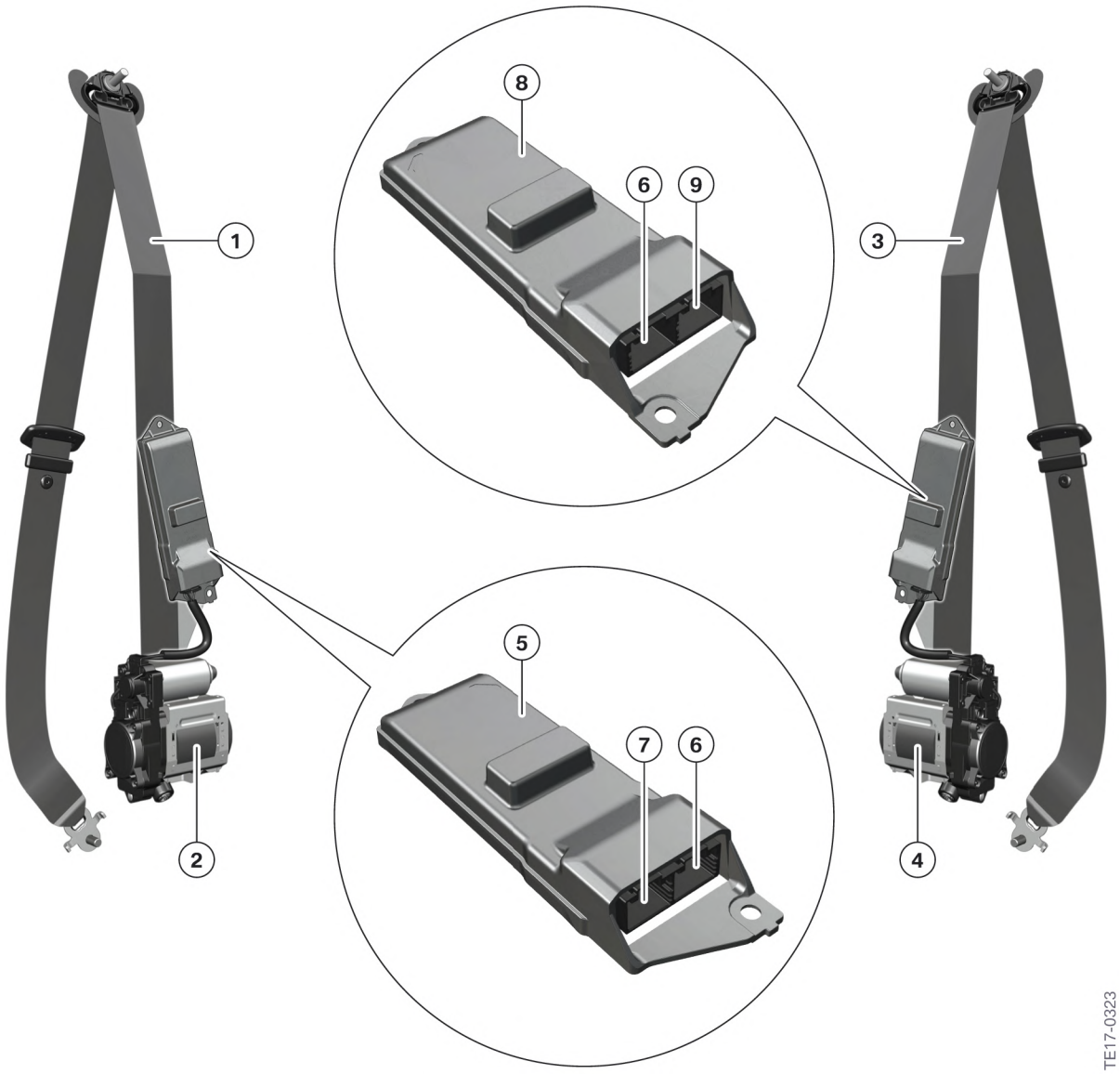
The automatic reel reduces the belt slack when the seat belt is fastened. An electric motor integrated in the inertia reel pulls in the seat belt with a predefined force. The seat belt therefore fits the occupant more snugly. This reduces the load forces acting on occupants in the event of a collision.

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.

When the seat belt is fastened (door closed), the control unit (EMALI or EMARE) activates the drive once for a certain period. The seat belt strap is retracted by means of the electric motor and the belt slack eliminated.

The automatic reel control unit is located externally and is installed in the B-pillar.

The following graphic shows the electromotive reel:



Overview of electromotive reel

Index	Explanation
1	Seat belt, right
2	Drive for right electromotive reel
3	Seat belt, left
4	Drive for left electromotive reel
5	Control unit for right electromotive reel (EMARE)
6	Plug connection to vehicle electrical system
7	Plug connection to drive for right electromotive reel
8	Control unit for left electromotive reel (EMALI)
9	Plug connection to drive for left electromotive reel

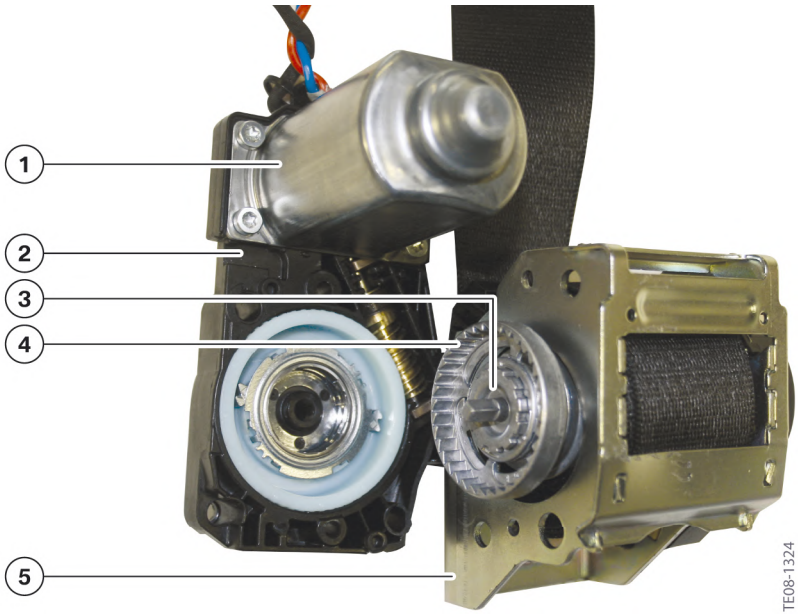
The control unit also activates the drive in critical driving situation or in the event of a possible accident. The duration is determined so that the drive is activated by the control unit up to the end of the critical driving situation.

The two control units (EMALI and EMARE) use the data from the following systems to calculate whether there is a critical driving situation (e.g. vehicle oversteer):

- **Integrated Chassis Management (ICM)**
  The ICM control unit provides the systems in the entire vehicle with information on the driving dynamics condition in the form of signals (e.g. data such as lateral and longitudinal acceleration, yaw rate).
- **Dynamic Stability Control (DSC)**
  The DSC control unit provides information such as speed and brake pressure.
- **Advanced Crash Safety Module (ACSM)**
  The Crash Safety Module sends a message on the status of the seat belt buckle contacts.

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.

8.4.1. Design



Electromotive reel

Index	Explanation
1	Electric motor
2	Transmission unit
3	Belt shaft
4	Clutch bell
5	Inertia reel

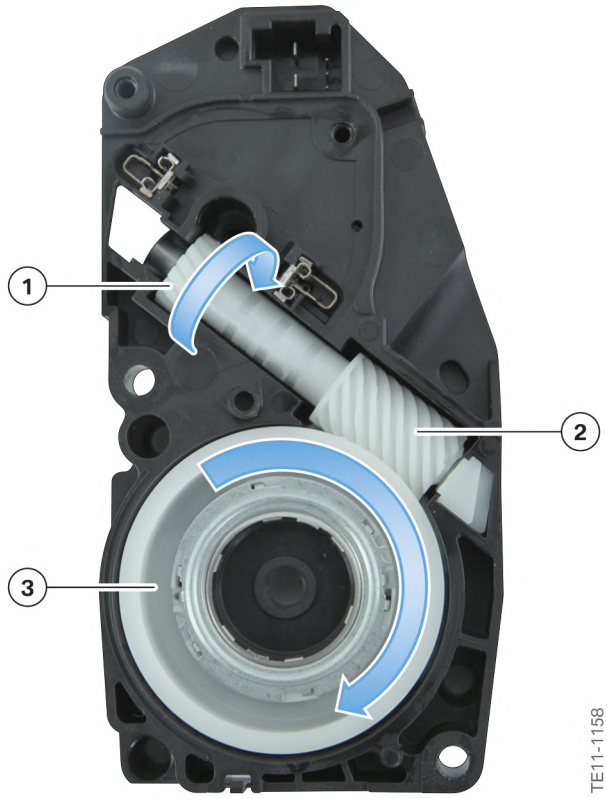
8.4.2. Function

If the electromotive reel is not in operation, the locking pawls are retracted. The clutch bell and the belt shaft are freely rotatable. The seat belt can be pulled out or reeled in.

If the electromotive reel is put into operation to reduce the belt slack, the electric motor is activated and moves the drive shaft with the worm gear.

The worm gear turns the drive gear with the clutch. The locking pawls extend and reach into the clutch bell.

The clutch bell drives the belt shaft. The seat belt is wound up and thus shortened. The seat belt is tensioned.



Function of electromotive reel

Index	Explanation
1	Sprocket of drive shaft
2	Worm gear
3	Driven gear with clutch

8.5. Reversible electric-driven reel

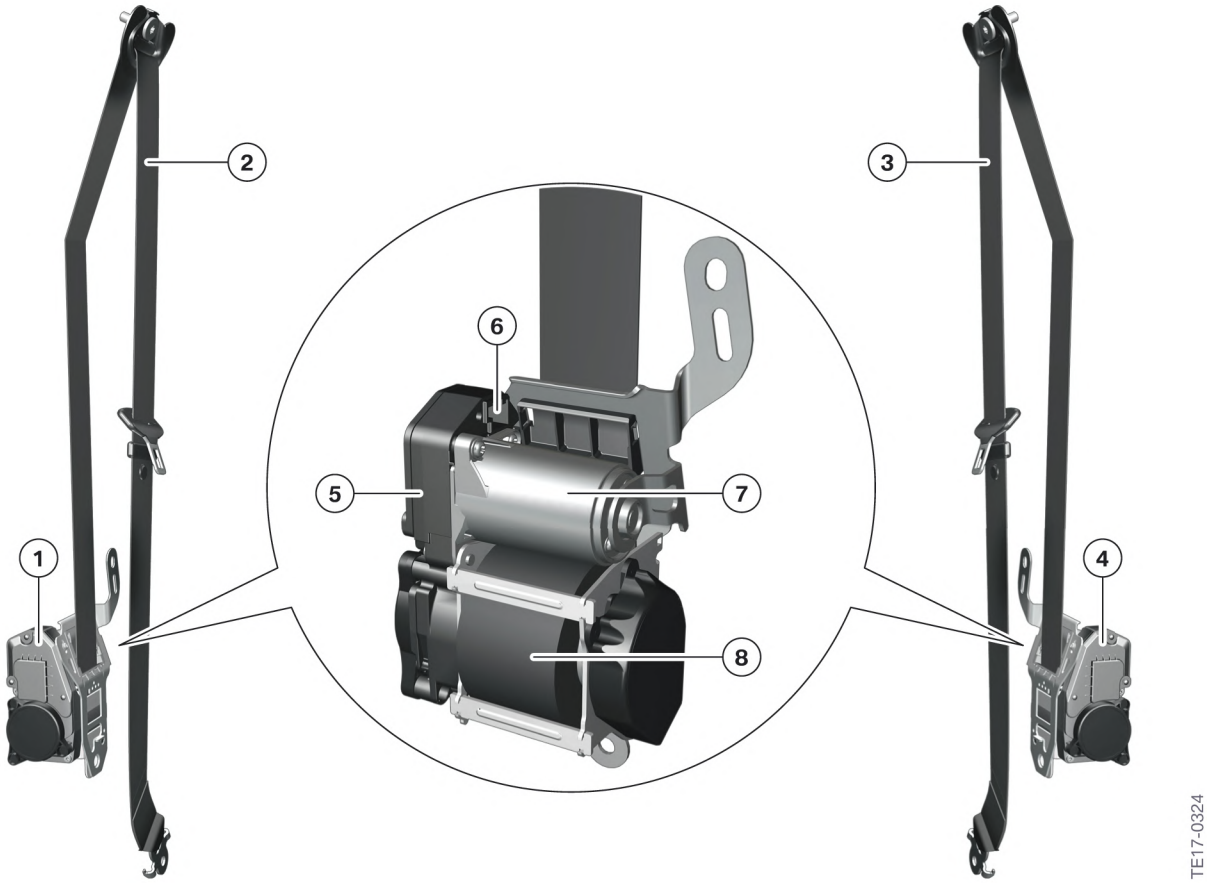
The control unit of the reversible electromotive reel left (REMALI) or right (REMARE) is used for activation of the drive. This is a further development of the electromotive reel from the F01/F02. The reversible electromotive reel was first used in the F30.

The inertia reel may be an automatic reel with linear force limiter like in the European version or a reel pretensioner with adaptive force limiter like in the US version.

The reversible electromotive reel control unit is integrated in the electromotive reel.

The following graphic shows the reversible electromotive reel:

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.



Overview of reversible electromotive reel

Index	Explanation
1	Seat belt, right
2	Reversible electromotive reel right with drive unit and integrated control unit (REMARE)
3	Seat belt, left
4	Reversible electromotive reel left with drive unit and integrated control unit (REMALI)
5	Lid (a printed circuit board and the gear housing with gear unit are located under this)
6	5-pin plug connection
7	Electric motor with gear unit (drive for reversible electromotive reel)
8	Automatic reel

The system of the reversible electromotive reel includes the following functions:

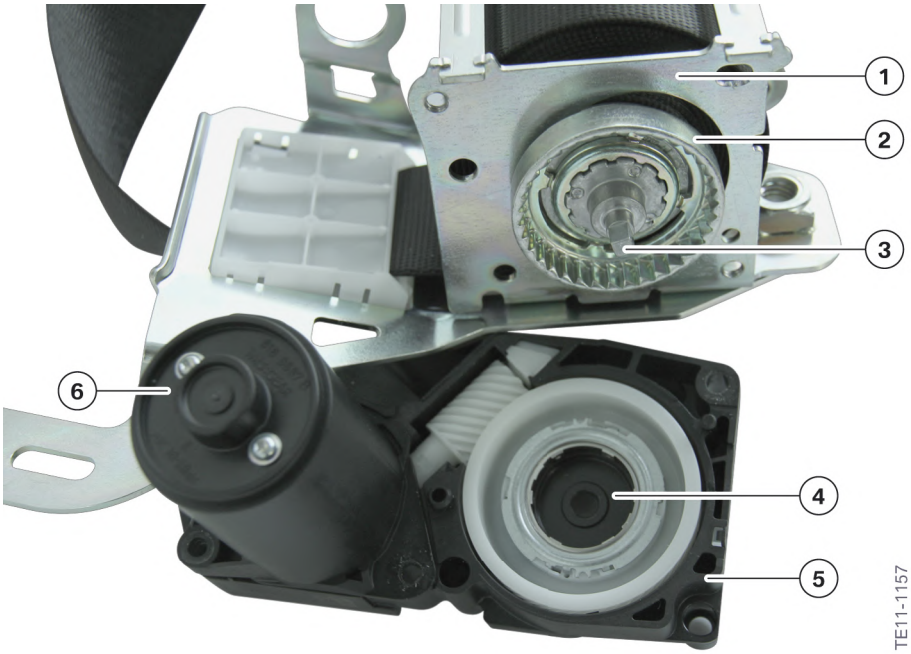
- Minimising belt slack

The seat belt has an optimum snug fit on the body of the driver or front passenger.

- Stabilisation of the seat position in a critical driving situation or in the event of a possible accident
  Pretensioning the seat belt to optimise the restraint system.

8.5.1. Design

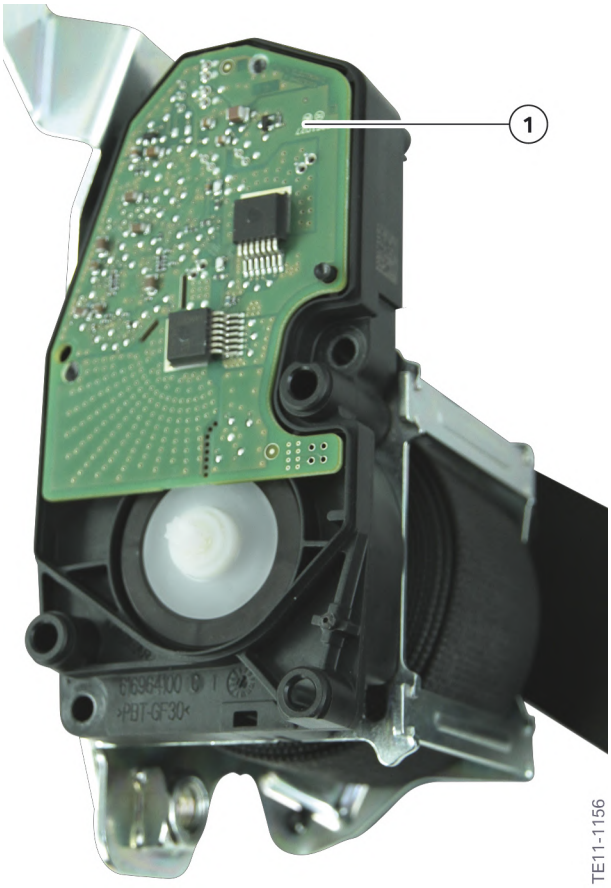
The reversible electromotive reel consists of the automatic reel, an electric motor with gear unit and an integrated control unit.



Reversible electromotive reel REMA

Index	Explanation
1	Automatic reel
2	Clutch bell
3	Belt shaft
4	Driven gear with clutch
5	Transmission unit
6	Electric motor

The REMA control unit is under a cover in the transmission unit of the REMA.



Reversible electromotive reel REMA

Index	Explanation
1	REMA control unit

8.5.2. Function

In the case of the "minimising belt slack" function, only the signal from the seat belt buckle switch and the driving speed are taken into account. After the seat belt is fastened, the belt slack is therefore minimised depending on the speed signal. In other words, the control unit (REMALI or REMARE) activates the drive once for a defined period as from an encodable driving speed. The seat belt strap is retracted by means of the electric motor and the belt slack eliminated.

The control unit also activates the drive in critical driving situation or in the event of a possible accident. The duration is determined so that the drive is activated by the control unit up to the end of the critical driving situation.

The two control units (REMALI and REMARE) use the data from the following systems, for example, to calculate whether there is a critical driving situation (e.g. vehicle oversteer):

- **Digital Motor Electronics (DME) or Digital Diesel Electronics (DDE)**

The Digital Motor Electronics or Digital Diesel Electronics supplies the signals of the accelerator pedal position.

- **Integrated Chassis Management (ICM) in combination with 4th generation ACSM**
  The ICM control unit provides the systems in the entire vehicle with information on the driving dynamics condition.
- **Dynamic Stability Control (DSC)**
  The DSC control unit provides information on the brake pressure.
- **Advanced Crash Safety Module (ACSM)**
  The Crash Safety Module sends a message on the status of the seat belt buckle contacts.

After evaluation of various signals and messages, the control unit detects whether pretensioning of the seat belt is necessary.

If the reversible motor-driven reel is not in operation, the locking pawls are retracted. The clutch bell and the belt shaft are freely rotatable. The seat belt can be pulled out or reeled in. If the reversible electromotive reel is put into operation to reduce the belt slack, the electric motor is activated and moves the drive shaft with the worm gear. The worm gear turns the drive gear with the clutch. The locking pawls extend and reach into the clutch bell.

8.6. Backrest lock

On seats with integrated seat belt system, it must be ensured that the backrest is correctly locked in order to guarantee fault-free functioning. The backrest lock must therefore be monitored. Monitoring of the backrest lock is performed by the driver's/front passenger's seat modules.

8.7. Seat belt warning light

A visual and audible warning is issued if the seat belt is not fastened or is unbuckled during the journey.

The following graphic shows the seat belt warning light in the instrument cluster:

SAFETY SYSTEMS.
8. SEAT BELT SYSTEMS.



G11/G12 seat belt warning light

SAFETY SYSTEMS.
9. SEAT OCCUPANCY DETECTION.

9.1. Visual and acoustic seat belt warning for driver's and front passenger belt

The signals of the 2 seat belt buckle switches (driver's seat belt, front passenger belt) are monitored separately for the seat belt warning. If the front passenger seat is detected as being occupied via the seat occupancy detection function, the front passenger belt must also be inserted in the seat belt buckle. The seat belt warning then goes out.

The following situations are taken into account for output of a seat belt warning:

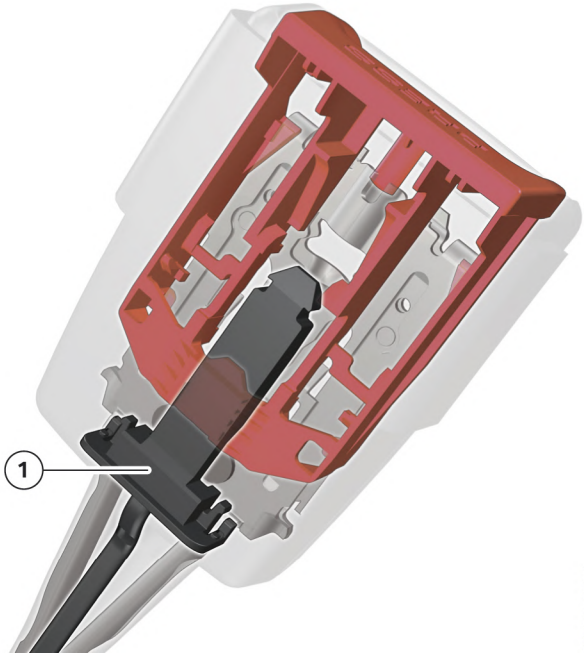
- **Front seat belt not fastened and distance travelled less than 200 m**
  Only a visual seat belt warning is output for a driving distance of less than 200 m (the seat belt warning light lights up). The vehicle can therefore be driven out of the garage, for example, without activating an acoustic seat belt warning.
- **Front seat belt not fastened and distance travelled more than 200 m**
  The seat belt warning light lights up (visual seat belt warning). The acoustic seat belt warning is switched on for approx. 90 s. The visual seat belt warning goes out when the seat belt is fastened. The acoustic seat belt warning is switched off. The seat belt warning light lights up if the seat belt is not fastened after the end of the acoustic seat belt warning.
- **Release of front seat belt while the vehicle is moving**
  The seat belt warning light lights up immediately if a seat belt buckle is opened while the vehicle is moving. A short acoustic signal sounds. After approx. 15 s, an acoustic seat belt warning is switched on additionally for approx. 90 s. The visual seat belt warning goes out when the seat belt is fastened. The acoustic seat belt warning is switched off. The seat belt warning light lights up if the seat belt is not fastened after the end of the acoustic seat belt warning.

9.2. Seat belt buckle switch

The seat belt buckle switches are located in the seat belt buckles of the driver and front passenger seat. Vehicles with the optional equipment EU-specific additional contents (OE 230) are also equipped with seat belt buckle switches on the seat bench.

The seat belt buckle switch detects whether the seat belt buckle tongue is in the seat belt buckle. The Crash Safety Module (ACSM) supplies power to the sensors and performs the evaluation. As from driving readiness, the seat belt buckle switch is permanently monitored and used for the visual and acoustic seat belt warnings and also to determine which restraint systems are to be triggered. The seat belt buckle switch is also used for the seat belt warning function.

SAFETY SYSTEMS.
9. SEAT OCCUPANCY DETECTION.



G11/G12 Seat belt buckle

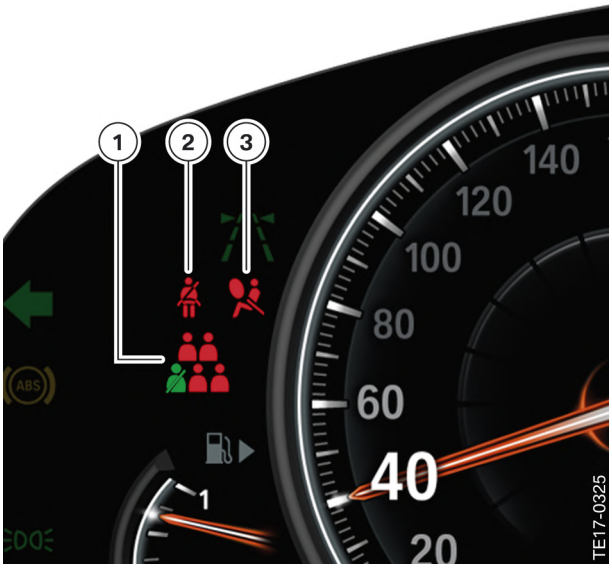
Index	Explanation
1	Hall effect sensor

The seat belt buckle switch is an hall effect sensor which is installed in all monitored seat belt buckles. A current flows through the hall effect sensor and through a perpendicular magnetic field. The sensor supplies an output voltage which is proportional to the product of magnetic flux density and current.

In all current vehicles, seat belt buckle switches are installed in the seat belt buckles in the rear passenger compartment in order to check the buckle status.

The driver can see by means of an indicator light whether the passengers on the front seats or in the rear passenger compartment have fastened their seat belts.

SAFETY SYSTEMS.
9. SEAT OCCUPANCY DETECTION.



Seat belt monitoring

Index	Explanation
1	Seat belt warning
2	Airbag system
3	Seat belt warning, rear passenger compartment

9.3. Seat occupancy mat

The European version of the vehicles is equipped with the seat occupancy mat; a load-sensing mat installed within the seat surface of the front passenger seat. The seat occupancy of the front passenger seat is detected via the seat occupancy mat. If the front passenger seat belt is not fastened, the visual and acoustic seat belt warning is activated. The seat occupancy mat cannot classify occupants and is therefore not used for automatic deactivation of the front passenger airbag.

Since the ACSM 4, the seat occupancy data is no longer evaluated in the seat occupancy mat but is now evaluated in the ACSM instead.

SAFETY SYSTEMS.
9. SEAT OCCUPANCY DETECTION.



TE17-0326

Seat occupancy mat

SAFETY SYSTEMS.
10. EMERGENCY CALL.

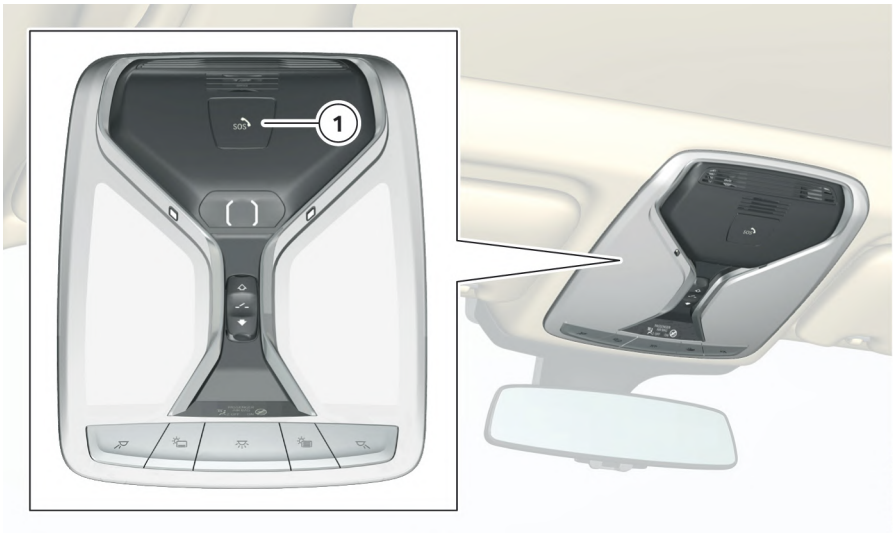
10.1. Introduction

The emergency call functions are country-specific. Prerequisites for an emergency call are an active SIM card and the availability of a mobile phone network.

The button has a direct connection to the telematics control unit.

By pressing the emergency call button, the driver can establish a connection to an emergency services control centre using the manual emergency call function.

In the case of the automatic emergency call, a call is automatically made to an emergency services control centre after an accident of corresponding severity.



G11/G12 roof function centre (FZD) with emergency call button

Index	Explanation
1	Emergency call button

10.2. Vehicles with intelligent emergency call equipment

Vehicles which are **not** equipped with the intelligent emergency call function have an emergency call button in the roof function centre FZD. The occupants can make an emergency call to an emergency coordination centre using the emergency call button (voice call **without** data transmission).

In the event of an accident with corresponding crash severity, the Crash Safety Module sends a message to the telematics control unit. The telematics control unit initiates an automatic emergency call to an emergency services control centre (voice call **without** data transmission).

10.3. Vehicles without intelligent emergency call function

The intelligent emergency call function always includes the option of a manual or automatic emergency call (voice call and data transmission). A manual emergency call can be made in a vehicle equipped with the intelligent emergency call function. A voice contact is established and data transmitted, e.g. the current position. The emergency call is made to a BMW Call Centre or an emergency coordination centre, depending on the national-market version.

In the event of an accident with corresponding crash severity, the Crash Safety Module sends a message to the telematics control unit. The telematics control unit initiates an emergency call.

With this automatic emergency call, the presumed number of people in the vehicle and sensor data, among other things, are transmitted to the BMW Call Centre, in order to identify the severity of the crash.

The information about the accident available at the call centre can be forwarded to the emergency coordination centre. The emergency coordination centre can then initiate appropriate measures.

Attempts are made at the same time to establish a voice contact with the occupants of the vehicle in order to obtain more detailed verbal information about the accident (e.g. condition of occupants).

The following table shows which functions are available for an emergency call with and without the equipment option:

Equipment	Manual emergency call	Automatic emergency call
Equipment **without** “intelligent emergency call”	• Voice call (emergency coordination centre)	• Voice call (emergency coordination centre)
Equipment **with** “intelligent emergency call”	• Voice call (BMW Call Centre) • Data transfer (location)	• Voice call (BMW Call Centre) • Data transfer (location) • Sensor data transmission

If, for various reasons, a connection to the BMW Call Centre cannot be established, the emergency call is automatically forwarded to the emergency coordination centre.

SAFETY SYSTEMS.
11. ACTIVE SAFETY.

Active safety is the term used to designate design measures on the vehicle which help to avoid accidents.

The most important aspects of active safety are:

- **Ride stability**
  This is the result of the chassis and suspension design with respect to drive concept, suspension and cornering stability. It also includes steering precision, braking power and braking stability.
- **Fitness safety**
  This describes the load on the occupants due to vehicle vibrations, noises from the chassis and suspension and engine as well as climatic influences.
- **Perception safety**
  This is improved by correspondingly good design of the lighting equipment on the vehicle as well as visibility (360° view, good view to the rear with the smallest possible blind spot).
- **Operating safety**
  This is achieved by good ergonomics and the logically correct arrangement of all control levers and switches, e.g. for lighting, wipers, climate control or infotainment. The elements must be as easy as possible to reach from the driver's position.

Further details on active safety can be found in the reference material ICT “Electrical system II – Assistance systems”.



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